A Conditional Full Conservative Derivation of the Two-Body 4PN Sector from the Unified 4D Toy Model

Trevor Norris

April 12, 2026

Abstract

We present the full conservative fourth post-Newtonian (4PN) assembly for the 4D toy-model program within the same declared closure hierarchy used at lower orders. The strict Schwarzschild one-body gate, exact quartic Legendre compiler, Hamiltonian-first generic-frame lift, canonical comparable-mass local residual, aligned-seed correction, and referee reconstruction together show that the entire local instantaneous 4PN sector already has an exact generic-frame ordinary representative. We then isolate the hereditary sector and show that its coefficient is not a new independent datum, but is fixed by the same canonically normalized orbital/worldtube STF quadrupole branch that controls the 2.5PN Burke–Thorne channel, through $C_{\text{tail}} = (GM/2c^3)\,\gamma^{\text{eff}}_{\text{quad}}$. Hence 4PN opens no new normalization constant: if $\gamma^{\text{eff}}_{\text{quad}} = 2G/(5c^5)$, then $C_{\text{tail}} = G^2M/(5c^8)$ and the full conservative 4PN local+tail structure matches the standard GR target. The remaining gap is therefore the same passive/outgoing quadrupole-normalization theorem on the moving-throat branch already isolated by the 2.5PN program.

1 Introduction and scope

1.1 Motivation: from full conservative 3PN and the 2.5PN narrowing to the local-plus-tail 4PN problem

The earlier papers in this program already froze a substantial conservative and reduction-theoretic backbone. On the parent-theory side, the unified 4D framework supplies the exact projection map, the exact projected continuity law with leakage, the exact longitudinal identity, the controlled quasi-static Poisson hook, and the coherent-defect / worldtube reduction used to define the brane-facing point-particle sector. On the post-Newtonian side, the lower-order ledger already carries the Newtonian, conservative 1PN, conservative 2PN, and conservative 3PN sectors within one declared closure hierarchy. In particular, the carried viability conditions remain

$$\kappa_\rho = 1, \qquad n = 5, \qquad \kappa_{\text{add}} = \frac{1}{2}, \qquad \kappa_{\text{PV}} = \frac{3}{2}, \qquad \beta_{\text{1PN}} = 3.$$

Just as importantly, the conservative 2PN and 3PN steps already sharpened the particle/throat ontology well beyond a pure scalar monopole picture. The carried structure now includes an odd dipole wake sector, a real $P_0 \oplus P_2$ mouth/support layer, and a separate geometry-closure channel, with the conservative 3PN middle block already assigned in the fixed ADM chart inside the same hierarchy.

The 2.5PN program then made the higher-order bridge much narrower. Its main structural lesson was that the surviving universal point-particle retarded route is the canonically normalized orbital/worldtube STF quadrupole branch. In the conservative bookkeeping used by the present series, that branch is the same higher-order lane that had already begun to appear as the grouped real

$$P_{20} \oplus P_{21} \oplus P_{22}$$

bundle. This narrowing changes the meaning of the next conservative step. After 3PN, the right question is no longer whether one can continue adding local coefficients somehow. The right question is whether the first conservative order that must be stated explicitly as a *local instantaneous sector plus a hereditary/tail sector* can still be closed inside the same hierarchy, and whether that hereditary sector opens a genuinely new datum or only reuses the already-known quadrupole-normalization gate.

That is the purpose of the present paper. At 4PN the conservative near-zone ledger through order c^{-8} must be written in the form

$$L_{\rm cons} = L_0 + c^{-2}L_1 + c^{-4}L_2 + c^{-6}L_3 + c^{-8}L_4 + O(c^{-10}),$$

with the theorem split

$$L^{\rm cons}_{\rm 4PN} = L^{\rm local}_{\rm 4PN} + L^{\rm tail}_{\rm 4PN}.$$

The local side must close the strict one-body 4PN Schwarzschild gate, the quartic ordinary/Hamiltonian compiler, the full local instantaneous comparable-mass generic-frame sector, and the fixed-chart uniqueness problem. The hereditary side must determine whether the 4PN tail coefficient is a new independent normalization datum or is instead controlled by the same STF quadrupole branch that already survived the 2.5PN audit.

A first structural lesson appears already at the strict one-body level. The exact isotropic Schwarzschild gate through 4PN is *not* reproduced by a single quartic continuation of the carried 3PN denominator-style self sector. The minimal one-body repair ledger is instead

$$\mu_{\rho,4} = \frac{1}{8}, \qquad d_4 = \frac{205}{16}, \qquad s_{34} = -\frac{15}{32}, \qquad s_{26} = -\frac{1}{16},$$

so the 4PN one-body problem is already not a one-parameter extension of the 3PN closure. But the one-body sector is still only the front door. The deeper algebraic question is whether the full local comparable-mass c^{-8} residual can be assigned exactly once the lower-order ledger is frozen, the fixed local target is imported, and the generic-frame chart is handled in the correct order.

The main positive result of the paper is that it can. The raw exchange-symmetric local generic-frame scaffold is already COM-complete blockwise; the decisive chart lesson is that the lift must be done Hamiltonian-first, because the fixed local Hamiltonian target respects the constant-coefficient scaffold degree ceilings while the translated ordinary target develops extra ν^4 tails in the middle and upper interaction blocks. In the Hamiltonian chart there is no span obstruction, the canonical comparable-mass local residual translates back to the ordinary chart by exact sign flip once the lower-order ledger and aligned seed convention are frozen, the aligned-seed ordinary correction admits an exact generic-frame lift after the structured seed enlargement recorded in the audit chain, and the full local instantaneous 4PN ordinary sector therefore has an exact generic-frame representative.

The hereditary side is narrower still. The exact bridge derived in the notes is

$$C_{\rm tail} = \frac{GM}{2c^3}\,\gamma^{\rm eff}_{\rm quad},$$

so the 4PN tail introduces no new independent normalization datum. Full GR matching of the hereditary coefficient is equivalent to the already-known 2.5PN normalization target

$$\gamma^{\rm eff}_{\rm quad} \stackrel{?}{=} \frac{2G}{5c^5}.$$

Thus the remaining full-4PN conditional step is not another open-ended coefficient search. It is the same passive/outgoing STF quadrupole normalization theorem that the 2.5PN program had already isolated.

The scope of the paper is therefore deliberately strong but intermediate. It is stronger than a notebook-level or target-matching statement because the entire local instantaneous 4PN sector is already assembled exactly and the hereditary interface has been reduced to one named normalization gate. It is weaker than a fully first-principles moving-throat PDE theorem because the passive/outgoing quadrupole normalization still awaits a completed branch derivation, and the microscopic origin of the grouped real conservative quadrupole data remains a deeper dynamical problem. The correct reading is the same as in the earlier full conservative 1PN, 2PN, and 3PN papers: a full assembly theorem *within a declared closure hierarchy*, not an assumption-free theorem of the fully solved bulk problem.

A second purpose of the paper is bookkeeping clarity. At 4PN the argument mixes carried exact lower-order structure, controlled one-body and COM reductions, fixed-chart import, protocol-level local seed choices, and one conditional hereditary interface that is inherited rather than newly opened. The reader should be able to see, line by line, which ingredients are exact, which depend on the declared reduction and chart choices, which live only inside the same response hierarchy that already succeeded below 4PN, and which are final theorem consequences. That separation is part of the result.

1.2 What this paper claims

The main claims of the paper are the following.

1. The strict one-body 4PN Schwarzschild gate is fixed exactly, and its minimal repair ledger already shows that 4PN is not a one-parameter continuation of the carried 3PN one-body closure.

2. The exact quartic perturbative Legendre-transform identity closes the chart bridge through 4PN and provides the unique compiler needed for any trustworthy local comparable-mass lift.

3. The raw exchange-symmetric generic-frame local scaffold is already COM-complete blockwise, so local 4PN never had an early span shortage at the level of kinematic image size.

4. The fixed local 4PN target must be lifted Hamiltonian-first, because the Hamiltonian degree ceilings match the constant-coefficient scaffold while the translated ordinary target develops extra ν^4 tails.

5. In the Hamiltonian chart there is no span obstruction: the generic-frame local interaction residual is fully reachable block by block, and one exact canonical comparable-mass local residual representative exists.

6. Once the lower-order ledger and aligned-seed convention are frozen, the canonical local residual translates back to the ordinary chart by exact sign flip,

$$\Delta H_4 = -\Delta L_4,$$

   and the aligned-seed ordinary correction admits an exact structured generic-frame lift.

7. The entire local instantaneous 4PN ordinary sector therefore has an exact generic-frame representative and can be reconstructed referee-style in the fixed local chart.

8. The hereditary 4PN coefficient is not a new independent datum; it is fixed by the same canonically normalized STF quadrupole branch already isolated by the 2.5PN program, namely

$$C_{\text{tail}} = \frac{GM}{2c^3}\,\gamma^{\text{eff}}_{\text{quad}}.$$

9. Therefore the remaining 4PN gap is exactly the same passive/outgoing quadrupole normalization problem that already survived the 2.5PN audit. In particular, the local 4PN bottleneck is no longer an algebraic coefficient search.

Stated differently, the point of the paper is not that the model can be tuned until a 4PN-looking ledger appears. The point is that, once the lower-order stack, the fixed local target, and the Hamiltonian-first chart logic are handled correctly, the full conservative c^{-8} problem narrows to one closed local theorem chain plus one already-named hereditary normalization gate.

1.3 What this paper does not claim

The negative side of the scope is equally important.

First, this paper does *not* claim an assumption-free theorem that begins with a fully solved moving-throat PDE and ends automatically with the complete conservative two-body 4PN dynamics. As in the earlier full conservative papers, the point-particle limit used here is a controlled reduction inside a declared hierarchy. The main text therefore presents a reduced-theory theorem chain, not a finished theorem of the full moving-throat bulk problem.

Second, this paper does *not* claim that the passive/outgoing quadrupole normalization has already been derived microscopically from the completed throat PDE. The hereditary bridge proved here is that the 4PN tail coefficient is controlled by the same canonically normalized STF quadrupole branch as the 2.5PN odd channel. The actual branch normalization on the true moving-throat solution remains outside the present theorem.

Third, this paper does *not* claim a final uniqueness or interpretation theorem for every aligned-seed local representative in the ordinary chart. What is established here is that the local instantaneous ledger is closed exactly inside the declared hierarchy and fixed chart once the aligned-seed convention is handled by the structured generic-frame lift. The deeper microscopic meaning of that aligned-seed sector is a separate follow-on problem.

Fourth, the paper does *not* claim any new dissipative or radiative theorem beyond the already narrowed quadrupole route. The scalar and dipole dangers had already been demoted above universal point-particle 2.5PN on the natural small-body branch, and the present paper does not reopen those channels.

Fifth, the paper is restricted to the nonspinning conservative two-body sector through order c^{-8}. It does not attempt a treatment of spin couplings, strong-field nonperturbative completion, generic many-body response, or higher-PN sectors beyond the present conservative 4PN scope. Nor does it claim to replace the broader moving-throat script-side program by a single closed PDE paper.

These non-claims are not qualifications to hide. They are part of the correct interpretation of the result. The paper should be read as establishing the sharpest honest conservative 4PN statement currently supported by the program: a full local-plus-tail assembly theorem within the declared hierarchy, with the remaining open work shifted from local algebra to microscopic quadrupole normalization on the true passive/outgoing branch.

1.4 Claim taxonomy

To keep the argument referee-safe, we retain the same bookkeeping discipline used in the earlier full conservative papers.

Exact. Identities that follow directly from the frozen lower-order ledger, the exact one-body 4PN target, the exact quartic compiler, the exact chart-translation theorems, symmetry bookkeeping, or the exact hereditary coefficient bridge once the target chart and basis choices have been fixed.

Controlled reduction. Steps that require an explicit reduced description, such as the carried 4D-to-brane and worldtube reductions, the strict one-body and COM reductions, the fixed local Hamiltonian import, or the grouped real quadrupole reduction inherited from the 2.5PN narrowing.

Protocol closure. Statements that depend on the same declared response hierarchy used successfully below 4PN rather than on a fully solved moving-throat PDE. In the present paper the most important examples are the carried self/static ledger, the Hamiltonian-aligned seed convention for the local ordinary chart, and the use of the already-isolated passive/outgoing STF quadrupole normalization as the hereditary interface datum.

Full assembly. The final theorem consequence after the previous ingredients are inserted. In the present paper this means that, once the lower-order ledger, the fixed local ADM target, the quartic compiler, the Hamiltonian-first lift, and the hereditary bridge are frozen, the final equality

$$L^{\rm cons}_{\rm 4PN} = L^{\rm local}_{\rm 4PN} + L^{\rm tail}_{\rm 4PN}$$

is exact algebra inside the declared hierarchy.

This taxonomy is not cosmetic. It prevents four logically distinct moves from being blurred together: carrying forward exact lower-order structure, using the declared reductions and chart choices, invoking the closure needed to define the local aligned-seed sector and the hereditary interface, and then stating the final theorem consequence that survives after those ingredients are frozen.

1.5 Roadmap

The next section gives an executive summary of the final theorem ledger. Section 3 then records the minimal carry-forward dictionary from the earlier 4D, 1PN, 2PN, 2.5PN, and 3PN papers and states the exact 4PN target. Section 4 begins the proof chain with the strict one-body Schwarzschild gate, the minimal repair ledger, the natural self/static seed, and the exact quartic Legendre compiler.

Section 5 then closes the local instantaneous theorem chain: the raw local scaffold, the fixed local target import, the Hamiltonian-chart generic-frame lift, the canonical comparable-mass residual slice, the ordinary translation, the aligned-seed correction and its generic-frame lift, and the exact local referee reconstruction. Section 6 isolates the hereditary bridge, proves that the tail coefficient is controlled by the same canonically normalized quadrupole branch already isolated at 2.5PN, and shows that 4PN opens no new independent normalization datum.

Section 7 states the final conditional conservative 4PN theorem in COM and generic-frame form. Section 8 then explains exactly how the present result interfaces with the 2.5PN program and with the moving-throat PDE: what is fixed now, what remains branch normalization, and why the correct next theorem gate is the passive/outgoing STF quadrupole normalization on the true branch. Section 9 separates frozen inputs, newly fixed quantities, and still-open objects; Section 10 records the role and limits of the referee archive; and Section 11 closes the paper.

The appendices collect the detailed one-body algebra, the quartic compiler, the local scaffold and target-import audits, the Hamiltonian-first lift, the aligned-seed correction and its structured generic-frame lift, the exact local referee reconstruction, the hereditary bridge algebra, and the verification ledger. They are essential to the theorem chain but would obscure the main logical structure if left in the body of the paper.

2 Executive summary of results

This paper closes the first conservative order of the program that must be stated explicitly as a sum of a full local instantaneous sector and a separate hereditary/tail sector. The earlier papers already froze the exact 4D reduction backbone, carried the conservative Newtonian+1PN+2PN+3PN ledger through its declared closure hierarchy, and narrowed the live higher-order universal point-particle bridge to the canonically normalized orbital/worldtube STF quadrupole branch. The present question is therefore not whether one can continue adding local 4PN coefficients somehow. It is whether the full nonspinning conservative two-body ledger through order c^{-8} can be assigned uniquely once the strict one-body gate, the exact quartic ordinary/Hamiltonian compiler, the fixed local target, the generic-frame lift, and the hereditary interface are all handled in the correct chart. The answer is now sharp: within the same declared closure hierarchy used successfully at 1PN, 2PN, and 3PN, the entire local instantaneous 4PN sector is already assembled exactly, and the only remaining conditional interface for a full conservative 4PN theorem is the same passive/outgoing STF quadrupole normalization already isolated by the 2.5PN program.

Headline result. The final theorem split is

$$L_{4\mathrm{PN}}^{\mathrm{cons}} = \underbrace{L_{4\mathrm{PN}}^{\mathrm{local}}}_{\text{closed exactly}} + \underbrace{L_{4\mathrm{PN}}^{\mathrm{tail}}}_{\text{fixed by the STF quadrupole bridge}},$$

with

$$C_{\mathrm{tail}} = \frac{GM}{2c^3}\,\gamma_{\mathrm{quad}}^{\mathrm{eff}}.$$

Thus 4PN opens no new independent normalization datum beyond the already-known quadrupole gap. Full GR recovery of the hereditary coefficient is equivalent to the single condition

$$\gamma_{\mathrm{quad}}^{\mathrm{eff}} = \frac{2G}{5c^5},$$

in which case

$$C_{\mathrm{tail}} = \frac{GM}{2c^3}\,\gamma_{\mathrm{quad}}^{\mathrm{eff}} = \frac{G^2M}{5c^8} = C_{\mathrm{tail}}^{\mathrm{GR}}.$$

What remains open is therefore no longer a local 4PN coefficient search. What remains open is the passive/outgoing STF quadrupole normalization on the true moving-throat branch.

Table 1 lists the load-bearing results established in the present paper and the role each plays in the theorem chain.

How the ledger closes. The proof chain has four positive gates and one decisive bookkeeping lesson. First, the strict one-body Schwarzschild target shows that 4PN is not a one-parameter continuation of the carried 3PN self sector. The quartic static gate, quartic denominator datum, and the two new self-sector slots in the U^3v^4 and U^2v^6 channels are all forced already at test-mass level. Second, the exact quartic Legendre compiler turns the local comparable-mass problem into a controlled ordinary/Hamiltonian bridge rather than a blind coefficient search. Third, the raw generic-frame local scaffold is already blockwise COM-complete, so the real issue is not early image shortage but the large COM-null sector and the correct choice of quotient chart. Fourth, once the local target is imported Hamiltonian-first, the entire comparable-mass local residual becomes exactly reachable, the canonical residual translates back to the ordinary chart by exact sign flip, and the aligned-seed ordinary correction admits its own exact structured generic-frame lift. The decisive bookkeeping lesson is therefore that the local 4PN lift is Hamiltonian-first. The translated ordinary target develops extra ν^4 tails in the middle and upper interaction blocks, whereas the fixed local Hamiltonian target matches the constant-coefficient scaffold degree ceilings exactly. Once this is respected, the local 4PN bottleneck is no longer algebraic span or chart ambiguity.

Table 1: Load-bearing results of the present 4PN paper.

Item	Result	Status
One-body gate	The strict isotropic Schwarzschild target through c^{-8} is fixed exactly, and the minimal repair ledger is $\mu_{\rho,4} = 1/8$, $d_4 = 205/16$, $s_{34} = -15/32$, $s_{26} = -1/16$.	Settled
Quartic compiler	The exact quartic perturbative Legendre-transform identity closes the ordinary/Hamiltonian bridge through 4PN and fixes the unique local compiler used by the Hamiltonian-first lift.	Settled
Raw local scaffold	The exchange-symmetric generic-frame local interaction scaffold has block sizes 52, 46, 29, 10, and 2, and its COM image already spans the full 15 local interaction slots blockwise.	Settled
Hamiltonian-chart lift	The local Hamiltonian target respects the scaffold degree ceilings $(4, 3, 3, 2, 2)$, the blockwise coefficient-space ranks are maximal 20/20, 12/12, 9/9, 4/4, and 2/2, and no Hamiltonian-chart span obstruction survives.	Exact closure
Canonical quotient slice	The 92-dimensional Hamiltonian-chart null family is pure COM-blind freedom. Fixing the canonical slice gives one sparse exact generic-frame Hamiltonian residual representative.	Exact quotient
Aligned-seed correction	The remaining ordinary-chart obstruction is purely a seed-alignment issue, and the correction is exactly generic-frame liftable in the structured seed spaces Q, $T \oplus (pq)T$, $S \oplus (pq)S$, $U \oplus (p^2q^2)U$, with $W = 0$.	Exact lift
Local referee reconstruction	The assembled generic-frame ordinary representative reproduces the full fixed-chart local ordinary 4PN target block by block under COM reduction.	Exact closure
Tail bridge	The GR tail has the exact Newtonian quadrupole shadow in the degree-2 G^4/r^4 block, and the hereditary coefficient satisfies $C_{\text{tail}} = (GM/2c^3)\,\gamma^{\text{eff}}_{\text{quad}}$.	Exact bridge
Remaining gap	The only conditional step left for a full conservative 4PN theorem is the passive/outgoing STF quadrupole normalization on the actual moving-throat branch.	Open

One-body gate and quartic compiler. The strict isotropic Schwarzschild one-body target is

$$\frac{L_{\text{4PN,1-body}}}{m} = \frac{7}{256}\frac{v^{10}}{c^8} + \frac{75}{128}\frac{Uv^8}{c^8} + \frac{59}{16}\frac{U^2v^6}{c^8} + \frac{203}{32}\frac{U^3v^4}{c^8} + \frac{31}{32}\frac{U^4v^2}{c^8} + \frac{1}{16}\frac{U^5}{c^8}.$$

A direct continuation of the carried 3PN denominator-style self sector does not reproduce this answer with only one new quartic datum. The exact minimal repair ledger is instead

$$\mu_{\rho,4} = \frac{1}{8}, \qquad d_4 = \frac{205}{16}, \qquad s_{34} = -\frac{15}{32}, \qquad s_{26} = -\frac{1}{16},$$

so the one-body sector already proves that 4PN is not merely the next scalar continuation of the earlier closure. Applying the exact quartic Legendre compiler then gives the strict one-body Hamiltonian gate

$$\frac{7}{256}p^{10}, \qquad \frac{45}{128}\frac{p^8}{r}, \qquad \frac{13}{8}\frac{p^6}{r^2}, \qquad \frac{105}{32}\frac{p^4}{r^3}, \qquad \frac{105}{32}\frac{p^2}{r^4}, \qquad -\frac{1}{16}\frac{1}{r^5},$$

which is the exact self/static Hamiltonian seed data used by the later local Hamiltonian-first lift.

Local scaffold, degree ceilings, and the Hamiltonian-first lesson. Symmetric promotion of the exact one-body target gives the natural local self/static seed

$$L_{4,\text{seed}} = \frac{7}{256}(m_A v_A^{10} + m_B v_B^{10}) + \frac{75 G m_A m_B}{128 r}(v_A^8 + v_B^8) + \cdots + \frac{G^5 m_A m_B}{16 r^5}(m_B^4 + m_A^4),$$

with the whole expression understood as the coefficient of c^{-8}. The comparable-mass local interaction scaffold built on top of this seed has raw block sizes

$$G/r : 52, \qquad G^2/r^2 : 46, \qquad G^3/r^3 : 29, \qquad G^4/r^4 : 10, \qquad G^5/r^5 : 2,$$

for a total of 139 exchange-symmetric generic-frame interaction directions. After COM projection these span exactly the full 15 local interaction slots, namely

$$G/r : 5, \qquad G^2/r^2 : 4, \qquad G^3/r^3 : 3, \qquad G^4/r^4 : 2, \qquad G^5/r^5 : 1.$$

So local 4PN never had an early kinematic shortage. The real issue is chart choice. In the Hamiltonian chart the imported local interaction blocks have ν-degree ceilings

$$(4, 3, 3, 2, 2),$$

whereas after quartic translation the ordinary target becomes

$$(4, 4, 4, 4, 2).$$

The extra ordinary-chart ν^4 tails in the middle and upper blocks are therefore a translation artifact rather than evidence of a true local span failure. This is why the correct local 4PN route is

$$\begin{gathered}\text{fixed local Hamiltonian target} \longrightarrow \text{generic-frame Hamiltonian lift} \\ \longrightarrow \text{ordinary translation only afterward}\end{gathered}.$$

Hamiltonian-chart completeness and exact ordinary reconstruction. Once the fixed local target is treated in the Hamiltonian chart, the blockwise coefficient-space ranks are maximal:

$$G/r : 20/20, \qquad G^2/r^2 : 12/12, \qquad G^3/r^3 : 9/9, \qquad G^4/r^4 : 4/4, \qquad G^5/r^5 : 2/2.$$

So the full interaction coefficient-space dimensions are

$$\text{rank} = 47, \qquad \text{nullity} = 92,$$

and there is no Hamiltonian-chart span obstruction. The null family is purely COM-blind algebraic freedom. Fixing the canonical quotient slice by setting all null coordinates to zero gives one sparse exact generic-frame Hamiltonian representative. Translating that representative back to the ordinary chart exposes the only genuine local obstruction left, namely the seed-alignment correction

$$\delta L_{4,\text{seed}} = L_{4,\text{seed}}^{(\text{aligned})} - L_{4,\text{seed}}^{(\text{natural})}.$$

That correction is not a failure of the generic-frame lift. It is an aligned-seed issue, and it is exactly generic-frame liftable in the structured seed spaces

$$Q, \qquad T \oplus (pq)T, \qquad S \oplus (pq)S, \qquad U \oplus (p^2q^2)U, \qquad W = 0.$$

Once the natural generic-frame seed, the structured aligned-seed correction, and the canonical translated local residual are assembled together, the local referee master verifies that the COM reduction of the resulting generic-frame ordinary candidate reproduces the full fixed-chart local ordinary 4PN target block by block. The entire local instantaneous 4PN ordinary sector is therefore closed exactly.

Tail bridge and the no-new-datum theorem. The standard conservative GR tail coefficient is

$$C_{\rm tail}^{\rm GR} = \frac{G^2M}{5c^8},$$

and the corresponding time-symmetric nonlocal Hamiltonian has local logarithmic shadow

$$F_{\rm tail}(t) = \frac{2}{5}\frac{G^2M}{c^8}\left(I_{ij}^{(3)}(t)\right)^2.$$

Using the Newtonian STF quadrupole

$$I_{ij} = \mu\,{\rm STF}(x_ix_j),$$

one finds the exact order-reduced shadow

$$\frac{F_{\rm tail}}{\mu} = \frac{16}{15}\,\nu\,\frac{U^4}{c^8}\left(12v^2 - 11\dot r^2\right), \qquad U \equiv \frac{GM}{r},$$

which occupies only the degree-2 G^4/r^4 block. The decisive hereditary identity is then

$$C_{\rm tail} = \frac{GM}{2c^3}\,\gamma_{\rm quad}^{\rm eff}.$$

This is the exact bridge to the already isolated 2.5PN quadrupole route. In particular, no new independent 4PN normalization datum opens. The same canonically normalized STF quadrupole branch that controls the 2.5PN Burke–Thorne channel also controls the 4PN hereditary coefficient. On the Burke–Thorne target branch,

$$\gamma_{\rm quad}^{\rm eff} = \frac{2G}{5c^5},$$

one recovers

$$C_{\rm tail}^{\rm toy} = \frac{G^2M}{5c^8} = C_{\rm tail}^{\rm GR}.$$

What the result fixes for the quadrupole program. The main positive interface result is that the local conservative 4PN problem is no longer waiting on another comparable-mass algebraic solve. The exact one-body gate, quartic compiler, raw local scaffold, Hamiltonian-first lift, canonical quotient slice, aligned-seed correction, and local referee reconstruction together remove the local uncertainty completely. The remaining open work is therefore much narrower than "derive 4PN somehow." What the completed moving-throat PDE still has to supply is the passive/outgoing STF quadrupole normalization on the true branch, equivalently the canonical low-frequency pair $(\overline{K}_0, \overline{K}_2)$ whose isotropic value fixes $\gamma_{\rm quad}^{\rm eff}$. Once that pair is determined on the natural branch, the normalized 2.5PN odd quadrupole coefficient closes, the normalized 4PN hereditary coefficient closes, and the present conditional 4PN theorem becomes unconditional inside the same hierarchy.

3 Carry-forward dictionary and theorem target

This section is intentionally downstream of the parent 4D action paper, the full conservative 1PN, 2PN, and 3PN assembly papers, and the 2.5PN quadrupole-normalization audit. We do not repeat those longer derivations here. Instead, we record only the imported objects, frozen lower-order values, chart conventions, and exact target statements that the present 4PN proof chain actually uses. The purpose of the section is fourfold: to freeze notation, to separate exact imports from controlled reductions, to make clear what is being carried rather than rederived, and to state the theorem target that organizes the sections that follow.

The guiding principle is the same one used in the earlier conservative papers. What is exact in the parent reduction framework should remain visibly exact; what enters only after a declared near-zone, small-body, COM, or grouped-branch assumption should remain visibly reduced; and what is fixed only inside the response hierarchy should remain visibly hierarchical rather than disguised as an assumption-free theorem of the fully solved moving-throat PDE. The present paper is therefore built on an imported dictionary rather than on a fresh rederivation of the full bulk problem.

3.1 Minimal imported 4D reduction dictionary

The parent theory lives on a $(4+1)$-dimensional spacetime with coordinates

$$X^M = (t, x, y, z, w), \qquad \mathbf{X} = (x, y, z, w) \in \mathbb{R}^4, \qquad \mathbf{x} = (x, y, z) \in \mathbb{R}^3.$$

The fundamental bulk objects carried into the present paper are the matter field $\psi(\mathbf{X}, t)$, its density

$$\rho(\mathbf{X}, t) = |\psi(\mathbf{X}, t)|^2,$$

the optional localized gauge field $A_M(\mathbf{X}, t)$ with localization profile $Z(w)$, and the effective throat geometry variables

$$a(t), \qquad L(t).$$

When electric charge is named explicitly in the inherited 4D notation, it is always the corrected branch quantity

$$q_* = \eta_Q e_*, \qquad q_{\text{eff}} = \frac{q_*}{\sqrt{Z_{\text{int}}}}, \qquad Z_{\text{int}} = \int_{-\infty}^{\infty} Z(w)\, dw,$$

not the gravity-side scalar coefficient κ_ρ that appears in the post-Newtonian ledger below.

A brane observer does not see the full bulk fields directly. Brane observables are introduced by a fixed normalized projection kernel $W(w)$,

$$\int_{-\infty}^{\infty} W(w)\, dw = 1,$$

through the projection map

$$\mathcal{P}_W[Q](t, \mathbf{x}) \equiv \int_{-\infty}^{\infty} W(w)\, Q(t, \mathbf{x}, w)\, dw.$$

In particular,

$$\rho_{\text{br}} = \mathcal{P}_W[\rho], \qquad \mathbf{J}_{\text{br}} = \mathcal{P}_W[(j^x, j^y, j^z)], \qquad \mathbf{v}_{\text{br}} = \frac{\mathbf{J}_{\text{br}}}{\rho_{\text{br}}} \quad (\rho_{\text{br}} > 0).$$

This projection operation is distinct from *reduction*. Projection defines what a brane observer measures; reduction refers to the further controlled elimination of the extra coordinate under an explicit ansatz or scaling regime.

The exact projected continuity law carried forward from the parent theory is

$$\partial_t \rho_{\text{br}} + \nabla_3 \cdot \mathbf{J}_{\text{br}} = S_{\text{leak}},$$

with leakage source

$$S_{\text{leak}} = -\left[W(w) j^w\right]_{-\infty}^{+\infty} + \int_{-\infty}^{\infty} W'(w) j^w\, dw.$$

After Helmholtz decomposition of the brane velocity,

$$\mathbf{v}_{\text{br}} = \nabla_3 \phi_{\text{br}} + \mathbf{v}_T, \qquad \nabla_3 \cdot \mathbf{v}_T = 0,$$

one obtains the exact longitudinal identity

$$\rho_{\rm br}\,\nabla_3^2\phi_{\rm br} = S_{\rm leak} - \partial_t\rho_{\rm br} - (\nabla_3\rho_{\rm br})\cdot(\nabla_3\phi_{\rm br} + \mathbf{v}_T).$$

Under the carried quasi-static near-zone assumptions this reduces to the Poisson hook

$$\nabla_3^2\phi_{\rm br} \simeq \frac{S_{\rm eff}}{\rho_0}, \qquad \Phi_N = \lambda_\Phi\,\phi_{\rm br},$$

which is the origin of the Newtonian scalar sector used throughout the lower-order particle reduction.

The second imported reduction step is the controlled worldtube/coherent-defect limit. For a compact defect of size $a_{\rm th} \ll r$ in a smooth external field, the center-of-mass reduction gives the usual point-particle force law

$$M_A\,\ddot{\mathbf{X}}_A = -M_A\nabla\Phi_{\rm ext}(\mathbf{X}_A) + O\!\left(\frac{a_{\rm th}^2}{r^2}\,\frac{\Phi_{\rm ext}}{r}\right),$$

so that the universal point-particle source at leading order is controlled by the worldline/worldtube multipoles rather than by arbitrary internal details of the defect. That worldtube dictionary is what later allows the higher-order universal bridge to be packaged in terms of the orbital/worldtube STF quadrupole rather than a generic internal support excitation.

3.2 Frozen conservative carry-forward ledger

The present paper imports the conservative Newtonian+1PN+2PN+3PN ledger as frozen input, not as open fit parameters. The surviving lower-order viability conditions are

$$\kappa_\rho = 1, \qquad n = 5, \qquad \kappa_{\rm add} = \frac{1}{2}, \qquad \kappa_{\rm PV} = \frac{3}{2}, \qquad \beta_{\rm 1PN} = 3.$$

The first four values are the scalar mass-dressing, optical/barotropic, added-mass, and adiabatic breathing-response coefficients frozen by the earlier program. The fifth is the resulting carried 1PN ledger value. The present paper is not allowed to repair a 4PN mismatch by retuning any of these numbers.

When referenced, the parity-even wake data inherited from the full conservative 1PN assembly are also treated as frozen branch data,

$$\alpha^2 = \frac{3}{4}, \qquad a_H = 0, \qquad K_{\rm vec} = \frac{2}{\pi^2},$$

not as live higher-order fit parameters.

The conservative 2PN and 3PN assemblies sharpen the ontology of the defect further. By the start of the present 4PN paper, the particle can no longer be treated as a pure scalar monopole with small corrections. The carried conservative structure already looks like a small throat-response theory with three distinct pieces:

1. a carried odd dipole wake sector,
2. a genuine even $P_0 \oplus P_2$ mouth/support layer,
3. and a separate geometry-closure channel.

Moreover, the full conservative 3PN paper has already fixed the first honest conservative grouped-real quadrupole front end in the fixed ADM chart. So the present 4PN work is not starting from a blank EFT and is not free to redefine the higher-order conservative payload. It is extending a lower-order ledger in which the grouped real quadrupole sector is already the live conservative structure, while the remaining universal dissipative route has already been narrowed at 2.5PN to the orbital/worldtube STF quadrupole branch.

What remains genuinely open from the lower-order side is not the existence of these channels but their full dynamical completion. The unresolved pole, compliance, and normalization data are not claimed as already derived from the fully solved moving-throat PDE. This is one reason the present paper continues to speak in terms of a declared closure hierarchy rather than an assumption-free first-principles theorem.

Two structural consequences of this imported ledger should be stated explicitly. First, the present paper is not allowed to repair a higher-order mismatch by changing lower-order conservative constants that were already fixed below 4PN. Second, the defect already carries enough internal support structure that the 4PN analysis must distinguish carefully between universal point-particle content, self/static carry-forward content, local comparable-mass lift data, and the separate hereditary quadrupole interface.

3.3 Why the 2.5PN notes control 4PN

The 2.5PN theorem audit is what makes the present 4PN problem sharply focused rather than diffuse. By the end of that audit, the scalar and dipole odd channels had already been demoted above universal point-particle 2.5PN on the strict small-body branch, and the surviving retarded quadrupole route had been isolated in the isotropic form

$$\mathcal{M}^R_{ab}(\omega) = \left[m_0 + m_2\omega^2 + m_4\omega^4 + i\Gamma_5\omega^5 + \cdots\right]\delta_{ab}, \qquad m_0 \neq 0, \quad \Gamma_5 > 0.$$

So 4PN should be read as the next *conservative* use of a quadrupole route that 2.5PN had already isolated, not as the opening of a new unrelated bridge.

In the branch language inherited from the narrowing, the canonically normalized STF quadrupole odd coefficient is written

$$\gamma^{\rm eff}_{\rm quad} = \mathcal{N}_Q \frac{a_{\rm th}^5}{27c_s^5} = \overline{\Gamma}_5 = 9\frac{\overline{K}_2^{5/2}}{\overline{K}_0^{3/2}}.$$

The key point for the present paper is that the hereditary 4PN coefficient is controlled by *that same* canonically normalized STF quadrupole branch. Thus the later hereditary calculation is not a fresh search for a new 4PN tail data structure. It is the conservative reuse of the already narrowed 2.5PN quadrupole normalization problem.

This is also why the Burke–Thorne target remains the only conditional interface that matters:

$$\gamma^{\rm eff}_{\rm quad} = \frac{2G}{5c^5}.$$

If this holds on the natural passive/outgoing STF quadrupole branch, then the 2.5PN odd coefficient and the 4PN hereditary coefficient close together. The present paper therefore treats the 2.5PN narrowing as a carried theorem gate, not as optional background motivation.

3.4 PN bookkeeping and invariant variables

The conservative expansion convention is

$$L_{\rm cons} = L_0 + c^{-2}L_1 + c^{-4}L_2 + c^{-6}L_3 + c^{-8}L_4 + O(c^{-10}).$$

So throughout this paper, "4PN" means the coefficient of c^{-8}.

In the generic frame, the basic two-body invariants remain

$$\mathbf{r} = \mathbf{x}_A - \mathbf{x}_B, \qquad r = |\mathbf{r}|, \qquad \mathbf{n} = \mathbf{r}/r,$$

$$v_A^2 = \mathbf{v}_A^2, \qquad v_B^2 = \mathbf{v}_B^2, \qquad v_{AB} = \mathbf{v}_A\cdot\mathbf{v}_B,$$
$$v_{An} = \mathbf{v}_A\cdot\mathbf{n}, \qquad v_{Bn} = \mathbf{v}_B\cdot\mathbf{n}.$$

The usual local pair potentials are

$$U_A = \frac{Gm_B}{r}, \qquad U_B = \frac{Gm_A}{r},$$

and in the strict one-body gate we write

$$U = \frac{GM}{r}.$$

In the COM sector, the reduced ordinary-Lagrangian basis uses the usual velocity invariants together with the symmetric mass ratio

$$d \equiv \dot{r}, \qquad \nu \equiv \frac{m_A m_B}{(m_A + m_B)^2}, \qquad v_\perp^2 \equiv v^2 - d^2.$$

The matching COM Hamiltonian basis uses the corresponding reduced momentum invariants. In the present paper the ordinary COM coefficients are denoted l_i, the COM Hamiltonian coefficients h_i, and the residuals beyond the chosen seed by Δl_i and Δh_i.

For the raw local generic-frame scaffold it is also convenient to use the invariant shorthand

$$a = v_A^2, \qquad b = v_B^2, \qquad c = \mathbf{v}_A\cdot\mathbf{v}_B, \qquad d = \mathbf{v}_A\cdot\mathbf{n}, \qquad e = \mathbf{v}_B\cdot\mathbf{n},$$
$$p = m_A, \qquad q = m_B.$$

These shorthands are purely local-scaffold variables. They are not the throat radius, electric charge, or grouped quadrupole labels.

3.5 COM versus generic-frame conventions

The settled 4PN convention is simple: use the COM basis whenever it simplifies the exact target or the reduced compiler map, but treat the *Hamiltonian chart* as the authoritative chart for the local generic-frame lift.

The reason for this convention is now settled. After the local/nonlocal split, the imported local 4PN Hamiltonian target has one pure free slot and fifteen interaction slots. Its strict test-mass limit matches the exact one-body 4PN Hamiltonian gate. More importantly, its interaction blocks have ν-degree ceilings

$$(4, 3, 3, 2, 2)$$

across the G/r, G^2/r^2, G^3/r^3, G^4/r^4, and G^5/r^5 blocks, and those ceilings match the constant-coefficient generic-frame scaffold exactly.

By contrast, after quartic translation the ordinary local target becomes

$$(4, 4, 4, 4, 2),$$

so the ordinary chart develops extra ν^4 tails in the G^2/r^2, G^3/r^3, and G^4/r^4 blocks. The correct 4PN sequence is therefore

$$\begin{aligned} \text{fixed local Hamiltonian target} &\longrightarrow \text{generic-frame Hamiltonian lift} \\ &\longrightarrow \text{ordinary translation only afterward.} \end{aligned}$$

The local lift is Hamiltonian-first not because the ordinary chart is illegitimate, but because the ordinary ν^4 tails are a quartic-translation artifact rather than evidence of a true local span failure.

A second settled convention follows from the canonical-slice result. In the Hamiltonian chart the remaining null family is purely COM-blind algebraic freedom. Once the canonical slice and aligned-seed convention are fixed, the later ordinary representative is no longer a live generic ambiguity. The fixed Hamiltonian chart is the authoritative local quotient statement.

3.6 Notation firewall

The notation firewall from the corrected 4D / EM / PN stack must be kept intact.

First, electric-charge notation is completely separate from the conservative 4PN coefficient ledger. When charge appears at all, it uses the corrected variables

$$\eta_Q = \pm 1, \qquad q_* = \eta_Q e_*, \qquad q_{\text{eff}} = \frac{q_*}{\sqrt{Z_{\text{int}}}},$$

while the historical gravity-side bare $q = 1$ remains renamed as

$$\kappa_\rho = 1.$$

Quantized circulation belongs to the magnetic/vortical sector, not to the electric-charge dictionary.

Second, the grouped labels $20, 21, 22$ always denote grouped real P_2 channels or their canonically normalized STF descendants. They are not tensor indices. Similarly, the later branch variables

$$\overline{K}_0, \qquad \overline{K}_2, \qquad \overline{\Gamma}_5, \qquad \gamma^{\text{eff}}_{\text{quad}}$$

are quadrupole-branch data, not COM slot coefficients.

Third, the generic-frame scaffold symbols

$$a, b, c, d, e, p, q$$

are only invariant shorthands for the local two-body algebra. In particular, the throat-radius scale entering the quadrupole-branch normalization is always written explicitly as

$$a_{\text{th}},$$

so that it cannot be confused with the shorthand

$$a = v_A^2.$$

If the throat-response source-weight vector

$$J = \left(\frac{4}{\sqrt{5}}, \frac{5}{4}, 0, 0, 0, 0\right)$$

appears later when the grouped branch is discussed, it is a throat-response source-weight vector rather than an electromagnetic current.

Finally, the three layers of notation used by the paper must remain separate: COM coefficients (l_i, h_i), grouped conservative/outgoing quadrupole data, and geometric throat variables (a, L). Most avoidable confusion at this stage is not physics but notation drift across those three ledgers.

3.7 Exact theorem target

The theorem target of the paper is not an unconstrained coefficient fit. It has a strict one-body front door, an imported fixed local target, and a hereditary bridge whose coefficient is already tied to the narrowed 2.5PN quadrupole route.

Strict one-body gate. The exact isotropic Schwarzschild target through c^{-8} is

$$\frac{L_{\text{4PN,1body}}}{m} = \frac{7}{256}v^{10} + \frac{75}{128}Uv^8 + \frac{59}{16}U^2v^6 + \frac{203}{32}U^3v^4 + \frac{31}{32}U^4v^2 + \frac{1}{16}U^5.$$

Any acceptable 4PN one-body/self sector must reproduce this answer exactly. The first task of the paper is therefore to show that the carried denominator-style package can be repaired minimally by the finite ledger

$$\mu_{\rho,4} = \frac{1}{8}, \qquad d_4 = \frac{205}{16}, \qquad s_{34} = -\frac{15}{32}, \qquad s_{26} = -\frac{1}{16},$$

rather than by an uncontrolled extension of the lower-order self sector.

Imported fixed local target. The local instantaneous 4PN target is imported only after the standard local/nonlocal split has been made. In reduced COM Hamiltonian form it has one pure free slot and fifteen interaction slots organized as

$$1 \oplus 5 \oplus 4 \oplus 3 \oplus 2 \oplus 1$$

across the free, G/r, G^2/r^2, G^3/r^3, G^4/r^4, and G^5/r^5 blocks. Its strict test-mass limit matches the one-body Hamiltonian gate exactly. The exact local theorem target is to show that, inside the declared hierarchy, this imported local Hamiltonian target admits a generic-frame representative, that its canonical comparable-mass residual translates back to the ordinary chart by exact sign flip,

$$\Delta H_4 = -\Delta L_4,$$

and that the remaining ordinary-chart seed-alignment correction is generic-frame liftable in the structured seed spaces

$$Q, \qquad T \oplus (pq)T, \qquad S \oplus (pq)S, \qquad U \oplus (p^2q^2)U, \qquad W = 0.$$

Equivalently, the local target is to prove the existence of one exact generic-frame ordinary representative of the form

$$L^{\text{gen}}_{4,\text{loc}} = L^{(\text{natural,gen})}_{4,\text{seed}} + \delta L^{(\text{gen})}_{4,\text{seed}} + \Delta L^{(\text{can,gen})}_{4,\text{loc}},$$

whose COM reduction reproduces the full fixed-chart local ordinary target block by block.

Hereditary bridge target. The remaining nonlocal target is the time-symmetric hereditary coefficient. In standard GR notation,

$$C^{\text{GR}}_{\text{tail}} = \frac{G^2 M}{5c^8},$$

and the corresponding local logarithmic shadow occupies only the degree-2 G^4/r^4 block. The exact bridge to the already narrowed quadrupole route is

$$C_{\text{tail}} = \frac{GM}{2c^3}\,\gamma^{\text{eff}}_{\text{quad}}.$$

This is the decisive theorem target on the hereditary side. It says that the 4PN tail sector is not introducing a new independent normalization constant. The only conditional interface is the same passive/outgoing STF quadrupole normalization already isolated by the 2.5PN program. On the Burke–Thorne branch,

$$\gamma^{\text{eff}}_{\text{quad}} = \frac{2G}{5c^5} \qquad \Longleftrightarrow \qquad \mathcal{N}^{\text{target}}_{Q} = \frac{54Gc_s^5}{5a_{\text{th}}^5 c^5},$$

and then automatically

$$C_{\text{tail}} = \frac{G^2 M}{5c^8} = C^{\text{GR}}_{\text{tail}}.$$

Fixed-chart theorem target. Within the same declared closure hierarchy used successfully at 1PN, 2PN, and 3PN, the goal is to show all of the following.

1. The exact one-body repair ledger closes the strict Schwarzschild gate.
2. The imported fixed local Hamiltonian target is fully reachable in the generic frame once the Hamiltonian chart is used first.

3. The canonical comparable-mass local residual translates back to the ordinary chart by exact sign flip once the lower-order ledger and aligned seed convention are fixed.

4. The seed-alignment correction is exactly liftable in the structured seed sector above, so the earlier ordinary-chart obstruction is revealed to be a seed issue rather than a local comparable-mass span failure.

5. The entire local instantaneous 4PN sector therefore has an exact generic-frame ordinary representative.

6. The hereditary coefficient is fixed by the same canonically normalized STF quadrupole branch already isolated by 2.5PN, so no new independent 4PN normalization datum opens.

Equivalently, the exact conditional full-assembly target to be proved in the rest of the paper is

$$\boxed{L_{4\mathrm{PN}}^{\mathrm{cons}} = \underbrace{L_{4\mathrm{PN}}^{\mathrm{local}}}_{\text{exactly closable in generic-frame ordinary form}} + \underbrace{L_{4\mathrm{PN}}^{\mathrm{tail}}}_{\text{fixed by } C_{\mathrm{tail}}=(GM/2c^3)\,\gamma_{\mathrm{quad}}^{\mathrm{eff}}}.}$$

This is the most compact statement of the theorem target. It says that once the strict one-body gate, the fixed local Hamiltonian target, the Hamiltonian-first generic-frame lift, the aligned-seed ordinary correction, and the hereditary coefficient bridge are all handled correctly, the remaining gap between the present hierarchy and a full conservative 4PN theorem is no longer "more 4PN coefficients." It is the already-known passive/outgoing STF quadrupole normalization on the true moving-throat branch.

The correct theorem-status reading is the same as in the earlier full conservative papers. The boxed statement above is a *full assembly target within a declared closure hierarchy*, not an assumption-free theorem of the fully solved moving-throat PDE. What remains open after the present paper is therefore not generic local 4PN algebra but the deeper moving-throat realization of the passive/outgoing quadrupole normalization, together with its final insertion into the complete referee/master chain.

4 One-body 4PN gate, self/static seed, and exact quartic compiler

The local 4PN chain begins by freezing the one-body answer and the unique order-4 ordinary/Hamiltonian compiler before any comparable-mass local lift is attempted. This is the same methodological move that organized the earlier 3PN paper, but the 4PN content is sharper. At this order the one-body continuation is no longer controlled by a single new denominator datum, and the quartic Legendre bridge is no longer a dispensable bookkeeping convenience. It is a theorem-level input. The purpose of the present section is therefore to record four things in one place: the exact Schwarzschild test-mass gate through c^{-8}, the minimal direct-continuation repair ledger relative to the carried 3PN self-sector packaging, the natural self/static two-body seed obtained by symmetric promotion of that gate, and the exact quartic perturbative Legendre-transform identity that governs every later local 4PN comparison.

4.1 Exact Schwarzschild test-mass target through 4PN

The strict test-mass gate is read from the exact isotropic Schwarzschild worldline Lagrangian

$$\frac{L_{\mathrm{exact}}}{m} = -c^2\sqrt{\left(\frac{1-u/2}{1+u/2}\right)^2 - (1+u/2)^4\frac{v^2}{c^2}}, \qquad u \equiv \frac{U}{c^2}. \tag{1}$$

Expanding (1) through order c^{-8} gives the unique admissible one-body 4PN coefficient

$$\frac{L_{\text{4PN,1-body}}}{m} = \frac{7}{256}\frac{v^{10}}{c^8} + \frac{75}{128}\frac{Uv^8}{c^8} + \frac{59}{16}\frac{U^2v^6}{c^8} + \frac{203}{32}\frac{U^3v^4}{c^8} + \frac{31}{32}\frac{U^4v^2}{c^8} + \frac{1}{16}\frac{U^5}{c^8}. \tag{2}$$

This coefficient is the exact test-mass answer that any acceptable local 4PN self sector must reproduce inside the present hierarchy. In particular, it freezes simultaneously the free v^{10} slot, the four mixed self/static channels

$$Uv^8, \qquad U^2v^6, \qquad U^3v^4, \qquad U^4v^2,$$

and the pure static U^5 gate. No later comparable-mass construction is allowed to modify these test-mass coefficients. If a proposed local 4PN self sector fails (2), it is excluded before any generic-frame or fixed-chart comparison even begins.

This is also the correct place to emphasize the methodological status of the one-body gate. The present paper does not infer the 4PN test-mass answer by continuing a lower-order pattern heuristically. It imports the exact Schwarzschild answer first and only afterward asks what continuation of the carried 3PN self-sector packaging is actually compatible with it. The full algebraic expansion is recorded in Appendix B; the main text keeps only the coefficient that the later local theorem chain must match exactly.

4.2 Minimal one-body repair ledger

To test whether the carried 3PN self-sector packaging continues directly, we extend the denominator-style one-body ansatz by writing

$$D(u) = 1 - 4u + 8u^2 + d_3u^3 + d_4u^4, \tag{3}$$

with the already frozen lower-order values

$$d_3 = -\frac{45}{4}, \qquad \mu_{\rho,3} = \frac{1}{4}, \qquad s_{24} = -\frac{1}{16}. \tag{4}$$

The most direct continuation of the 3PN one-body package is then

$$\begin{aligned} L_{\text{cand}} = \; & -c^2(1-u)\sqrt{1 - \frac{v^2/c^2}{D(u)}} - \frac{U^2}{2c^2} + \frac{U^3}{4c^4} - \frac{\mu_{\rho,3}U^4}{2c^6} + \frac{\mu_{\rho,4}U^5}{2c^8} \\ & + s_{24}\frac{U^2v^4}{c^6} + s_{34}\frac{U^3v^4}{c^8} + s_{26}\frac{U^2v^6}{c^8}. \end{aligned} \tag{5}$$

Comparing (5) with the exact gate (2) shows that a quartic denominator plus a quartic static gate is *not* enough. Two genuinely new self-sector slots survive at 4PN, namely the

$$\frac{U^3v^4}{c^8} \qquad \text{and} \qquad \frac{U^2v^6}{c^8}$$

channels. The minimal direct-continuation repair ledger is therefore

$$\boxed{\mu_{\rho,4} = \frac{1}{8}, \qquad d_4 = \frac{205}{16}, \qquad s_{34} = -\frac{15}{32}, \qquad s_{26} = -\frac{1}{16}.} \tag{6}$$

The interpretation is immediate. Relative to the carried 3PN one-body package, 4PN opens

1. a quartic static gate, represented by $\mu_{\rho,4}$,
2. a quartic denominator datum, represented by d_4,
3. and two genuinely new self-sector slots, represented by s_{34} and s_{26}.

So 4PN is already not a one-parameter continuation of the 3PN one-body closure. The admissible one-body front end of the local 4PN theorem is the full four-parameter ledger

$$(\mu_{\rho,4}, d_4, s_{34}, s_{26}),$$

not a single extra denominator coefficient. The detailed repair algebra is recorded in Appendix C.

4.3 Natural 4PN self/static seed

Once the exact one-body target is frozen, the natural local self/static seed is obtained by direct symmetric promotion of the six one-body channels appearing in (2). The resulting two-body seed is

$$
\begin{aligned}
L_{4,\mathrm{seed}} = {} & \frac{7}{256}(m_A v_A^{10} + m_B v_B^{10}) + \frac{75 G m_A m_B}{128 r}(v_A^8 + v_B^8) + \frac{59 G^2 m_A m_B}{16 r^2}(m_B v_A^6 + m_A v_B^6) \\
& + \frac{203 G^3 m_A m_B}{32 r^3}(m_B^2 v_A^4 + m_A^2 v_B^4) \\
& + \frac{31 G^4 m_A m_B}{32 r^4}(m_B^3 v_A^2 + m_A^3 v_B^2) + \frac{G^5 m_A m_B}{16 r^5}(m_B^4 + m_A^4),
\end{aligned} \tag{7}
$$

understanding the whole block as the coefficient of c^{-8} in the full conservative expansion.

The role of (7) is exactly the same one played by the corresponding natural seed below 4PN. It isolates the universal one-body/self/static content before any genuine comparable-mass local residual is solved. No comparable-mass cross information is inserted here. Each term is simply the unique exchange-symmetric two-body promotion of the corresponding one-body channel, and the strict test-mass limit of the seed reduces back to (2) exactly.

This seed should therefore be read as a theorem-level starting point, not as a convenient ansatz. It is the correct local front end because the one-body gate forces it. Later generic-frame comparable-mass data must be measured *relative* to it. In particular, the subsequent local comparable-mass problem is not allowed to hide one-body repair information inside the residual. That information has already been frozen here. The later Hamiltonian-first lift will modify neither the strict one-body content nor the self/static promotion principle; it will only determine how the remaining comparable-mass local content is represented consistently across charts.

4.4 Exact quartic Legendre-transform identity

The second non-negotiable input is the exact quartic perturbative Legendre-transform identity. Let

$$
L = L_0 + \varepsilon L_1 + \varepsilon^2 L_2 + \varepsilon^3 L_3 + \varepsilon^4 L_4, \tag{8}
$$

with quadratic L_0 and constant Newtonian mass matrix M. Define

$$
v_0 = M^{-1} p, \qquad A_0 = (\partial_v L_1)|_{v_0}, \qquad B_0 = (\partial_v L_2)|_{v_0}, \qquad D_0 = (\partial_v L_3)|_{v_0}, \tag{9}
$$

$$
C_0 = (\partial_v^2 L_1)|_{v_0}, \qquad E_0 = (\partial_v^2 L_2)|_{v_0}, \qquad T_0 = (\partial_v^3 L_1)|_{v_0}. \tag{10}
$$

Then the exact perturbative Legendre coefficients through quartic order are

$$
H_1 = -L_1(v_0), \tag{11}
$$

$$
H_2 = -L_2(v_0) + \frac{1}{2} A_0^T M^{-1} A_0, \tag{12}
$$

$$
H_3 = -L_3(v_0) + A_0^T M^{-1} B_0 - \frac{1}{2} A_0^T M^{-1} C_0 M^{-1} A_0, \tag{13}
$$

$$
\begin{aligned}
H_4 = {} & -L_4(v_0) + A_0^T M^{-1} D_0 + \frac{1}{2} B_0^T M^{-1} B_0 - B_0^T M^{-1} C_0 M^{-1} A_0 \\
& - \frac{1}{2} A_0^T M^{-1} E_0 M^{-1} A_0 + \frac{1}{2} A_0^T M^{-1} C_0 M^{-1} C_0 M^{-1} A_0 + \frac{1}{6} T_0[M^{-1} A_0, M^{-1} A_0, M^{-1} A_0].
\end{aligned} \tag{14}
$$

Equations (11)–(14) are the exact ordinary/Hamiltonian bridge used throughout the local 4PN chain. Once the lower-order ledger through 3PN is frozen, the genuinely new 4PN block is not free to enter the Hamiltonian arbitrarily. It enters through the rigid quartic compiler above.

This has two immediate consequences. First, applying (11)–(14) to the strict one-body Lagrangian gate reproduces the one-body 4PN Hamiltonian seed used later by the Hamiltonian-first local lift. So the Hamiltonian one-body gate is not an independent heuristic ansatz; it is a direct theorem-level output of the exact Schwarzschild Lagrangian gate together with the quartic compiler. Second, no trustworthy local comparable-mass 4PN solve can be formulated before this compiler is frozen, because every later translation between fixed-chart Hamiltonian targets and ordinary generic-frame representatives depends on it.

The detailed order-by-order inversion and derivation of (11)–(14) are placed in Appendix D. For the main text, the important lesson is structural. The exact one-body gate, the minimal repair ledger, the natural self/static seed, and the quartic compiler already determine the admissible self-sector front end of the local 4PN theorem. What remains after this section is therefore not a search for one-body or chart data, but the genuinely comparable-mass local lift and the separate hereditary quadrupole bridge.

5 The local instantaneous 4PN theorem chain

This section closes the entire *local instantaneous* 4PN sector. Its role is narrower than the full paper and sharper than the earlier executive summary. The hereditary/tail contribution is kept completely separate; the only question addressed here is whether the strict one-body gate fixed in Section 4, together with the exact quartic compiler (14), is enough to determine a full generic-frame local ordinary representative of the two-body 4PN sector. The answer is yes. The key lesson is that the local problem never had an early span shortage. The raw exchange-symmetric scaffold is already large enough. What matters instead is chart choice, quotienting of the COM-blind null sector, and correct treatment of the aligned ordinary seed.

The logic therefore runs through eight exact steps. First, one builds the raw local comparable-mass scaffold relative to the natural self/static seed (7) and checks its COM image. Second, one imports the fixed local target in the Hamiltonian chart and reads off the true polynomial degree ceilings. Third, one proves that the Hamiltonian-chart scaffold is polynomially complete against the imported local target. Fourth, one chooses a canonical representative inside the resulting COM-blind null family. Fifth, one translates that representative back to the ordinary chart, where the comparable-mass residual obeys an exact sign-flip theorem. Sixth, one shows that the remaining ordinary obstruction is only a seed-alignment correction. Seventh, one lifts that correction in a minimal structured seed sector. Finally, one assembles the natural seed, the lifted seed-alignment correction, and the canonical translated residual into one exact generic-frame ordinary local representative. At that point the local theorem is closed.

5.1 Raw local scaffold and COM completeness

With the strict one-body front end already frozen, the first local question is whether the exchange-symmetric generic-frame comparable-mass scaffold has enough room to support a full 4PN interaction residual. Using the invariant variables

$$a = v_A^2, \qquad b = v_B^2, \qquad c = \mathbf{v}_A\cdot\mathbf{v}_B, \qquad d = \mathbf{v}_A\cdot\mathbf{n}, \qquad e = \mathbf{v}_B\cdot\mathbf{n}, \qquad p = m_A, \qquad q = m_B,$$

and imposing strict vanishing on the one-body branch, the raw local interaction families have block sizes

$$\dim\mathcal{Q} = 52, \qquad \dim\mathcal{T} = 46, \qquad \dim\mathcal{S} = 29, \qquad \dim\mathcal{U} = 10, \qquad \dim\mathcal{W} = 2, \tag{15}$$

where the notation follows the standard block decomposition

$$\frac{G}{r}\,\mathcal{Q}, \qquad \frac{G^2}{r^2}\,\mathcal{T}, \qquad \frac{G^3}{r^3}\,\mathcal{S}, \qquad \frac{G^4}{r^4}\,\mathcal{U}, \qquad \frac{G^5}{r^5}\,\mathcal{W}.$$

So before any quotienting or target import the local generic-frame interaction scaffold already contains

$$52 + 46 + 29 + 10 + 2 = 139 \tag{16}$$

exchange-symmetric directions.

Projecting to the center-of-mass chart does not reveal a shortage. It reveals a large null family. The induced COM interaction-slot structure is

$$\#^{\text{int}}_{\text{COM}} = (5, 4, 3, 2, 1), \qquad 5 + 4 + 3 + 2 + 1 = 15, \tag{17}$$

corresponding respectively to the

$$\frac{G}{r}, \qquad \frac{G^2}{r^2}, \qquad \frac{G^3}{r^3}, \qquad \frac{G^4}{r^4}, \qquad \frac{G^5}{r^5}$$

interaction blocks. The blockwise COM image ranks are already maximal at this slot-counting level:

$$52 \to 5, \qquad 46 \to 4, \qquad 29 \to 3, \qquad 10 \to 2, \qquad 2 \to 1. \tag{18}$$

Equivalently, the raw scaffold spans the full COM interaction image block by block, with total COM-nullity

$$(52 - 5) + (46 - 4) + (29 - 3) + (10 - 2) + (2 - 1) = 124. \tag{19}$$

The interpretation is important. Local 4PN never suffered from an early image shortage. Already at the raw-scaffold stage the question is not whether there is enough room to span the reduced interaction sector. The question is which chart carries the correct polynomial degree ceilings and which quotient slice extracts a canonical representative from the very large COM-blind family.

5.2 Fixed local target import and the Hamiltonian-first lesson

The next step is to import the fixed-chart *local* 4PN target. At this order the conservative theorem splits into a local instantaneous sector and a separate hereditary/tail sector, so the target imported here is deliberately the local ADM Hamiltonian sector only. This target passes the strict one-body gate: a direct application of the quartic compiler (14) to the one-body Lagrangian gate (2) yields the exact reduced one-body Hamiltonian coefficients

$$\frac{7}{256}p^{10}, \qquad \frac{45}{128}\frac{p^8}{r}, \qquad \frac{13}{8}\frac{p^6}{r^2}, \qquad \frac{105}{32}\frac{p^4}{r^3}, \qquad \frac{105}{32}\frac{p^2}{r^4}, \qquad -\frac{1}{16}\frac{1}{r^5}, \tag{20}$$

which the imported local target reproduces exactly in the strict test-mass limit.

The imported COM Hamiltonian target consists of one free slot plus the same fifteen interaction slots counted above, so it is already structurally matched to the scaffold. The decisive issue is instead the *polynomial degree ceiling* in the symmetric mass ratio ν. In the Hamiltonian chart the local target has blockwise ceilings

$$(G/r,\ G^2/r^2,\ G^3/r^3,\ G^4/r^4,\ G^5/r^5) = (4, 3, 3, 2, 2), \tag{21}$$

whereas translating the same fixed local target back to the ordinary chart by (14) produces

$$(G/r,\ G^2/r^2,\ G^3/r^3,\ G^4/r^4,\ G^5/r^5) = (4, 4, 4, 4, 2) \tag{22}$$

in the same block order. So the quartic compiler itself creates the extra ordinary-chart ν^4 tails in the middle and upper interaction blocks

$$\frac{G^2}{r^2}, \qquad \frac{G^3}{r^3}, \qquad \frac{G^4}{r^4},$$

while leaving the top static block unchanged.

This is the central structural lesson of the whole local chain. The fixed local 4PN lift must be performed *Hamiltonian-first*. In the Hamiltonian chart the constant-coefficient generic-frame scaffold matches the imported target exactly at the level of degree ceilings. In the translated ordinary chart the same target develops extra polynomial tails that do not belong to the original constant-coefficient comparable-mass scaffold. So the correct order of work is not

$$\text{ordinary target} \longrightarrow \text{generic-frame lift},$$

but rather

$$\boxed{\begin{gathered}\text{fixed local Hamiltonian target} \;\longrightarrow\; \text{generic-frame Hamiltonian lift} \\ \longrightarrow \text{ordinary translation only afterward}\end{gathered}} \tag{23}$$

That statement is no longer a heuristic design preference. It is the exact chart lesson forced by the quartic compiler.

5.3 Hamiltonian-chart generic-frame local span theorem

Once the Hamiltonian chart is chosen, the local comparable-mass existence question becomes exact. The COM interaction residual is decomposed into explicit ν-polynomial coefficient space:

$$\begin{aligned}
\frac{G}{r}: &\quad 5 \text{ slots} \times (\nu, \nu^2, \nu^3, \nu^4), \\
&\quad \Rightarrow 20 \text{ coeffs}, \\
\frac{G^2}{r^2}: &\quad 4 \text{ slots} \times (\nu, \nu^2, \nu^3), \\
&\quad \Rightarrow 12, \\
\frac{G^3}{r^3}: &\quad 3 \text{ slots} \times (\nu, \nu^2, \nu^3), \\
&\quad \Rightarrow 9, \\
\frac{G^4}{r^4}: &\quad 2 \text{ slots} \times (\nu, \nu^2), \\
&\quad \Rightarrow 4, \\
\frac{G^5}{r^5}: &\quad 1 \text{ slot} \times (\nu, \nu^2), \\
&\quad \Rightarrow 2,
\end{aligned} \tag{24}$$

so the full interaction coefficient space has dimension

$$20 + 12 + 9 + 4 + 2 = 47. \tag{25}$$

The exact blockwise rank test in the Hamiltonian chart is then

$$\begin{aligned}
\frac{G}{r}: &\quad 52 \to 20, &&\quad \text{nullity } 32, \\
\frac{G^2}{r^2}: &\quad 46 \to 12, &&\quad \text{nullity } 34, \\
\frac{G^3}{r^3}: &\quad 29 \to 9, &&\quad \text{nullity } 20, \\
\frac{G^4}{r^4}: &\quad 10 \to 4, &&\quad \text{nullity } 6, \\
\frac{G^5}{r^5}: &\quad 2 \to 2, &&\quad \text{nullity } 0.
\end{aligned} \tag{26}$$

Therefore the total interaction coefficient-space rank is maximal,

$$\text{rank} = 47, \qquad \text{nullity} = 32 + 34 + 20 + 6 + 0 = 92. \tag{27}$$

This proves the Hamiltonian-chart local span theorem: every interaction coefficient of the imported fixed-chart COM Hamiltonian residual lies in the image of the constant-coefficient exchange-symmetric generic-frame scaffold. There is no Hamiltonian-chart span obstruction, no missing middle block, and no need for additional seed dressing before the comparable-mass Hamiltonian lift is performed. The only ambiguity is the 92-direction COM-blind null family. That ambiguity is algebraic. It is not a failure of local existence.

5.4 Canonical comparable-mass local residual slice

Because the Hamiltonian-chart image is complete, the next question is not existence but quotienting. The generic-frame invariants satisfy the usual COM relations

$$\mathcal{C}_1 = pa + qc, \qquad \mathcal{C}_2 = pc + qb, \qquad \mathcal{C}_3 = pd + qe, \tag{28}$$

$$\mathcal{C}_4 = ab - c^2, \qquad \mathcal{C}_5 = ae - cd, \qquad \mathcal{C}_6 = bd - ce. \tag{29}$$

The exact ideal-membership audit shows that the 4PN Hamiltonian-chart nullities split as

$$Q : 32, \qquad T : 34, \qquad S : 20, \qquad U : 6, \qquad W : 0, \tag{30}$$

and moreover that

1. all Q-block null vectors lie in the full COM ideal $\langle \mathcal{C}_1, \ldots, \mathcal{C}_6 \rangle$,
2. all T-, S-, and U-block null vectors already lie in the linear momentum ideal $\langle \mathcal{C}_1, \mathcal{C}_2, \mathcal{C}_3 \rangle$,
3. and the top static W block has zero nullity.

So the entire 92-direction ambiguity is purely COM-blind algebraic freedom. No physical local datum is hiding there once the COM target is fixed.

The canonical quotient slice is therefore defined by the exact Gauss–Jordan lift with all null coordinates set to zero. This choice gives one sparse generic-frame Hamiltonian representative of the comparable-mass local residual,

$$\Delta H^{\text{can}}_{4,\text{loc}} = \frac{Gpq}{r} Q_{\text{can}} + \frac{G^2 pq}{r^2} T_{\text{can}} + \frac{G^3 pq}{r^3} S_{\text{can}} + \frac{G^4}{r^4} U_{\text{can}} + \frac{G^5}{r^5} W_{\text{can}}, \tag{31}$$

with nonzero scaffold-direction counts

$$Q : 11, \qquad T : 12, \qquad S : 9, \qquad U : 4, \qquad W : 2. \tag{32}$$

The explicit block formulas are lengthy and belong in the appendix. The main text only needs the structural conclusion: the Hamiltonian-chart local comparable-mass existence problem is solved by one exact canonical representative.

5.5 Ordinary translation and the aligned-seed issue

The exact quartic compiler then transports the canonical comparable-mass Hamiltonian residual back to the ordinary chart. Once the lower-order ledger is held fixed, the translation is rigid and remarkably simple:

$$\boxed{\Delta L^{\text{can}}_{4,\text{loc}} = -\Delta H^{\text{can}}_{4,\text{loc}}.} \tag{33}$$

This is the exact local 4PN sign-flip theorem. It shows that the canonical comparable-mass residual itself is not the source of the earlier ordinary-chart obstruction. That obstruction sits elsewhere.

What remains misaligned in the ordinary chart is the seed. The correct ordinary seed is not simply the natural seed obtained by symmetric promotion of the one-body gate. It is the *aligned* seed selected by the Hamiltonian-first translation. The resulting seed-alignment correction is

$$\delta L_{4,\mathrm{seed}} = L_{4,\mathrm{seed}}^{(\mathrm{aligned})} - L_{4,\mathrm{seed}}^{(\mathrm{natural})}. \tag{34}$$

The aligned seed already fixes the full free block of the local ordinary target exactly, while the translated canonical comparable-mass residual fixes the true interaction residual by (33). Therefore the entire remaining ordinary-chart mismatch is a *seed-alignment* issue. It is not a failure of the local generic-frame comparable-mass lift.

This distinction is conceptually important. The Hamiltonian chart has already solved the genuine comparable-mass local problem. The ordinary chart only has to repair the seed so that the exact translated residual can sit on the correct free and self/static background. Once that is recognized, the old ordinary obstruction becomes narrow and intelligible.

5.6 Generic-frame lift of the aligned-seed ordinary correction

The seed-alignment correction (34) does not require a new comparable-mass interaction scaffold. It requires a minimal structured enlargement of the *seed sector*. Writing the original local generic-frame block families as

$$\mathcal{Q}, \qquad \mathcal{T}, \qquad \mathcal{S}, \qquad \mathcal{U}, \qquad \mathcal{W},$$

the exact lift statement is

$$Q_\delta \in \mathcal{Q}, \qquad T_\delta \in \mathcal{T} \oplus (pq)\,\mathcal{T}, \qquad S_\delta \in \mathcal{S} \oplus (pq)\,\mathcal{S}, \qquad U_\delta \in \mathcal{U} \oplus (p^2q^2)\,\mathcal{U}, \qquad W_\delta = 0. \tag{35}$$

In these structured seed spaces the coefficient-space ranks are again maximal:

$$Q : 20/20, \qquad T : 16/16, \qquad S : 12/12, \qquad U : 8/8. \tag{36}$$

So the aligned-seed correction is exactly generic-frame liftable once the seed sector, rather than the residual sector, is dressed in this specific way.

A canonical sparse lift is again obtained by setting all structured null coordinates to zero. One such representative uses

$$Q : 13, \qquad T : 14, \qquad S : 12, \qquad U : 8, \qquad W : 0 \tag{37}$$

nonzero scaffold directions. The important point is not the particular sparse choice. It is the theorem-level structural message: the ordinary seed-alignment correction is not an inexplicable extra obstruction. It is an exactly liftable generic-frame object inside a narrow and intelligible structured seed sector.

5.7 Exact local referee reconstruction

At this point all three local ingredients are available in generic-frame ordinary form.

First, the natural generic-frame local seed is the direct symmetric promotion of the one-body gate,

$$Q_{\mathrm{nat}} = \frac{75}{128}(a^4 + b^4), \qquad T_{\mathrm{nat}} = \frac{59}{16}(qa^3 + pb^3), \qquad S_{\mathrm{nat}} = \frac{203}{32}(q^2a^2 + p^2b^2), \tag{38}$$

$$U_{\rm nat}=\frac{31}{32}(q^3a+p^3b),\qquad W_{\rm nat}=\frac{1}{16}(q^4+p^4). \tag{39}$$

Second, the aligned-seed correction has the exact structured lift described in (35). Third, the canonical comparable-mass local residual has already been translated to the ordinary chart by the sign-flip identity (33).

These pieces assemble into one exact generic-frame ordinary local candidate,

$$\boxed{L^{(\rm gen)}_{4,\rm loc}=L^{(\rm natural,gen)}_{4,\rm seed}+\delta L^{(\rm gen)}_{4,\rm seed}+\Delta L^{(\rm can,gen)}_{4,\rm loc}.} \tag{40}$$

Let $\Pi_{\rm COM}$ denote exact reduction to the center-of-mass chart. The end-to-end referee identity is then

$$\Pi_{\rm COM}\left[L^{(\rm gen)}_{4,\rm loc}\right]=L^{(\rm target,ord)}_{4,\rm loc}, \tag{41}$$

with equality holding slot by slot in all five local interaction blocks:

$$Q:5\ \text{slots},\qquad T:4\ \text{slots},\qquad S:3\ \text{slots},\qquad U:2\ \text{slots},\qquad W:1\ \text{slot}.$$

In other words, the local referee suite verifies not merely formal existence but full end-to-end reconstruction of the fixed-chart local ordinary target.

5.8 Final exact local theorem

The local theorem can now be stated without qualification. Within the declared closure hierarchy used throughout the lower conservative program, the complete local instantaneous two-body 4PN sector is fixed exactly. More precisely:

1. the strict one-body 4PN Schwarzschild gate (2) and its natural self/static two-body promotion (7) are frozen exactly;
2. the raw exchange-symmetric generic-frame local scaffold is already blockwise COM-complete, so local 4PN has no early span shortage;
3. the correct fixed-chart import is Hamiltonian-first, because only the Hamiltonian chart respects the true degree ceilings of the local target;
4. in that Hamiltonian chart there is no span obstruction, and one exact canonical generic-frame comparable-mass residual representative exists;
5. the canonical residual translates back to the ordinary chart by exact sign flip;
6. the remaining ordinary mismatch is only the aligned-seed correction, which is itself exactly generic-frame liftable in the structured seed spaces (35);
7. and the assembled candidate (40) reproduces the full fixed-chart local ordinary target exactly under COM reduction.

Equivalently, the full local instantaneous sector may be written in the exact assembled form

$$L^{\rm local}_{\rm 4PN}=L^{(\rm natural,gen)}_{4,\rm seed}+\delta L^{(\rm gen)}_{4,\rm seed}+\Delta L^{(\rm can,gen)}_{4,\rm loc},\qquad \Pi_{\rm COM}\left[L^{\rm local}_{\rm 4PN}\right]=L^{(\rm target,ord)}_{4,\rm loc}. \tag{42}$$

So the local instantaneous 4PN bottleneck is no longer algebraic span, chart ambiguity, or generic-frame existence. All of that is already closed. What remains for the full conservative 4PN theorem is not more local algebra. It is the separate hereditary/tail bridge, whose unique compatible coefficient is controlled by the same passive/outgoing STF quadrupole normalization already isolated by the 2.5PN program.

6 Tail / hereditary 4PN bridge

With the local instantaneous sector already fixed exactly, the remaining 4PN question is not another generic-frame existence solve. The 2.5PN audit has already reduced the universal higher-order retarded route to the canonically normalized orbital/worldtube STF quadrupole branch, while the local referee chain has already closed the full instantaneous 4PN sector. The hereditary side is therefore sharply narrower than the local side ever was. The purpose of this section is to make that narrowing explicit. First, we freeze the standard conservative GR tail functional and its universal local logarithmic shadow. Second, we show that the hereditary coefficient is tied exactly to the same canonically normalized STF quadrupole coefficient that already controls the 2.5PN Burke–Thorne channel. Third, we isolate the only remaining model-side ambiguity as a single scalar transport factor rather than a new tensor channel, generic-frame lift, or normalization family. The resulting conclusion is the main structural message of the section: 4PN opens no new independent normalization datum beyond the already-known passive/outgoing STF quadrupole gate.

6.1 Exact GR tail structure

The conservative 4PN dynamics in GR splits into a local instantaneous sector plus a time-symmetric nonlocal tail sector. On the action side the hereditary piece may be written as

$$S_{\text{tail}} = \frac{G^2 M}{5c^8}\,\mathrm{Pf}_{2s/c} \iint \frac{dt\,dt'}{|t-t'|} I_{ij}^{(3)}(t) I_{ij}^{(3)}(t'), \tag{43}$$

while on the Hamiltonian side the corresponding time-symmetric nonlocal term is

$$H_{\text{4PN}}^{\text{tail,sym}}(t) = -\frac{G^2 M}{5c^8} I_{ij}^{(3)}(t)\,\mathrm{Pf}_{2s/c} \int_{-\infty}^{+\infty} \frac{dv}{|v|} I_{ij}^{(3)}(t+v). \tag{44}$$

So the universal GR hereditary coefficient is

$$C_{\text{tail}}^{\text{GR}} = \frac{G^2 M}{5c^8}. \tag{45}$$

Equations (43)–(45) show that the conservative hereditary 4PN sector is a single STF-quadrupole functional with one scalar coefficient. In particular, the tail side is not a second local-generic-frame lift problem. Its representation is already fixed at the level of the canonical STF quadrupole.

The same nonlocal functional has a local logarithmic shadow. In the present normalization that shadow is controlled by

$$F_{\text{tail}}(t) = \frac{2}{5}\frac{G^2 M}{c^8}\left(I_{ij}^{(3)}(t)\right)^2. \tag{46}$$

This is the local object that later appears when one compares the hereditary sector with the local instantaneous 4PN blocks.

6.2 Universal Newtonian quadrupole shadow

Because the hereditary coefficient first enters at 4PN, its local logarithmic shadow is fixed already by the Newtonian order-reduced STF quadrupole. Writing

$$I_{ij} = \mu\,\mathrm{STF}(x_i x_j), \qquad M \equiv m_A + m_B, \qquad \mu \equiv \frac{m_A m_B}{M}, \qquad \nu \equiv \frac{\mu}{M}, \tag{47}$$

one finds the exact order-reduced third derivative

$$I_{ij}^{(3)} = -\frac{2GM\mu}{r^3}\left(4x_{\langle i}v_{j\rangle} - 3\frac{\dot r}{r}x_{\langle i}x_{j\rangle}\right), \tag{48}$$

and therefore

$$(I^{(3)}_{ij})^2 = \frac{8}{3}\frac{G^2M^2\mu^2}{r^4}\left(12v^2 - 11\dot{r}^2\right). \tag{49}$$

Substituting (49) into (46) gives the exact local logarithmic tail shadow

$$\boxed{\frac{F_{\rm tail}}{\mu} = \frac{16}{15}\,\nu\,\frac{U^4}{c^8}\left(12v^2 - 11\dot{r}^2\right), \qquad U \equiv \frac{GM}{r}.} \tag{50}$$

Equation (50) makes the structural point transparent. The hereditary shadow lives only in the degree-2 local G^4/r^4 block, i.e. in the U block of the local 4PN bookkeeping. That is the exact reason the hereditary side is much smaller than the full local lift problem. The local theorem had to reconstruct an entire generic-frame interaction sector. The tail theorem only has to attach one scalar coefficient to one already-fixed STF quadrupole representation.

6.3 Exact bridge to the 2.5PN quadrupole coefficient

The 2.5PN narrowing already fixed the correct representation and normalization language for the surviving radiative branch. On that canonical passive/outgoing STF quadrupole branch the carried coefficient is written

$$\gamma^{\rm eff}_{\rm quad} = \mathcal{N}_Q\,\frac{a^5_{\rm th}}{27c_s^5} = \overline{\Gamma}_5 = 9\frac{\overline{K}_2^{5/2}}{\overline{K}_0^{3/2}}, \tag{51}$$

where $a_{\rm th}$ is the throat-radius scale carried by the quadrupole branch. In GR the corresponding Burke–Thorne coefficient is

$$\gamma_{\rm GR} = \frac{2G}{5c^5}. \tag{52}$$

The crucial arithmetic observation is that the 4PN GR hereditary coefficient is related exactly to the 2.5PN Burke–Thorne coefficient by

$$\boxed{C^{\rm GR}_{\rm tail} = \frac{GM}{2c^3}\,\gamma_{\rm GR}.} \tag{53}$$

Indeed,

$$\frac{GM}{2c^3}\,\frac{2G}{5c^5} = \frac{G^2M}{5c^8} = C^{\rm GR}_{\rm tail}.$$

So the hereditary coefficient is not merely analogous to the already-isolated quadrupole odd coefficient; it is fixed by the same coefficient through the exact bridge (53).

This observation upgrades directly from the GR reference coefficient to the canonically normalized branch carried by the toy-model program. The unique compatible conservative hereditary coefficient is

$$\boxed{C_{\rm tail} = \frac{GM}{2c^3}\,\gamma^{\rm eff}_{\rm quad}.} \tag{54}$$

Equation (54) is the cleanest structural statement of the whole hereditary bridge. Once the STF quadrupole branch is canonically normalized, the conservative 4PN tail coefficient follows automatically. There is no second normalization problem hiding on the hereditary side.

6.4 Toy-model branch form and the remaining transport factor

The same bridge can be written in a form that isolates the only remaining model-side ambiguity. Let

$$\gamma_{\rm toy} \equiv \hat{m}_0^2\Gamma_5 \tag{55}$$

be the canonically normalized STF quadrupole coefficient inherited from the 2.5PN audit. Then the hereditary side may be parameterized by one scalar transport factor $\Theta_{\rm tail}$ through

$$\boxed{\alpha_{\rm tail}^{\rm toy} = \Theta_{\rm tail}\frac{GM}{2c_s^3}\gamma_{\rm toy}.} \tag{56}$$

On the branch with $c_s = c$ and

$$\gamma_{\rm toy} = \gamma_{\rm quad}^{\rm eff}, \tag{57}$$

Eq. (56) reduces to the exact bridge (54) when

$$\Theta_{\rm tail} = 1. \tag{58}$$

This is the sharpest way to say what is and is not still open. The representation/source problem has already been solved by the 2.5PN quadrupole audit. The local 4PN existence problem has already been solved by the local referee chain. The only possible remaining hereditary mismatch is one scalar monopole-scattering transport factor.

6.5 Why no new independent 4PN normalization datum opens

The hereditary theorem may now be stated without further search language. Inside the same closure hierarchy used throughout the lower-order conservative program, the following statements hold:

1. the unique compatible conservative hereditary coefficient is the bridge coefficient (54),
2. therefore the 4PN tail sector introduces no new independent quadrupole-normalization datum,
3. and full GR matching of the hereditary coefficient is equivalent to the already-known 2.5PN normalization target

$$\gamma_{\rm quad}^{\rm eff} = \frac{2G}{5c^5}. \tag{59}$$

The same recovery condition may be written in the normalization language carried by the moving-throat program as

$$\mathcal{N}_Q^{\rm target} = \frac{54Gc_s^5}{5a_{\rm th}^5c^5}. \tag{60}$$

Substituting (59) into (54) gives

$$C_{\rm tail} = \frac{GM}{2c^3}\,\frac{2G}{5c^5} = \frac{G^2M}{5c^8} = C_{\rm tail}^{\rm GR}. \tag{61}$$

So the 4PN hereditary bridge closes automatically once the already-isolated passive/outgoing STF quadrupole normalization is fixed on the natural branch.

What remains open is correspondingly narrow. The present section does not claim a first-principles moving-throat PDE derivation of the passive/outgoing quadrupole normalization, nor does it claim the final insertion of that normalized tail module into a single end-to-end referee/master script. But the bottleneck is now fully exposed: the local 4PN sector is already closed, the hereditary coefficient is fixed by the same invariant quadrupole branch as 2.5PN, and the remaining full-4PN gap is therefore the same passive/outgoing quadrupole-normalization theorem already identified by the 2.5PN program.

7 Final conditional conservative 4PN theorem

The preceding sections have separated the 4PN problem into two logically distinct pieces. The first is the full local instantaneous sector, whose one-body gate, quartic compiler, Hamiltonian-chart lift, canonical quotient slice, aligned-seed repair, and referee reconstruction have already been closed exactly inside the declared hierarchy. The second is the hereditary/tail sector, whose coefficient is not an independent new tensor datum but a scalar bridge to the same canonically normalized STF quadrupole branch that was already isolated by the 2.5PN audit. The purpose of the present section is therefore not to introduce new machinery, but to state the final theorem in the sharpest possible form and record exactly what remains conditional.

Theorem 1 (Final conditional conservative 4PN theorem)**.** *Work within the same declared closure hierarchy used in the Newtonian,* 1*PN,* 2*PN, and* 3*PN papers, together with the carried* 2.5*PN narrowing to the canonically normalized orbital/worldtube STF quadrupole branch. Then the full conservative two-body* 4*PN coefficient admits the exact split*

$$L_{4\mathrm{PN}}^{\mathrm{cons}} = L_{4\mathrm{PN}}^{\mathrm{local}} + L_{4\mathrm{PN}}^{\mathrm{tail}}, \tag{62}$$

with the following properties.

(i) The local instantaneous sector $L_{4\mathrm{PN}}^{\mathrm{local}}$ is already fixed exactly inside the declared hierarchy. In particular, there exists an exact generic-frame ordinary representative whose COM reduction reproduces the full fixed-chart local ordinary 4*PN target slot by slot.*

(ii) The hereditary coefficient is fixed by the exact bridge

$$C_{\mathrm{tail}} = \frac{GM}{2c^3}\,\gamma_{\mathrm{quad}}^{\mathrm{eff}}, \tag{63}$$

where $\gamma_{\mathrm{quad}}^{\mathrm{eff}}$ is the canonically normalized passive/ outgoing STF quadrupole coefficient already isolated by the 2.5*PN program.*

(iii) Consequently, 4*PN opens no new independent normalization datum beyond that already-known quadrupole coefficient. Full conservative GR recovery at* 4*PN is equivalent to the single scalar condition*

$$\gamma_{\mathrm{quad}}^{\mathrm{eff}} = \frac{2G}{5c^5}, \tag{64}$$

in which case

$$C_{\mathrm{tail}} = \frac{GM}{2c^3}\,\gamma_{\mathrm{quad}}^{\mathrm{eff}} = \frac{G^2M}{5c^8} = C_{\mathrm{tail}}^{\mathrm{GR}}. \tag{65}$$

Therefore the only remaining conditional interface for a full conservative 4*PN GR theorem is the passive/outgoing STF quadrupole normalization on the actual moving-throat branch.*

Proof. The proof is the synthesis of the earlier sections.

First, the strict one-body Schwarzschild gate, the exact quartic perturbative Legendre compiler, the raw exchange-symmetric generic-frame scaffold, the fixed local Hamiltonian target, and the Hamiltonian-first generic-frame lift together show that the local instantaneous 4PN problem has no remaining span obstruction. The canonical Hamiltonian residual is exactly reachable in the Hamiltonian chart, its ordinary translation obeys the exact sign-flip theorem, and the only ordinary-chart obstruction is an aligned-seed issue rather than a failure of the local generic-frame lift itself. That aligned-seed correction is then exactly liftable in the structured seed spaces

$$Q, \qquad T \oplus (pq)T, \qquad S \oplus (pq)S, \qquad U \oplus (p^2q^2)U, \qquad W = 0,$$

so the full fixed-chart local ordinary 4PN target is reproduced exactly after COM reduction. This establishes item (i).

Second, the hereditary side is already much narrower than the local side. The conservative GR tail is a single STF-quadrupole functional with coefficient $C_{\rm tail}^{\rm GR} = G^2M/(5c^8)$, and its Newtonian local shadow lives only in the degree-2 G^4/r^4 block. The 2.5PN narrowing had already shown that the only surviving universal higher-order branch is the canonically normalized passive/outgoing STF quadrupole route. Matching the universal GR hereditary coefficient to that carried quadrupole coefficient gives the exact bridge

$$C_{\rm tail} = \frac{GM}{2c^3}\,\gamma_{\rm quad}^{\rm eff},$$

which proves item (ii).

Finally, because the hereditary coefficient depends only on the same scalar quadrupole normalization already isolated by the 2.5PN program, no second independent 4PN normalization datum can open. GR recovery of the hereditary sector is therefore equivalent to the single condition $\gamma_{\rm quad}^{\rm eff} = 2G/(5c^5)$, which immediately yields $C_{\rm tail} = G^2M/(5c^8)$. This proves item (iii) and the final claim. □

Corollary (no new 4PN normalization datum). Within the present hierarchy, the 4PN conservative sector introduces no new independent normalization problem beyond the already-known passive/outgoing STF quadrupole normalization. Equivalently, once the moving-throat PDE fixes $\gamma_{\rm quad}^{\rm eff}$, both the 2.5PN Burke–Thorne coefficient and the conservative 4PN tail coefficient are fixed simultaneously. This follows immediately from (63): the same scalar branch normalization that determines the odd quadrupole coefficient at 2.5PN also determines the conservative hereditary coefficient at 4PN.

Interpretation. The theorem above is the correct endpoint of the present paper. The local instantaneous 4PN sector is no longer provisional, symbolic, or span-limited: it is already closed exactly inside the declared hierarchy. The hereditary side is likewise no longer diffuse: it is controlled by one scalar bridge to the canonically normalized STF quadrupole branch. Thus the remaining open problem is not a generic 4PN algebraic search, but the completed moving-throat PDE realization of the passive/outgoing quadrupole normalization.

Practical reading rule. The correct conservative 4PN verdict is therefore:

The local instantaneous sector is already exact. The full conservative 4PN sector is conditionally complete, and the sole remaining condition is $\gamma_{\rm quad}^{\rm eff} = 2G/(5c^5)$, i.e. the same passive/outgoing STF quadrupole normalization already isolated by the 2.5PN program.

8 Interface to the 2.5PN program and the moving-throat PDE

From the viewpoint of the present paper, the 2.5PN audit and the conditional 4PN theorem are not two unrelated higher-order stories. They are the retarded and conservative faces of one and the same quadrupole program. The 2.5PN work already narrowed the universal higher-order bridge to the canonically normalized orbital/worldtube STF quadrupole branch; the present 4PN paper shows that the full local instantaneous algebra is already closed and that the hereditary coefficient depends on that same branch through one scalar bridge. The purpose of this section is to state that interface cleanly, record what the present 4PN result contributes to it, and identify the exact piece that the completed moving-throat PDE must still supply.

The strategic message is simple. The next theorem gate is no longer another local 4PN generic-frame solve, another seed search, or another free normalization family. It is the passive/outgoing STF quadrupole normalization on the natural moving-throat branch. Once that branch is fixed, the odd 2.5PN Burke–Thorne coefficient and the conservative 4PN hereditary coefficient close together.

8.1 Why the same STF quadrupole branch controls 2.5PN and 4PN

The 2.5PN audit already showed that the frozen conservative hierarchy cannot produce a GR-like local radiation-reaction theorem by itself and that, on the strict small-body branch, the scalar and dipole odd channels are pushed above universal point-particle 2.5PN. The surviving universal route is the canonically normalized orbital/worldtube STF quadrupole branch packaged in the isotropic grouped-real P_2 language. On the passive/outgoing compact branch, the normalized outgoing $\ell = 2$ fingerprint is

$$\widehat{Y}_2^{(\text{out})}(\omega) = 1 + \frac{a_{\text{th}}^2\omega^2}{9c_s^2} + \frac{4a_{\text{th}}^4\omega^4}{81c_s^4} + i\,\frac{a_{\text{th}}^5\omega^5}{27c_s^5} + O(\omega^6), \tag{66}$$

where a_{th} is the throat-radius scale on the moving-throat branch. On the isotropic grouped branch the conservative and outgoing low-frequency data collapse to common coefficients,

$$D_{20,n} = D_{21,n} = D_{22,n} = D_n, \qquad N_{20,n} = N_{21,n} = N_{22,n} = N_n, \tag{67}$$

so the static outgoing prefactor is

$$P_0 = \frac{N_0}{D_0}. \tag{68}$$

The leading odd quadrupole coefficient is then controlled by the same static prefactor,

$$\Gamma_5 = P_0\,\frac{a_{\text{th}}^5}{27c_s^5}, \qquad \gamma_{\text{quad}}^{\text{eff}} = \hat{m}_0{}^2\Gamma_5 = \hat{m}_0{}^2 P_0\,\frac{a_{\text{th}}^5}{27c_s^5}, \tag{69}$$

with $\hat{m}_0$ the source-map normalization factor on the natural branch. Equivalently, in the invariant-pair language already isolated by the quadrupole normalization package,

$$\gamma_{\text{quad}}^{\text{eff}} = 9\,\frac{\overline{K}_2^{5/2}}{\overline{K}_0^{3/2}}. \tag{70}$$

This is exactly the same scalar coefficient that enters the conservative hereditary bridge at 4PN. The present paper has shown that the hereditary coefficient is not an independent new tensor datum, but rather

$$C_{\text{tail}} = \frac{GM}{2c^3}\,\gamma_{\text{quad}}^{\text{eff}}. \tag{71}$$

So the 2.5PN odd channel and the 4PN conservative tail are controlled by one and the same canonically normalized STF quadrupole branch. The difference between the two orders is not a difference of branch, but only a difference in which coefficient of that branch is being read: the odd ω^5 coefficient for 2.5PN and the conservative hereditary coefficient for 4PN.

This is the main conceptual simplification produced by the combined 2.5PN and 4PN work. The higher-order bridge is no longer split into an apparently "dissipative" quadrupole problem and a separate "conservative tail" problem. It is one normalization problem viewed in two different PN sectors.

8.2 What the present 4PN result fixes for the quadrupole program

The positive interface result of the present paper is not that it computes a new branch parameter. Its actual contribution is sharper: it proves that no such new 4PN-specific parameter is needed. The local instantaneous 4PN sector is already closed exactly inside the declared hierarchy, and the hereditary coefficient has been reduced to the scalar bridge (71). Therefore the quadrupole program no longer has two conceptually independent open fronts. The only live front is the normalization of the passive/outgoing STF quadrupole branch itself.

The implication chain may be written in the compact form

$$\hat{m}_0^2 P_0 = \frac{54 G c_s^5}{5 a_{\rm th}^5 c^5} \Longleftrightarrow \gamma_{\rm quad}^{\rm eff} = \frac{2G}{5c^5} \Longleftrightarrow C_{\rm tail} = \frac{G^2 M}{5c^8}. \tag{72}$$

The first equality is the invariant moving-throat normalization target inherited from the reduced 2.5PN quadrupole package; the middle equality is the canonical quadrupole-normalization statement; and the final equality is the GR value of the conservative 4PN hereditary coefficient. Thus the present 4PN paper fixes the conservative end of the chain completely: once the moving-throat side lands on the target value of $\hat{m}_0^2 P_0$, the hereditary 4PN sector is no longer open.

In particular, the role of the present paper in the broader quadrupole program is now very specific. The 2.5PN audit had already shown that, on the isotropic real-P_2 branch, the grouped conservative coefficients through $O(\omega^4)$ form the front end from which the odd Burke–Thorne coefficient is read. The present 4PN paper proves that the same normalized branch also fixes the conservative hereditary coefficient and that 4PN opens no second normalization datum beyond that already-known branch. So the old question "what extra 4PN information is still needed after the 2.5PN audit?" now has a very short answer: none beyond the same branch normalization.

This also clarifies what the present 4PN theorem does *not* leave open. It does not leave a local-algebra bottleneck. It does not leave an unfixed local generic-frame existence question. And it does not leave a separate tail normalization family. The remaining problem is genuinely a moving-throat branch problem, not a missing post-Newtonian bookkeeping problem.

8.3 What the full moving-throat PDE still has to supply

What remains open is therefore sharp and narrow. The completed moving-throat PDE must still supply the actual passive/outgoing quadrupole normalization on its natural branch. In the grouped-real P_2 language, this means that the PDE must determine the conservative operator moments

$$D_A^{(\rm cons)}(\omega) = D_{A,0} + D_{A,2}\omega^2 + D_{A,4}\omega^4 + O(\omega^6), \tag{73}$$

together with the outgoing-transfer moments

$$N_A(\omega) = N_{A,0} + N_{A,2}\omega^2 + N_{A,4}\omega^4 + O(\omega^6), \tag{74}$$

for the grouped lanes $A \in \{20, 21, 22\}$. On the natural isotropic branch, these data must satisfy the grouped isotropy gate

$$a_2 = b_2 = a_4 = b_4 = 0, \tag{75}$$

so that the outgoing prefactor collapses to the scalar object $P_0 = N_0/D_0$ of (68). Equivalently, in the invariant-pair language, the PDE must determine the canonical pair $(\overline{K}_0, \overline{K}_2)$ on the passive/outgoing STF quadrupole branch.

The point is that the remaining PDE task is no longer diffuse. It is not "derive all of higher-order radiation reaction from scratch." It is to compute one isotropic branch and evaluate one invariant normalization product. In the most compact form, the outstanding theorem gate is

$$\boxed{\hat{m}_0^2 P_0 = \frac{54Gc_s^5}{5a_{\mathrm{th}}^5 c^5}} \tag{76}$$

or, equivalently,

$$\boxed{\gamma_{\mathrm{quad}}^{\mathrm{eff}} = \frac{2G}{5c^5}.} \tag{77}$$

Once this is true on the actual moving-throat branch, both the normalized 2.5PN odd quadrupole coefficient and the normalized 4PN conservative tail coefficient close simultaneously.

The current reduced moving-throat program sharpens this still further. On the natural isotropic grouped-P_2 one-pole branch, if the geometry lane is static through $O(\omega^4)$, then the conservative precursor is forced into the minimal form

$$\widehat{Y}_Q^{\mathrm{cons}}(\omega) = \frac{3}{4} + \frac{1}{4}\frac{1}{1-\omega^2/\Omega_Q^2}, \tag{78}$$

and the explicit Family-1 support/source side is no longer the active bottleneck. A convenient reduced defect packaging is then

$$N_Q := \frac{\overline{K}_0}{\overline{K}_0^{\mathrm{target}}}, \qquad \frac{\overline{\Gamma}_5}{\overline{\Gamma}_5^{\mathrm{target}}} = \chi_Q N_Q, \qquad \hat{m}_0^2 \chi_Q N_Q = 1, \tag{79}$$

where N_Q measures the conservative grouped-quadrupole normalization and χ_Q measures the outgoing $\ell = 2$ DtN normalization. On the canonical compact outgoing branch one finds the reduced closure $\chi_Q = 1$, so in the strict point-particle limit the condition becomes $N_Q = 1$. This is a sharpened reduced guide to the PDE problem, not yet a theorem that the completed moving-throat PDE must realize the canonical branch rather than a deformed one.

Two additional tasks remain attached to that PDE-side normalization problem, but both are now secondary rather than diffuse. First, the moving-throat PDE should still explain the deeper microscopic origin of the grouped conservative quadrupole data that appear on the isotropic branch. Second, once the normalization is fixed, the normalized hereditary module should be inserted into a full release-grade referee/master package so that the conditional 4PN statement becomes a fully assembled theorem package rather than only a theorem ledger. Neither of these is a new 4PN-specific parameter search; both are assembly problems downstream of the same quadrupole branch.

Correct next theorem gate. The correct next step after the present paper is therefore not more local 4PN algebra. It is to derive the passive/outgoing STF quadrupole normalization on the moving-throat branch, or equivalently to determine $(\overline{K}_0, \overline{K}_2)$ — or the invariant product $\hat{m}_0^2 P_0$ — on the natural isotropic branch. Once that gate is closed, the normalized 2.5PN odd quadrupole coefficient and the normalized 4PN hereditary coefficient become simultaneously fixed inside the same hierarchy.

9 Fixed versus open quantities

The conditional theorem established above closes the full local instantaneous 4PN ledger and fixes the hereditary bridge, but it does not erase the status boundary that runs through the whole series. As in the earlier conservative papers, one must keep separate what is exact in the carried reduction backbone, what is frozen as lower-order input, what is fixed by the present 4PN audit, and what still belongs to the unsolved moving-throat completion. The purpose of this section is to record that boundary in one place. Its role is not to weaken the theorem, but to state its true strength: the conservative 4PN coefficient problem is no longer open-ended, and the remaining open work has collapsed to one already-known quadrupole-normalization interface.

9.1 Frozen carry-forward inputs

The present paper does not begin from a blank effective theory. It imports the exact 4D parent reduction backbone already frozen by the earlier papers: the projection map, projected continuity with leakage, the exact longitudinal identity, the quasi-static Poisson hook, and the coherent-defect / worldtube particle reduction. These formulas remain the structural dictionary that turns the bulk ontology into the near-zone two-body language used throughout the post-Newtonian series.

The lower-order conservative ledger is carried in exactly the same way. The frozen Newtonian and 1PN constants remain

$$\kappa_\rho = 1, \qquad n = 5, \qquad \kappa_{\rm add} = \frac{1}{2}, \qquad \kappa_{\rm PV} = \frac{3}{2}, \qquad \beta_{\rm 1PN} = 3, \tag{80}$$

and the full conservative Newtonian+1PN, 2PN, and 3PN assemblies are already treated as fixed lower-order output inside the same declared closure hierarchy. In particular, the exact 1PN EIH match, the exact 2PN ADM match, and the full conservative 3PN chart-level closure are imported here as frozen theorem input rather than reopened as free data.

The 2.5PN narrowing is likewise a frozen input. The surviving universal higher-order route had already been reduced to the orbital/worldtube STF quadrupole branch, packaged in the grouped real language as

$$P_{20} \oplus P_{21} \oplus P_{22}. \tag{81}$$

That means the present 4PN paper is not searching for a new higher-order carrier. It is built from the start on the statement that the only universal retarded interface already lives on the canonically normalized STF quadrupole branch isolated by the 2.5PN audit.

9.2 Quantities fixed by the present 4PN audit

The first new fixed data are the strict one-body 4PN repair parameters. The exact Schwarzschild gate freezes the four-parameter continuation ledger

$$\mu_{\rho,4} = \frac{1}{8}, \qquad d_4 = \frac{205}{16}, \qquad s_{34} = -\frac{15}{32}, \qquad s_{26} = -\frac{1}{16}. \tag{82}$$

So the carried 3PN self-sector packaging does *not* extend to 4PN with one new scalar knob. The exact one-body front end already fixes a quartic static datum, a quartic denominator datum, and two genuinely new self-sector slots.

The second new fixed object is the exact order-4 ordinary/Hamiltonian bridge. Once the quartic compiler is frozen, the local 4PN chart problem is no longer a matter of heuristic translation. Combined with the imported local ADM target, it yields the Hamiltonian-first local lift theorem: the raw exchange-symmetric local scaffold is already complete blockwise in the Hamiltonian chart, the canonical comparable-mass local residual exists there exactly, and the ordinary translation obeys the exact sign-flip rule

$$\Delta H_4 = -\Delta L_4. \tag{83}$$

So the ordinary-chart obstruction is not a failure of the local generic-frame lift itself. It is an aligned-seed issue, and that issue is also fixed by the present audit. In particular, the aligned ordinary local seed is generically liftable once the seed sector is enlarged in the exact structured way

$$Q, \qquad T \oplus (pq)T, \qquad S \oplus (pq)S, \qquad U \oplus (p^2q^2)U, \qquad W = 0. \tag{84}$$

Therefore the whole fixed-chart local ordinary 4PN target now has an exact generic-frame representative. In theorem language, the entire local instantaneous sector is already fixed inside the declared hierarchy.

The third new fixed result is the hereditary bridge itself. The tail side does not open a new tensor family or a second normalization ladder. Once the canonically normalized STF quadrupole coefficient is denoted by $\gamma_{\rm quad}^{\rm eff}$, the exact hereditary coefficient is

$$C_{\rm tail} = \frac{GM}{2c^3}\,\gamma_{\rm quad}^{\rm eff}. \tag{85}$$

So the full conservative 4PN theorem split may now be written as

$$L_{\rm 4PN}^{\rm cons} = L_{\rm 4PN}^{\rm local} + L_{\rm 4PN}^{\rm tail}, \qquad C_{\rm tail} = \frac{GM}{2c^3}\,\gamma_{\rm quad}^{\rm eff}, \tag{86}$$

and the meaning of (86) is now sharp: the entire local instantaneous sector is already exact, and the hereditary side is already reduced to the same quadrupole branch that controlled the 2.5PN Burke–Thorne channel. In particular, (85) proves that 4PN opens no new independent normalization datum beyond that branch.

Object	**Status**	**Role at 4PN**
Parent reduction backbone	frozen	Exact/reduced dictionary inherited from the 4D parent theory
Lower-order Newtonian+1PN+2PN+3PN ledger	frozen	Fixed conservative input and chart conventions
2.5PN STF quadrupole narrowing	frozen	Fixes the only surviving universal higher-order branch
$(\mu_{\rho,4}, d_4, s_{34}, s_{26})$	fixed here	Exact one-body 4PN repair ledger
Quartic compiler	fixed here	Exact ordinary/Hamiltonian bridge through 4PN
Hamiltonian-first local lift and $\Delta H_4 = -\Delta L_4$	fixed here	Exact comparable-mass local chart theorem
Structured aligned-seed lift	fixed here	Exact ordinary-chart seed correction repair
$L_{\rm 4PN}^{\rm local}$	fixed here	Complete local instantaneous 4PN sector
$C_{\rm tail} = \frac{GM}{2c^3}\gamma_{\rm quad}^{\rm eff}$	fixed here	Hereditary bridge; no new 4PN datum
$\gamma_{\rm quad}^{\rm eff} = 2G/(5c^5)$	open	Sole remaining normalization condition

9.3 Quantities that remain symbolic or open

What remains open is no longer algebraic coefficient matching. The remaining open problem is the deeper moving-throat realization of already isolated branch data. The only live conditional theorem gate for a full conservative 4PN statement is

$$\gamma_{\rm quad}^{\rm eff} \stackrel{?}{=} \frac{2G}{5c^5}. \tag{87}$$

Equivalently, in the reduced moving-throat normalization language inherited from the 2.5PN package, the open condition is

$$\hat{m}_0^2 P_0 \stackrel{?}{=} \frac{54Gc_s^5}{5a_{\rm th}^5 c^5}, \tag{88}$$

with $a_{\rm th}$ the throat-radius scale on the natural branch. If (87) or, equivalently, (88) holds, then the GR hereditary coefficient follows automatically. If it fails, the present theorem remains conditional.

This is the main open theorem quantity. Two further open layers should be kept conceptually separate from it. First, the present paper does not yet derive from a completed moving-throat PDE the actual grouped-lane conservative and outgoing data on the true branch. In the grouped-real language, those still open branch quantities are the conservative operator moments

$$D_A^{(\mathrm{cons})}(\omega) = D_{A,0} + D_{A,2}\omega^2 + D_{A,4}\omega^4 + O(\omega^6), \tag{89}$$

together with the outgoing-transfer moments

$$N_A(\omega) = N_{A,0} + N_{A,2}\omega^2 + N_{A,4}\omega^4 + O(\omega^6), \tag{90}$$

for $A \in \{20, 21, 22\}$. On the natural isotropic branch, these data must collapse to the scalar prefactor $P_0 = N_0/D_0$ that appears in (88). So the remaining PDE task is not to search for a new 4PN coefficient family. It is to compute one already-isolated quadrupole branch and evaluate one already-isolated invariant normalization product.

Second, a few objects remain open only at the level of deeper microscopic interpretation rather than at the level of coefficient matching. The present paper does not yet provide a first-principles moving-throat derivation of the conservative grouped real quadrupole data beneath (89); it does not yet prove a final uniqueness/interpretation theorem for every aligned-seed local representative; and it does not attempt any dissipative or radiative completion beyond the already narrowed quadrupole route. These are genuine open problems, but they are no longer algebraic obstacles to the local 4PN theorem chain.

Finally, there is a practical assembly distinction that should remain visible. Once the normalization target is fixed on the actual branch, the normalized hereditary module still has to be inserted into a fully assembled release-grade referee/master package. That remaining step is an end-to-end assembly task, not a new coefficient-discovery problem.

9.4 Interpretation

The correct reading rule is now short. What should be treated as fixed carry-forward 4PN data are the one-body repair ledger (82), the quartic compiler, the Hamiltonian chart lift and sign-flip theorem (83), the structured aligned-seed lift (84), the existence of an exact generic-frame representative for the full local instantaneous sector, and the hereditary bridge (85). What should *not* be treated as fixed yet is the final passive/outgoing quadrupole normalization on the actual moving-throat branch.

So the main status verdict of the whole paper may be summarized in one sentence:

The only live conditional interface left at 4PN is the already-known passive/outgoing STF quadrupole-normalization gate, $\gamma_{\mathrm{quad}}^{\mathrm{eff}} = 2G/(5c^5)$, equivalently $\hat{m}_0^2 P_0 = 54Gc_s^5/(5a_{\mathrm{th}}^5 c^5)$.

That is the practical meaning of the present theorem package. The local instantaneous 4PN problem is no longer open. The hereditary side no longer opens any new datum. The remaining open work is concentrated in one already identified branch-normalization theorem that is inherited from the 2.5PN program rather than created afresh by 4PN.

10 Verification and reproducibility ledger

The derivation in this paper is analytic rather than computational. The accompanying symbolic archive is therefore not presented as a black-box generator of the theorem. Its role is narrower and more auditable: it serves as a referee ledger for the exact one-body gates, compiler identities, local target imports, generic-frame lift statements, aligned-seed corrections, referee reconstructions, and hereditary bridge formulas used in the main text. The archive verifies the paper as written *within the declared closure hierarchy*; it does not replace any analytic step, and it does not by itself upgrade the present work into an assumption-free theorem of a fully solved moving-throat bulk PDE.

At 4PN the verification story is naturally split into three layers. The first layer is the inherited lower-order archive carried forward from the 4D, 1PN, 2PN, 2.5PN, and 3PN packages, which freezes the reduction backbone, notation firewall, and already solved coefficient ledger used unchanged here. The second layer is the dedicated dual-CAS 4PN referee suite, consisting of a stagewise SymPy ledger together with a grouped Mathematica mirror. Its job is to check the new theorem chain specific to the present paper: the strict one-body gate, the quartic Legendre compiler, the local scaffold and fixed-chart import, the Hamiltonian-first generic-frame lift, the aligned-seed correction, the local referee reconstruction, and the hereditary/tail bridge. The third layer is the conditional end-to-end replay, which ties the already-frozen 2.5PN quadrupole narrowing to the exact local 4PN closure and the hereditary bridge, and thereby verifies that the only remaining 4PN interface is the same passive/outgoing STF quadrupole normalization already isolated on the 2.5PN side.

10.1 Role of the 4PN referee suite

The present referee suite should be read as a symbolic reproducibility archive, not as an independent source of physical assumptions. Its function is to make each load-bearing algebraic step re-runnable in a transparent way. In particular, the suite is designed to verify the statements that matter most to a referee reading the paper as a theorem chain:

- whether the exact Schwarzschild one-body gate really forces the 4PN repair ledger $\mu_{\rho,4} = 1/8,\ d_4 = 205/16,\ s_{34} = -15/32,\ s_{26} = -1/16$;
- whether the quartic perturbative Legendre-transform identity used to move between ordinary and Hamiltonian descriptions is exact;
- whether the imported local ADM target, the Hamiltonian-first lift, and the canonical comparable-mass slice really close the full local instantaneous sector;
- whether the aligned ordinary seed correction is generically liftable in the structured seed spaces claimed in the main text; and
- whether the hereditary coefficient really reduces to the single bridge relation

$$C_{\rm tail} = \frac{GM}{2c^3}\,\gamma^{\rm eff}_{\rm quad}, \tag{91}$$

  so that the only remaining 4PN gate is the already-known quadrupole normalization.

So the archive is not being asked to prove a different theorem from the one stated in the paper. It is being asked to verify that the paper's own exact, reduced, and closure-level statements close where they are claimed to close.

10.2 Core artifacts in the present package

In the published supplementary archive, the dedicated 4PN package is organized around a stagewise SymPy ledger together with a grouped Mathematica mirror. The SymPy side carries the full fine-grained theorem chain, while the Mathematica side independently rechecks the same load-bearing gates in a smaller set of bundle-level scripts.

Table 2: Core 4PN referee artifacts and their intended roles in the theorem chain.

Artifact	Intended role in the 4PN chain
`4pn_onebody_audit.py`	Exact one-body 4PN Schwarzschild gate and repair ledger.
`4pn_quartic_legendre_audit.py`	Exact quartic Legendre compiler through order c^{-8}.
`4pn_local_scaffold_audit.py`	Raw local generic-frame scaffold, COM completeness, and pivot-spanning local basis.
`4pn_local_target_import_audit.py`	Fixed-chart local ADM target import and the blockwise local degree-ceiling audit.
`4pn_local_hamiltonian_to_ordinary_audit.py`	Reduced COM Hamiltonian-to-ordinary translation of the fixed local target.
`4pn_hamiltonian_chart_generic_frame_lift_audit.py`	Hamiltonian-chart generic-frame lift of the local comparable-mass residual.
`4pn_hamiltonian_chart_canonical_slice_audit.py`	Canonical comparable-mass residual slice in the Hamiltonian chart.
`4pn_generic_frame_ordinary_translation_audit.py`	Ordinary-chart residual translation and aligned-seed setup.
`4pn_generic_frame_aligned_seed_lift_audit.py`	Generic-frame lift of the aligned-seed ordinary correction.
`4pn_local_referee_master_sympy_audit.py`	End-to-end local referee reconstruction of the full fixed-chart ordinary target.
`4pn_tail_bridge_audit.py`	Tail-side bridge setup and universal quadrupole local shadow.
`4pn_tail_hereditary_bridge_audit.py`	Exact hereditary coefficient bridge to the canonically normalized STF quadrupole branch.
`4pn_conditional_referee_master_audit.py`	Conditional full conservative 4PN replay tying the local closure to the inherited 2.5PN quadrupole-normalization gate.

The package should be read together with the carried lower-order referee archive. In particular, the final conditional replay imports the already-frozen 2.5PN master audit rather than rederiving the quadrupole narrowing from scratch, because the point of the 4PN package is precisely to show that no new independent normalization datum opens beyond that carried interface.

The grouped Mathematica mirror lives beside the stagewise SymPy archive and is organized around the following seven scripts:

```
4pn_onebody_mathematica_audit.wl
4pn_quartic_legendre_mathematica_audit.wl
4pn_local_scaffold_target_mathematica_audit.wl
4pn_hamiltonian_chain_mathematica_audit.wl
4pn_ordinary_chain_mathematica_audit.wl
4pn_tail_bridge_mathematica_audit.wl
4pn_conditional_referee_master_mathematica_audit.wl
```

These files do not replace the finer stage structure on the SymPy side; their role is to provide an independent second-engine replay of the same one-body, compiler, local-chain, hereditary-bridge, and conditional-closure statements.

10.3 Stage audits replayed by the master runner

The stage structure mirrors the theorem chain in the main text and is organized so that failure in one lane cannot be hidden by compensating changes in another. At a practical level the archive is best read in four tranches.

One-body and compiler front end. The first tranche consists of the one-body and compiler audits. These scripts expand the exact Schwarzschild target, solve the minimal 4PN repair ledger, and freeze the quartic ordinary/Hamiltonian compiler that every later local comparison depends on.

Local scaffold and fixed-chart import. The second tranche consists of the raw local scaffold, target-import, and reduced translation audits. Their job is to verify that the raw exchange-symmetric local scaffold is already COM-complete block by block, that the fixed local ADM target is imported in the correct chart, and that the Hamiltonian chart rather than the translated ordinary chart carries the correct local degree ceilings.

Generic-frame local reconstruction. The third tranche consists of the Hamiltonian-chart lift, canonical slice, ordinary translation, aligned-seed lift, and local referee master audits. Together they verify the full Hamiltonian-first local theorem chain: existence of a generic-frame Hamiltonian representative, exact ordinary sign flip, structured liftability of the aligned-seed correction, and final reconstruction of one exact generic-frame ordinary representative whose COM reduction reproduces the complete fixed-chart local ordinary target.

Hereditary and conditional interface lane. The final tranche consists of the tail bridge, hereditary bridge, and conditional master replay. This lane verifies the universal Newtonian quadrupole shadow of the GR tail, the exact coefficient bridge (91), and the interface theorem stating that the full conservative 4PN theorem is conditional only on the same passive/outgoing STF quadrupole normalization already isolated by the 2.5PN program.

This modular staging is intentional. It keeps the one-body gate, the local Hamiltonian-first lift, the aligned-seed repair, and the hereditary bridge separate, so that a failure in one sector cannot be disguised as a different kind of local or tail compensation.

10.4 What the suite verifies

The suite verifies the load-bearing symbolic statements of the paper sector by sector.

First, the early audits verify the exact one-body and compiler backbone. They check the strict Schwarzschild 4PN target, confirm that the carried 3PN self-sector packaging is insufficient by itself, solve the minimal repair ledger, and then prove the quartic perturbative Legendre-transform identity needed to move cleanly between ordinary and Hamiltonian descriptions at 4PN.

Second, the local middle audits verify the entire comparable-mass chart-level closure chain. They check the exact local scaffold and COM completeness, the fixed local target import, the Hamiltonian-first degree-ceiling lesson, the Hamiltonian-chart generic-frame lift, the canonical residual slice, the exact ordinary sign-flip rule

$$\Delta H_4 = -\Delta L_4, \tag{92}$$

and the structured lift of the aligned-seed ordinary correction. In the language of the main text, this is the stage chain that verifies that the entire local instantaneous 4PN sector has an exact generic-frame ordinary representative inside the declared hierarchy.

Third, the local referee master verifies the end-to-end local assembly itself. It rebuilds the exact reduced COM local ordinary target from the imported local ADM Hamiltonian, verifies the one-body gate and natural seed, replays the canonical residual translation and aligned-seed correction, and confirms that the assembled generic-frame candidate reduces back to the full fixed-chart local ordinary target slot by slot. This is the direct executable check of the main local theorem claim.

Fourth, the tail-side audits verify the narrow hereditary bridge. They confirm that the GR tail contributes a universal Newtonian quadrupole shadow, and that on the canonically normalized STF quadrupole branch the coefficient is exactly

$$C_{\rm tail} = \frac{GM}{2c^3}\,\gamma^{\rm eff}_{\rm quad}. \tag{93}$$

So the hereditary side is not an independent 4PN coefficient hunt. It is an interface theorem tying the conservative hereditary coefficient to the same quadrupole normalization that already controls the 2.5PN odd sector.

Finally, the conditional master replay verifies the paper's sharpest global status statement: once the inherited 2.5PN narrowing, the exact local 4PN closure, and the hereditary bridge are replayed together, the only remaining full-4PN gate is the already-known passive/outgoing quadrupole normalization condition,

$$\gamma^{\rm eff}_{\rm quad} \stackrel{?}{=} \frac{2G}{5c^5}, \tag{94}$$

or, equivalently on the moving-throat side,

$$\hat{m}_0^2 P_0 \stackrel{?}{=} \frac{54Gc_s^5}{5a_{\rm th}^5 c^5}. \tag{95}$$

The executable archive therefore verifies the same conclusion as the analytic paper: no new independent normalization datum opens at 4PN.

Two reproducibility conventions are worth recording explicitly. First, the archive is stagewise rather than monolithic, so each load-bearing symbolic claim is attached to a named audit rather than hidden inside one opaque master file. Second, the local and hereditary lanes are kept separate all the way to the conditional replay, which makes the theorem split $L^{\rm cons}_{\rm 4PN} = L^{\rm local}_{\rm 4PN} + L^{\rm tail}_{\rm 4PN}$ visible in the executable package itself rather than only in the prose of the paper.

10.5 What the suite does not verify

The suite does *not* replace the analytic argument in the main text, and it does not solve the full moving-throat PDE. In particular, the archive does not by itself derive the passive/outgoing STF quadrupole normalization on the actual moving-throat branch; it does not prove from first principles that (94) or (95) holds; and it does not upgrade the present work into an assumption-free theorem of the full bulk dynamics.

It also does not attempt to settle every deeper microscopic interpretation issue that remains outside the present 4PN scope. The archive is not a proof of the full moving-throat origin of all grouped conservative data, it is not a final uniqueness theorem for every aligned-seed local representative, and it is not a general radiative or higher-PN completion package. Those questions remain outside the stated theorem target of this paper.

So the correct reading rule is simple. The referee archive verifies that the paper's exact local theorem chain and hereditary bridge close where claimed, and that the whole conservative 4PN problem has been reduced to one already-known quadrupole-normalization gate. What it does not verify is the final PDE-side realization of that gate. That distinction is exactly the one on which the conditional status of the present 4PN theorem rests.

11 Conclusions

This paper closes the conservative two-body 4PN problem in the same precise sense that the earlier full papers closed the conservative 1PN, 2PN, and 3PN sectors: not as an assumption-free theorem of a fully solved moving-throat PDE, but as an exact algebraic assembly within a declared closure hierarchy. The gain is larger than one more coefficient list. At 4PN the entire local instantaneous sector is now fixed exactly, while the hereditary sector has collapsed to a single scalar bridge to the same canonically normalized passive/outgoing STF quadrupole branch that had already been isolated by the 2.5PN program. In that sharpened sense, the conservative 4PN bottleneck is no longer an unresolved coefficient search. It is one already-known normalization interface.

11.1 What has been established

The proof line established in the paper may be stated compactly.

1. The strict one-body Schwarzschild gate through c^{-8} is fixed exactly, and its minimal repair ledger is not one-dimensional but

$$\mu_{\rho,4} = \frac{1}{8}, \qquad d_4 = \frac{205}{16}, \qquad s_{34} = -\frac{15}{32}, \qquad s_{26} = -\frac{1}{16}. \tag{96}$$

   So the carried 3PN self-sector package does not extend to 4PN by a single denominator coefficient or a single static gate alone.

2. The exact quartic ordinary/Hamiltonian compiler is rigid, and once the lower-order ledger is frozen the correct comparable-mass lift is Hamiltonian-first. In the local chart this gives the exact sign-flip rule

$$\Delta H_4 = -\Delta L_4, \tag{97}$$

   so the comparable-mass local residual is most naturally fixed on the Hamiltonian side and only translated back to the ordinary chart afterward.

3. In that Hamiltonian chart there is no local span obstruction. The raw exchange-symmetric generic-frame scaffold is already complete against the full fixed-chart local interaction target, the canonical comparable-mass residual is exactly reachable, and the only ordinary-chart obstruction is an aligned-seed issue rather than a failure of the local generic-frame lift itself.

4. That aligned-seed obstruction is not a new free datum. It is exactly liftable in the structured seed spaces

$$Q, \qquad T \oplus (pq)T, \qquad S \oplus (pq)S, \qquad U \oplus (p^2q^2)U, \qquad W = 0, \tag{98}$$

   so one exact generic-frame ordinary representative of the full local instantaneous sector exists inside the declared hierarchy.

5. Consequently, the local instantaneous 4PN theorem is already exact in the form

$$L^{\text{local}}_{4\text{PN}} = L^{(\text{natural,gen})}_{4,\text{seed}} + \delta L^{(\text{gen})}_{4,\text{seed}} + \Delta L^{(\text{can,gen})}_{4,\text{loc}}, \qquad \Pi_{\text{COM}}\left[L^{\text{local}}_{4\text{PN}}\right] = L^{(\text{target,ord})}_{4,\text{loc}}. \tag{99}$$

   So the local instantaneous 4PN bottleneck is no longer algebraic span, chart ambiguity, or generic-frame existence.

6. The hereditary side is now equally sharp. Its unique compatible coefficient is

$$C_{\text{tail}} = \frac{GM}{2c^3}\,\gamma^{\text{eff}}_{\text{quad}}, \tag{100}$$

   where $\gamma^{\text{eff}}_{\text{quad}}$ is the canonically normalized passive/outgoing STF quadrupole coefficient already isolated by the 2.5PN program. Therefore 4PN opens no new independent normalization datum.

When these facts are assembled, the full conservative 4PN theorem takes the exact split

$$\boxed{L^{\rm cons}_{\rm 4PN} = L^{\rm local}_{\rm 4PN} + L^{\rm tail}_{\rm 4PN}, \qquad C_{\rm tail} = \frac{GM}{2c^3}\,\gamma^{\rm eff}_{\rm quad}.} \tag{101}$$

Full conservative GR recovery is therefore equivalent to the single scalar condition

$$\gamma^{\rm eff}_{\rm quad} = \frac{2G}{5c^5}, \tag{102}$$

which immediately yields

$$C_{\rm tail} = \frac{G^2 M}{5c^8} = C^{\rm GR}_{\rm tail}. \tag{103}$$

Conceptually, the paper does three things at once. First, it closes the entire local instantaneous 4PN sector exactly inside the declared hierarchy. Second, it proves that the hereditary side is not a second independent higher-order search, but a scalar bridge to the same canonically normalized STF quadrupole branch already identified by the 2.5PN audit. Third, it sharpens the bookkeeping of what is fixed, what is closure-level, and what still has to be supplied by the completed moving-throat PDE. The accompanying referee archive supports this reading in an auditable way: the one-body, compiler, local-lift, aligned-seed, referee-master, and tail-bridge audits collectively verify that the exact theorem chain closes where claimed inside the same hierarchy.

11.2 What remains open

The negative statement is equally important.

First, the paper does not solve the fully dynamical moving-throat PDE and then recover the 4PN theorem as an automatic bulk consequence. The present result remains a full conservative derivation only within the declared reduction and closure hierarchy already used by the lower-order papers. That status is not a weakness of the theorem; it is the correct statement of its scope.

Second, the paper does not yet derive from first principles the passive/ outgoing STF quadrupole normalization on the actual moving-throat branch. The remaining universal target may be stated in either of the equivalent forms

$$\boxed{\gamma^{\rm eff}_{\rm quad} = \frac{2G}{5c^5}, \qquad \hat{m}_0^2 P_0 = \frac{54 G c_s^5}{5 a_{\rm th}^5 c^5}.} \tag{104}$$

The present 4PN theorem removes the conservative algebraic uncertainty *beneath* this bridge, but it does not by itself prove that the physical moving-throat branch realizes (104).

Third, the deeper microscopic derivation of the grouped conservative quadrupole data also remains open. On the moving-throat side, the relevant grouped bundle still has to be computed as an actual branch output,

$$D_A(\omega) = D_{A0} + D_{A2}\omega^2 + D_{A4}\omega^4 + O(\omega^6), \qquad N_A(\omega) = N_{A0} + N_{A2}\omega^2 + N_{A4}\omega^4 + O(\omega^6), \tag{105}$$

with $A \in \{20, 21, 22\}$ and with the isotropic branch determined rather than assumed. These quantities are not leftover algebraic residues in the present 4PN theorem. They are dynamical response data of the microscopic mouth/support/mixed-sector system.

Fourth, the paper does not yet prove a final uniqueness or interpretation theorem for every aligned-seed local representative. What has been proved is the exact existence of a full local generic-frame representative and the structured liftability of the aligned-seed correction. That is sufficient for the present conservative theorem, but it is not the last possible microscopic word on the ordinary-chart seed organization.

Finally, larger sectors remain outside the honest scope of the present paper: dissipative leakage and radiation reaction beyond the already narrowed quadrupole route, spin couplings, strong-field completion, and higher-PN work beyond the solved conservative 4PN scope. What remains open is therefore no longer a local coefficient-matching problem. It is the deeper moving-throat realization of data that are already algebraically assigned.

11.3 Interface to the next calculation

This gives the project a much cleaner handoff than before. The next job is no longer to reopen the local 4PN generic-frame algebra, to search for another hidden seed parameter, or to treat the hereditary sector as a fresh independent normalization family. That work is substantially finished in the present chart. The next job is the moving-throat grouped-quadrupole calculation itself.

On the conservative side, the physical branch must first determine whether the natural grouped real P_2 isotropy gate actually holds,

$$\boxed{a_2 = 0, \qquad b_2 = 0,} \tag{106}$$

and, equivalently on the outgoing side,

$$\boxed{a(P_0) = 0, \qquad b(P_0) = 0.} \tag{107}$$

If these first gates fail, then the minimal isotropic passive/outgoing quadrupole branch is already dead before the final normalization test is even attempted. If they pass, then the next task is simply to evaluate the single invariant normalization product in (104).

Equivalently, the completed moving-throat PDE must determine the grouped-bundle coefficients in (105) together with the source-map factor $\hat{m}_0$, collapse them to the isotropic branch if that branch is realized, and then test whether

$$\hat{m}_0^2 P_0 = \frac{54 G c_s^5}{5 a_{\text{th}}^5 c^5}. \tag{108}$$

At that point the odd 2.5PN Burke–Thorne coefficient and the conservative 4PN hereditary coefficient close together; before that point, both remain conditional on the same microscopic normalization theorem.

So the correct next calculation is sharply defined. One must compute the canonically normalized grouped quadrupole data on the physical moving-throat branch, decide whether the isotropic passive/outgoing branch is actually realized there, and then evaluate the single invariant normalization product in (108). That is the only place where the present theorem still needs to be sharpened from a closure-level conservative result into a fully microscopic one.

The strongest honest endpoint of the paper is therefore also the strongest honest starting point for the next one: within the declared hierarchy, the local instantaneous 4PN bottleneck is gone, the hereditary bridge is exact, and the only serious universal question left open is the passive/outgoing normalization of the already identified STF quadrupole branch.

Acknowledgments

This work was carried out by an independent researcher. Generative-AI tools were used during the development of the manuscript to help translate conceptual ideas into mathematical form, assist with symbolic manipulations, prototype and revise companion Mathematica and SymPy verification scripts, and improve the exposition. The conceptual direction, hypotheses, and overall structure of the work were directed by the author. The resulting arguments and calculations were checked for internal consistency using the accompanying symbolic scripts, but the work should be regarded as exploratory and would benefit from review by subject-matter experts. Any errors, omissions, or misunderstandings remain the author's.

A Carry-forward formulas from the earlier papers

This appendix records the specific formulas and status statements imported from the parent 4D paper and the full conservative 1PN, 2PN, 2.5PN, and 3PN packages and used without rederivation in the present 4PN work. Its role is technical rather than historical. We do not repeat the full parent action, the full lower-order proofs, or the longer appendix-side derivations that fixed the wake, breathing, support, grouped-quadrupole, and geometry-completion data. Instead, we collect only the load-bearing formulas that the present 4PN paper actually uses: the exact parent reduction identities, the frozen Newtonian+1PN+2PN ledger, the 2.5PN narrowing to the canonically normalized STF quadrupole branch, and the 3PN front end that the 4PN one-body and local theorem chains continue.

The claim-status boundary is the same one used throughout the earlier papers. Exact projection identities from the parent 4D theory are kept visibly exact. The near-zone Poisson hook, the zero-mode Maxwell law, and the worldtube point-particle limit are carried only as controlled reductions. The conservative Newtonian+1PN+2PN+3PN ledgers are treated as frozen outputs of the earlier program within its declared closure hierarchy, not as live parameters to be retuned in the present paper. The 2.5PN package is used in the same way: not as a finished moving-throat PDE theorem, but as the theorem audit that narrowed the surviving higher-order universal route to the orbital/worldtube STF quadrupole branch.

A.1 Parent 4D reduction dictionary and Newtonian+1PN carried data

The parent theory lives on a $(4+1)$-dimensional spacetime with coordinates

$$X^M = (t, x, y, z, w), \qquad \mathbf{X} = (x, y, z, w) \in \mathbb{R}^4, \qquad \mathbf{x} = (x, y, z) \in \mathbb{R}^3.$$

The bulk variables carried into the present paper are the matter field $\psi(\mathbf{X}, t)$, its density

$$\rho(\mathbf{X}, t) = |\psi(\mathbf{X}, t)|^2,$$

the optional localized gauge field $A_M(\mathbf{X}, t)$ with localization profile $Z(w)$, and the effective throat geometry variables

$$a(t), \qquad L(t).$$

When electric charge is named explicitly in this downstream notation, it is always the corrected branch quantity

$$q_* = \eta_Q e_*, \qquad q_{\text{eff}} = \frac{q_*}{\sqrt{Z_{\text{int}}}}, \qquad Z_{\text{int}} = \int_{-\infty}^{\infty} Z(w)\, dw,$$

not the gravity-side scalar coefficient κ_ρ that appears in the post-Newtonian ledger below.

Brane observables are defined by a fixed normalized projection kernel $W(w)$,

$$\int_{-\infty}^{\infty} W(w)\, dw = 1, \qquad \mathcal{P}_W[Q](t, \mathbf{x}) \equiv \int_{-\infty}^{\infty} W(w)\, Q(t, \mathbf{x}, w)\, dw.$$

In particular,

$$\rho_{\text{br}} = \mathcal{P}_W[\rho], \qquad \mathbf{J}_{\text{br}} = \mathcal{P}_W[(j^x, j^y, j^z)], \qquad \mathbf{v}_{\text{br}} = \frac{\mathbf{J}_{\text{br}}}{\rho_{\text{br}}} \quad (\rho_{\text{br}} > 0).$$

Projection is not the same thing as reduction. Projection defines what a brane observer measures; reduction refers to the further elimination of the extra coordinate under an explicit ansatz or scaling regime.

From exact bulk continuity,

$$\partial_t \rho + \partial_i j^i = 0, \qquad i \in \{x, y, z, w\},$$

one obtains the exact open-system brane continuity law

$$\partial_t \rho_{\rm br} + \nabla_3 \cdot \mathbf{J}_{\rm br} = S_{\rm leak},$$

with leakage source

$$S_{\rm leak} = -\big[W(w) j^w\big]_{-\infty}^{+\infty} + \int_{-\infty}^{\infty} W'(w) j^w \, dw.$$

After Helmholtz decomposition of the brane velocity,

$$\mathbf{v}_{\rm br} = \nabla_3 \phi_{\rm br} + \mathbf{v}_T, \qquad \nabla_3 \cdot \mathbf{v}_T = 0,$$

projection continuity becomes the exact longitudinal identity

$$\rho_{\rm br} \, \nabla_3^2 \phi_{\rm br} = S_{\rm leak} - \partial_t \rho_{\rm br} - (\nabla_3 \rho_{\rm br}) \cdot (\nabla_3 \phi_{\rm br} + \mathbf{v}_T).$$

Under the usual quasi-static near-zone simplifications this becomes the carried Poisson hook

$$\nabla_3^2 \phi_{\rm br} \simeq \frac{S_{\rm eff}}{\rho_0}, \qquad \Phi_N = \lambda_\Phi \, \phi_{\rm br},$$

which is the origin of the Newtonian scalar sector used in the lower-order particle reduction. The corresponding coherent-defect / small-body worldtube limit gives the point-particle force law

$$M_A \, \ddot{\mathbf{X}}_A = -M_A \nabla \Phi_{\rm ext}(\mathbf{X}_A) + O\!\left(\frac{a_{\rm th}^2}{r^2} \frac{\Phi_{\rm ext}}{r}\right),$$

with the first finite-size correction quadrupolar and therefore suppressed in the controlled small-worldtube regime. For completeness, the same parent paper also carries a controlled zero-mode Maxwell reduction. Under the far-field brane assumptions

$$A_w \approx 0, \qquad \partial_w A_\mu \approx 0, \qquad J^w \approx 0, \qquad F_{\mu w} \approx 0,$$

integration over w gives the effective brane Maxwell law

$$\partial_\mu F^{\mu\nu} = \mu_0^{\rm eff} J^\nu_{\rm eff}, \qquad \mu_0^{\rm eff} = \frac{\mu_0}{Z_{\rm int}}.$$

This reduction suppresses the mixed channels only in the controlled far-field limit. It does not erase $(A_w, J^w, F_{\mu w})$ from the microscopic ontology.

The Newtonian point-particle block carried into every later paper is therefore

$$L_0 = \frac{1}{2} m_A v_A^2 + \frac{1}{2} m_B v_B^2 + \frac{G m_A m_B}{r}.$$

At 1PN the scalar/optical one-body reduction fixes the lower-order viability conditions

$$\kappa_\rho = 1, \qquad n = 5, \qquad \kappa_{\rm add} = \frac{1}{2}, \qquad \kappa_{\rm PV} = \frac{3}{2}, \qquad \beta_{\rm 1PN} = 3.$$

In the parity-even wake sector, the constructive isotropic Fourier-space closure freezes

$$\alpha^2 = \frac{3}{4}, \qquad a_H = 0, \qquad K_{\rm vec} = \frac{2}{\pi^2},$$

which in turn give the EIH cross coefficients

$$C_\parallel = -\frac{7}{2}, \qquad C_L = -\frac{1}{2}.$$

With these values frozen, the full carried Newtonian+1PN two-body block is

$$\begin{aligned} L_1 =& \frac{1}{8}(m_A v_A^4 + m_B v_B^4) + \frac{G m_A m_B}{r} \left[\frac{3}{2}(v_A^2 + v_B^2) - \frac{7}{2} v_{AB} - \frac{1}{2} v_{An} v_{Bn}\right] \\ & - \frac{G^2 m_A m_B (m_A + m_B)}{2 r^2}. \end{aligned}$$

Nothing in the present 4PN paper is allowed to retune these lower-order values. They are carry-forward anchors rather than open coefficients.

A.2 2PN carried data and the $P_0 \oplus P_2$ support hierarchy

The full conservative 2PN paper closes the comparable-mass c^{-4} ledger against the standard generic-frame ADM target within the same declared closure hierarchy, but the present 4PN paper imports from it only the load-bearing formulas and the response-side structure that sharpened the ontology of the defect. At the strict one-body level, the exact isotropic Schwarzschild gate through 2PN is packaged by

$$D_{\text{target}}(u) = 1 - 4u + 8u^2 + O(u^3), \qquad u \equiv \frac{U}{c^2},$$

and the pure-static 2PN gate fixes

$$\lambda_\rho = \frac{1}{2}, \qquad L_{2,\text{static}} = \frac{G^3 m_A m_B (m_A^2 + m_B^2)}{4r^3}.$$

The full carried 2PN coefficient inserted into the Newtonian+1PN ledger is

$$\begin{aligned}
L_{2,\text{full}} = {} & \frac{m_A v_A^6 + m_B v_B^6}{16} \\
& + \frac{G m_A m_B}{r} \Big[\frac{7}{8}(v_A^4 + v_B^4) - \frac{7}{4} v_{AB}(v_A^2 + v_B^2) - \frac{1}{4} v_{An} v_{Bn} (v_A^2 + v_B^2) \\
& \qquad + \frac{11}{8} v_A^2 v_B^2 + \frac{1}{4} v_{AB}^2 - \frac{5}{8}(v_A^2 v_{Bn}^2 + v_B^2 v_{An}^2) + \frac{3}{2} v_{AB} v_{An} v_{Bn} + \frac{3}{8} v_{An}^2 v_{Bn}^2 \Big] \\
& + \frac{G^2 m_A m_B}{r^2} \Big[\Big(2m_B + \frac{11}{8} m_A\Big) v_A^2 + \Big(2m_A + \frac{11}{8} m_B\Big) v_B^2 \\
& \qquad - \frac{15}{4}(m_A + m_B) v_{AB} + \frac{15}{8}(m_A v_{An}^2 + m_B v_{Bn}^2) \Big] \\
& + \frac{G^3 m_A m_B}{4r^3}(m_A^2 + 5 m_A m_B + m_B^2).
\end{aligned}$$

So by the start of the present paper, the full conservative Newtonian+1PN+2PN ledger is already frozen in the declared hierarchy.

Conceptually, the decisive ontology change at 2PN is that the defect is no longer described adequately as a pure scalar monopole. The conservative cross sector already splits into three response lanes:

1. a carried odd dipole wake sector,
2. a genuine even $P_0 \oplus P_2$ mouth/support layer,
3. and a separate geometry-closure channel.

The appendix-side zero-frequency throat data make this explicit. The odd dipole residues are frozen to

$$R_{1\perp} = \frac{7}{2}, \qquad R_{10} = 4,$$

while the canonical even support residues are

$$R_0 = R_{20} = R_{21} = R_{22} = 1.$$

The same solved static even operator fixes the source-weight vector

$$J = \left(\frac{4}{\sqrt{5}}, \frac{5}{4}, 0, 0, 0, 0 \right), \qquad J \cdot J = \frac{381}{80},$$

and the pure-potential geometry-closure deficit

$$\Delta_{\text{geom}} = \frac{281}{80}.$$

The monopole wall add-on

$$\Delta K_{00} = \frac{109}{280}$$

is kept distinct from $\Delta_{\rm geom}$; they are not the same object and should not be merged in later bookkeeping. What remains open after the conservative 2PN theorem is not these zero-frequency residues but the fully dynamical completion. In particular, the pole or compliance scales

$$\Omega_{1\perp},\ \Omega_{10},\ \Omega_0,\ \Omega_{20},\ \Omega_{21},\ \Omega_{22},\ \Omega_g$$

are not claimed as already derived from the completed moving-throat PDE.

A.3 2.5PN narrowing to the grouped real P_2 quadrupole branch

The 2.5PN theorem audit is what makes the present 4PN problem sharply focused rather than diffuse. Its decisive benchmark shows that the frozen conservative hierarchy by itself cannot generate a nonzero local 2.5PN reaction theorem, while a minimal retarded quadrupole odd term lands on the standard Burke–Thorne / Iyer–Will family member

$$\alpha = 4, \qquad \beta = 5.$$

For the present conservative 4PN paper, however, the important carry-forward message is narrower: after the scalar, dipole, and quadrupole channel audits, the live higher-order universal point-particle route collapses to the orbital/worldtube STF quadrupole branch

$$\boxed{I^{\rm orb}_{ij} = \mu x_{\langle i}x_{j\rangle}.}$$

The odd-channel hierarchy on the narrow branch is

$$\text{scalar } \sim i\omega, \qquad \text{dipole } \sim i\omega^3, \qquad \text{quadrupole } \sim i\omega^5,$$

and the quadrupole normalization target is

$$\boxed{\hat{m}_0^2\Gamma_5 = \frac{2G}{5c^5}.}$$

In grouped real language, the conservative front end is organized as

$$Y_{20}(\omega) = 1 + u_2^{(20)}\omega^2 + u_4^{(20)}\omega^4 + O(\omega^6),$$
$$Y_{21}(\omega) = 1 + u_2^{(21)}\omega^2 + u_4^{(21)}\omega^4 + O(\omega^6),$$
$$Y_{22}(\omega) = 1 + u_2^{(22)}\omega^2 + u_4^{(22)}\omega^4 + O(\omega^6).$$

It is convenient to repackage these grouped triples into trace/anomaly variables. At $O(\omega^2)$ one defines

$$\bar{u}_2 = \frac{u_2^{(20)} + 2u_2^{(21)} + 2u_2^{(22)}}{5}, \qquad a_2 = \frac{2u_2^{(20)} - u_2^{(21)} - u_2^{(22)}}{10}, \qquad b_2 = \frac{u_2^{(21)} - u_2^{(22)}}{2},$$

and at $O(\omega^4)$

$$\bar{u}_4 = \frac{u_4^{(20)} + 2u_4^{(21)} + 2u_4^{(22)}}{5}, \qquad a_4 = \frac{2u_4^{(20)} - u_4^{(21)} - u_4^{(22)}}{10}, \qquad b_4 = \frac{u_4^{(21)} - u_4^{(22)}}{2}.$$

The exact inverse map is

$$u_2^{(20)} = \bar{u}_2 + 4a_2, \qquad u_2^{(21)} = \bar{u}_2 - a_2 + b_2, \qquad u_2^{(22)} = \bar{u}_2 - a_2 - b_2,$$

$$u_4^{(20)} = \bar{u}_4 + 4a_4, \qquad u_4^{(21)} = \bar{u}_4 - a_4 + b_4, \qquad u_4^{(22)} = \bar{u}_4 - a_4 - b_4.$$

So the grouped-real isotropy gate on the minimal branch is simply

$$a_2 = b_2 = a_4 = b_4 = 0.$$

On that branch,

$$\bar{u}_4 = 4\bar{u}_2^2, \qquad \Omega_Q^2 = \frac{1}{4\bar{u}_2}, \qquad \overline{K}_0^{\rm target} = \frac{2G}{45c^5\bar{u}_2^{5/2}}.$$

For later use, the normalized outgoing $\ell = 2$ moment package can be written as

$$\widehat{Y}_2(k) = 1 + \frac{a_{\rm th}^2 k^2}{9} + \frac{4a_{\rm th}^4 k^4}{81} + i\frac{a_{\rm th}^5 k^5}{27} + \cdots,$$

with the moment identities

$$m_4 = \frac{4m_2^2}{m_0}, \qquad \Gamma_5^{\rm PDE} = 9\frac{m_2^{5/2}}{m_0^{3/2}}.$$

These formulas are not the present paper's theorem. Their role here is to make explicit why the hereditary 4PN bridge can be controlled by the same narrow quadrupole-normalization problem that already survived the 2.5PN audit.

A.4 3PN carried data and the frozen local front end

The full conservative 3PN paper is the immediate lower-order predecessor of the present work. What the 4PN paper imports from it is not just the fact of a lower-order match, but the exact one-body repair ledger, the fixed-chart local front end, and the sharpened statement of what remained open after the conservative 3PN bottleneck was removed.

At the strict one-body level, the exact isotropic Schwarzschild target through c^{-6} is

$$\frac{L_{\text{3PN,1-body}}}{m} = \frac{5}{128}v^8 + \frac{11}{16}Uv^6 + \frac{47}{16}U^2v^4 + \frac{13}{8}U^3v^2 - \frac{1}{8}U^4.$$

Trying to continue the lower-order denominator strategy with only one new cubic denominator datum fails, so the exact one-body repair ledger opened at 3PN is

$$\mu_{\rho,3} = \frac{1}{4}, \qquad d_3 = -\frac{45}{4}, \qquad s_{24} = -\frac{1}{16}.$$

These are exactly the lower-order self-sector values that the present 4PN one-body continuation carries into the repair ansatz.

At the full conservative 3PN level, the comparable-mass residual in COM form is fixed by the split

$$\Delta\hat{L}_3^{\rm GR} = \underbrace{\Delta l_1 v^8}_{\text{compiler image of the free COM Hamiltonian}} + \underbrace{L_{P_2}^{\rm mid}}_{\text{exact grouped real } P_2 \text{ closure}} + \underbrace{\Delta l_{15}^{(g)} U^4}_{\text{unique geometry completion}}.$$

Within the same declared closure hierarchy as the 1PN and 2PN papers, the earlier paper also fixed the exact sign and determinant statements

$$\Delta H_3 = -\Delta L_3, \qquad \det M_{\rm mid} = -\frac{4}{27},$$

so that the grouped-real quadrupole middle block and the unique geometry completion are already frozen inputs by the time the present 4PN work begins.

The carry-forward lesson for the present paper is therefore very specific. After the full conservative 3PN assembly, what remained open was no longer algebraic 3PN bookkeeping. The remaining bridge problem had become: derive the grouped real P_2 constitutive law and the unique geometry completion microscopically from the moving-throat dynamics, then insert that data into the already narrowed 2.5PN outgoing quadrupole-normalization problem. The present 4PN paper starts exactly from that status boundary. It carries the frozen 3PN front end, extends the local conservative ledger through c^{-8}, and then meets the same narrowed quadrupole-normalization interface from the conservative side.

Putting the carried formulas together, the lower-order conservative ladder used without rederivation in the present paper is

$$L_{\rm cons}^{(<4\rm PN)} = L_0 + c^{-2}L_1 + c^{-4}L_{2,\rm full} + c^{-6}L_3 + O(c^{-8}),$$

where L_0, L_1, and $L_{2,\rm full}$ are the explicit blocks recorded above, while L_3 is the fully assigned fixed-chart conservative 3PN block of the earlier paper. The present 4PN theorem chain is not allowed to alter any part of this carried ledger. It begins only after these lower-order data, identities, and grouped-branch status statements have already been frozen.

B Exact 4PN one-body Schwarzschild expansion

This appendix records the exact test-mass Schwarzschild expansion used in Section 4.1. Its role is purely foundational. We keep the full isotropic worldline series in one place so that the carried 1PN, 2PN, and 3PN one-body coefficients and the new 4PN coefficient can all be read off from one exact formula before any self-sector continuation or comparable-mass local lift is attempted.

B.1 Exact isotropic worldline form

On the strict test-mass branch, with

$$U = \frac{GM}{r}, \qquad u_{\rm sym} \to 0,$$

and in isotropic Schwarzschild coordinates, the exact line element is

$$ds^2 = -\left(\frac{1-u/2}{1+u/2}\right)^2 c^2dt^2 + (1+u/2)^4 d\mathbf{x}^2, \qquad u \equiv \frac{U}{c^2}. \tag{109}$$

Therefore the exact worldline Lagrangian is

$$\frac{L_{\rm exact}}{m} = -c^2\sqrt{\left(\frac{1-u/2}{1+u/2}\right)^2 - (1+u/2)^4\frac{v^2}{c^2}}, \qquad v^2 \equiv \mathbf{v}^2. \tag{110}$$

This is the unique one-body source for the local conservative post-Newtonian series used throughout the paper. No self-sector continuation is allowed to modify the coefficients obtained by expanding (110).

B.2 Expansion through 4PN

Expanding (110) in powers of c^{-2} at fixed U and v^2 gives

$$\begin{aligned}\frac{L_{\rm exact}}{m} = & -c^2 + \left(\frac{1}{2}v^2 + U\right) + \frac{1}{c^2}\Big[\frac{1}{8}v^4 + \frac{3}{2}Uv^2 - \frac{1}{2}U^2\Big] \\ & + \frac{1}{c^4}\Big[\frac{1}{16}v^6 + \frac{7}{8}Uv^4 + 2U^2v^2 + \frac{1}{4}U^3\Big] \\ & + \frac{1}{c^6}\Big[\frac{5}{128}v^8 + \frac{11}{16}Uv^6 + \frac{47}{16}U^2v^4 + \frac{13}{8}U^3v^2 - \frac{1}{8}U^4\Big] \\ & + \frac{1}{c^8}\Big[\frac{7}{256}v^{10} + \frac{75}{128}Uv^8 + \frac{59}{16}U^2v^6 \\ & \quad + \frac{203}{32}U^3v^4 + \frac{31}{32}U^4v^2 + \frac{1}{16}U^5\Big] + O(c^{-10}).\end{aligned} \tag{111}$$

Equation (111) is the complete exact one-body expansion through 4PN. In particular, the 4PN coefficient is read off without ambiguity as

$$\begin{aligned}\frac{L_{\rm 4PN,1\text{-}body}}{m} & \\ & = \frac{7}{256}v^{10} \\ & \quad + \frac{75}{128}Uv^8 + \frac{59}{16}U^2v^6 \\ & \quad + \frac{203}{32}U^3v^4 + \frac{31}{32}U^4v^2 + \frac{1}{16}U^5.\end{aligned} \tag{112}$$

This is the exact gate used in Section 4.1 and nowhere in the later local chain is there any freedom to alter it.

B.3 Order-by-order one-body coefficients carried into the 4PN chain

If one drops the irrelevant additive rest-mass term $-mc^2$, then the strict one-body coefficients imported into the conservative hierarchy are simply the successive coefficients of (111). Writing

$$\begin{aligned}L_{\rm exact} = & -mc^2 + L_{\rm N,1\text{-}body} + \frac{1}{c^2}L_{\rm 1PN,1\text{-}body} + \frac{1}{c^4}L_{\rm 2PN,1\text{-}body} \\ & + \frac{1}{c^6}L_{\rm 3PN,1\text{-}body} + \frac{1}{c^8}L_{\rm 4PN,1\text{-}body} + O(c^{-10}),\end{aligned}$$

one has

$$\frac{L_{\rm N,1\text{-}body}}{m} = \frac{1}{2}v^2 + U, \tag{113}$$

$$\frac{L_{\rm 1PN,1\text{-}body}}{m} = \frac{1}{8}v^4 + \frac{3}{2}Uv^2 - \frac{1}{2}U^2, \tag{114}$$

$$\frac{L_{\rm 2PN,1\text{-}body}}{m} = \frac{1}{16}v^6 + \frac{7}{8}Uv^4 + 2U^2v^2 + \frac{1}{4}U^3, \tag{115}$$

$$\frac{L_{\rm 3PN,1\text{-}body}}{m} = \frac{5}{128}v^8 + \frac{11}{16}Uv^6 + \frac{47}{16}U^2v^4 + \frac{13}{8}U^3v^2 - \frac{1}{8}U^4, \tag{116}$$

and finally the 4PN coefficient (112). The first three of these are the carried exact test-mass anchors from the earlier papers; the last one is the new order-c^{-8} input that the present paper adds.

The practical lesson is the one emphasized in the main text. The exact Schwarzschild series fixes the one-body front end before any denominator-style continuation of the self sector is attempted. The later repair ledger $(\mu_{\rho,4}, d_4, s_{34}, s_{26})$ is therefore not a guess about the Schwarzschild answer; it is the minimal continuation required to reproduce the already frozen coefficient (112) exactly.

C One-body repair algebra

Appendix B freezes the exact isotropic Schwarzschild one-body target through total order c^{-8}. The natural next question is whether the carried 3PN denominator/self/static packaging can simply be pushed one order higher, or whether the strict test-mass gate already forces genuinely new 4PN one-body data. The purpose of the present appendix is to answer that question algebraically and slot by slot.

The outcome is sharper than at 3PN. A quartic denominator datum is needed to repair the U^4v^2 slot, a quartic static datum is needed to repair the pure U^5 slot, and two additional explicit self-sector invariants are needed because the U^3v^4 and U^2v^6 slots remain wrong even after the first two repairs are imposed. So the full one-body 4PN repair ledger is already a four-parameter ledger at the strict Schwarzschild gate.

C.1 Quartic denominator datum

The first test is the minimal quartic continuation of the carried denominator-side self sector,

$$L_{\rm red} = -mc^2(1-u)\sqrt{1-\frac{v^2/c^2}{D(u)}}, \qquad u \equiv \frac{U}{c^2}, \qquad D(u) = 1-4u+8u^2+d_3u^3+d_4u^4+O(u^5). \tag{117}$$

Expanding (117) through total order c^{-8} gives

$$\begin{aligned}\frac{L_{\rm red}}{m} = &-c^2+\frac{1}{2}v^2+U+\frac{1}{c^2}\left(\frac{1}{8}v^4+\frac{3}{2}Uv^2\right)+\frac{1}{c^4}\left(\frac{1}{16}v^6+\frac{7}{8}Uv^4+2U^2v^2\right)\\ &+\frac{1}{c^6}\left(\frac{5}{128}v^8+\frac{11}{16}Uv^6+3U^2v^4+\left(-4-\frac{d_3}{2}\right)U^3v^2\right)\\ &+\frac{1}{c^8}\left(\frac{7}{256}v^{10}+\frac{75}{128}Uv^8+\frac{15}{4}U^2v^6+\left(4-\frac{d_3}{4}\right)U^3v^4\right.\\ &\left.+\left(-32-\frac{7d_3}{2}-\frac{d_4}{2}\right)U^4v^2\right)+O(c^{-10}).\end{aligned} \tag{118}$$

Two features of (118) are immediate. First, the carried denominator logic already reproduces the universal 4PN coefficients of v^{10} and Uv^8 automatically. Second, once the carried 3PN denominator datum

$$\boxed{d_3 = -\frac{45}{4}} \tag{119}$$

is imposed, the quartic denominator coefficient d_4 can influence only the U^4v^2 slot. The 4PN denominator-side coefficient then becomes

$$\left(\frac{L_{\rm red}}{m}\right)_{c^{-8},\,d_3=-45/4} = \frac{7}{256}v^{10}+\frac{75}{128}Uv^8+\frac{15}{4}U^2v^6+\frac{109}{16}U^3v^4+\left(\frac{59}{8}-\frac{d_4}{2}\right)U^4v^2. \tag{120}$$

Matching the exact Schwarzschild target (112) in the U^4v^2 channel gives

$$\frac{59}{8}-\frac{d_4}{2} = \frac{31}{32} \qquad \Longrightarrow \qquad \boxed{d_4 = \frac{205}{16}.} \tag{121}$$

So the quartic denominator datum is fixed exactly by the U^4v^2 slot.

This also shows why the denominator-side continuation is not enough by itself. Even before the static gate is discussed, the carried denominator sector still predicts

$$\frac{15}{4}U^2v^6, \qquad \frac{109}{16}U^3v^4, \tag{122}$$

whereas the exact Schwarzschild target requires

$$\frac{59}{16}U^2v^6, \qquad \frac{203}{32}U^3v^4. \tag{123}$$

So the quartic denominator correction repairs only the U^4v^2 gate. It does not repair the new U^3v^4 or U^2v^6 channels.

C.2 Quartic static gate

The pure static slot gives an independent exact gate. Extend the carried local mass-scaling sector by one quartic static coefficient $\mu_{\rho,4}$,

$$\frac{L_{\text{static,1body}}}{m} = U - \frac{1}{2}\frac{U^2}{c^2} + \frac{1}{4}\frac{U^3}{c^4} - \frac{\mu_{\rho,3}}{2}\frac{U^4}{c^6} + \frac{\mu_{\rho,4}}{2}\frac{U^5}{c^8} + O(c^{-10}), \tag{124}$$

with the carried 3PN value

$$\boxed{\mu_{\rho,3} = \frac{1}{4}.} \tag{125}$$

The exact Schwarzschild target requires the pure static 4PN term

$$\frac{1}{16}\frac{U^5}{c^8}. \tag{126}$$

So the quartic static gate is fixed immediately by

$$\frac{\mu_{\rho,4}}{2} = \frac{1}{16} \qquad \Longrightarrow \qquad \boxed{\mu_{\rho,4} = \frac{1}{8}.} \tag{127}$$

As at lower order, the important point is structural rather than cosmetic: the pure static slot remains a separate exact gate of its own. It is not absorbed into the denominator datum and it is not a bookkeeping convenience.

C.3 Two new self-sector slots and why a quartic denominator plus a quartic static gate is not enough

At this point the minimal direct-continuation one-body candidate is

$$\begin{aligned}\frac{L_{\text{cand}}}{m} = \frac{L_{\text{red}}}{m} - \frac{1}{2}\frac{U^2}{c^2} + \frac{1}{4}\frac{U^3}{c^4} - \frac{\mu_{\rho,3}}{2}\frac{U^4}{c^6} + \frac{\mu_{\rho,4}}{2}\frac{U^5}{c^8} \\ + s_{24}\frac{U^2v^4}{c^6} + s_{34}\frac{U^3v^4}{c^8} + s_{26}\frac{U^2v^6}{c^8},\end{aligned} \tag{128}$$

where the first five displayed corrections restore the carried lower-order static ledger together with the new quartic static gate, and the last two terms are the minimal new explicit 4PN self-sector slots. Subtracting (128) from the exact target and imposing the already frozen lower-order values

$$\boxed{d_3 = -\frac{45}{4}, \qquad \mu_{\rho,3} = \frac{1}{4}, \qquad s_{24} = -\frac{1}{16}} \tag{129}$$

gives the exact residual

$$\begin{aligned}\left(\frac{L_{\text{4PN,1-body}}}{m} - \frac{L_{\text{cand}}}{m}\right)_{c^{-8}} = \left(\frac{1}{16} - \frac{\mu_{\rho,4}}{2}\right)U^5 + \left(-\frac{205}{32} + \frac{d_4}{2}\right)U^4v^2 \\ + \left(-\frac{15}{32} - s_{34}\right)U^3v^4 + \left(-\frac{1}{16} - s_{26}\right)U^2v^6.\end{aligned} \tag{130}$$

So the one-body repair algebra again separates cleanly by slot.

The first two coefficients in (130) vanish once (121) and (127) are imposed. But even after those two exact repairs, the candidate with no new 4PN self slots still leaves the uncancelled discrepancy

$$\left(\frac{L_{\text{4PN,1-body}}}{m} - \frac{L_{\text{cand}}}{m}\right)_{\substack{c^{-8},\, d_4=205/16,\\ \mu_{\rho,4}=1/8,\, s_{34}=s_{26}=0}} = -\frac{15}{32}U^3v^4 - \frac{1}{16}U^2v^6. \tag{131}$$

So a quartic denominator plus a quartic static gate is still not enough. Two new self-sector slots survive exactly at 4PN. They are repaired by the conditions

$$-\frac{15}{32} - s_{34} = 0 \qquad \Longrightarrow \qquad \boxed{s_{34} = -\frac{15}{32},} \tag{132}$$

after which the remaining U^2v^6 discrepancy is fixed by

$$-\frac{1}{16} - s_{26} = 0 \qquad \Longrightarrow \qquad \boxed{s_{26} = -\frac{1}{16}.} \tag{133}$$

Collecting everything, the exact 4PN one-body repair ledger is

$$\boxed{\mu_{\rho,4} = \frac{1}{8}, \qquad d_4 = \frac{205}{16}, \qquad s_{34} = -\frac{15}{32}, \qquad s_{26} = -\frac{1}{16}.} \tag{134}$$

This is the cleanest strict one-body lesson of the whole 4PN analysis. The 4PN gate does *not* say "continue the 3PN denominator idea by one new coefficient." It says that four distinct data are already needed at the strict test-mass level:

1. a quartic static gate,
2. a quartic denominator datum,
3. and two genuinely new self-sector slots.

That is exactly why the full 4PN program had to begin with a one-body audit before the comparable-mass local lift was even well posed. Once the exact target and its minimal repair ledger are frozen, the next step is to promote the strict one-body structure into the natural two-body self/static seed used in the local theorem chain.

D Exact quartic Legendre-transform identity at 4PN

Section 4.4 recorded the compact quartic ordinary-Lagrangian → Hamiltonian compiler used throughout the local 4PN construction. The purpose of the present appendix is to freeze that identity in its full perturbative form, in the same spirit in which the earlier 3PN paper froze the exact cubic compiler. At 4PN the first genuinely new compiler content is the *quartic-order* extension. Once that identity is available, the later reduced COM translations and generic-frame Hamiltonian-first lifts are controlled algebraically rather than guessed.

The role of this appendix is therefore very specific. It does not solve the full comparable-mass problem by itself. What it does is fix the exact ordinary Lagrangian → Hamiltonian compiler through order ε^4 with $\varepsilon = c^{-2}$. Once that compiler is frozen, the remaining local 4PN comparison against the fixed-chart Hamiltonian target becomes a finite algebraic residual problem.

D.1 Perturbative setup

Write the conservative ordinary Lagrangian as

$$L(v) = L_0(v) + \varepsilon L_1(v) + \varepsilon^2 L_2(v) + \varepsilon^3 L_3(v) + \varepsilon^4 L_4(v), \qquad \varepsilon = c^{-2}, \tag{135}$$

with quadratic zeroth-order block

$$L_0(v) = \frac{1}{2} v^T M v, \tag{136}$$

where the Newtonian mass matrix M is constant at the order relevant for the perturbative inversion. Throughout this appendix, gradients with respect to v are taken as column vectors and Hessians are taken as matrices.

The canonical momentum is

$$p \equiv \frac{\partial L}{\partial v} = Mv + \varepsilon \frac{\partial L_1}{\partial v} + \varepsilon^2 \frac{\partial L_2}{\partial v} + \varepsilon^3 \frac{\partial L_3}{\partial v} + \varepsilon^4 \frac{\partial L_4}{\partial v}. \tag{137}$$

Define the Newtonian momentum-space velocity

$$v_0 = M^{-1} p, \tag{138}$$

and the derivative data evaluated at v_0,

$$A_0 \equiv \left.\frac{\partial L_1}{\partial v}\right|_{v_0}, \qquad B_0 \equiv \left.\frac{\partial L_2}{\partial v}\right|_{v_0}, \qquad D_0 \equiv \left.\frac{\partial L_3}{\partial v}\right|_{v_0}, \tag{139}$$

$$C_0 \equiv \left.\frac{\partial^2 L_1}{\partial v^2}\right|_{v_0}, \qquad E_0 \equiv \left.\frac{\partial^2 L_2}{\partial v^2}\right|_{v_0}, \qquad T_0 \equiv \left.\frac{\partial^3 L_1}{\partial v^3}\right|_{v_0}. \tag{140}$$

Here T_0 is the symmetric trilinear form defined by the third derivatives of L_1. For vectors x, y, z, the notation $T_0[x, y, z]$ denotes full contraction, while $T_0[x, y, \cdot]$ denotes the vector obtained by leaving one slot uncontracted.

The perturbative momentum inversion is written as

$$v = v_0 + \varepsilon v_1 + \varepsilon^2 v_2 + \varepsilon^3 v_3 + O(\varepsilon^4). \tag{141}$$

For the quartic Hamiltonian coefficient H_4, only v_1, v_2, and v_3 are needed explicitly.

D.2 Order-by-order momentum inversion

Insert (141) into (137) and expand around v_0. To the order needed for H_4, the Taylor expansions of the derivative blocks are

$$\frac{\partial L_1}{\partial v}(v) = A_0 + \varepsilon C_0 v_1 + \varepsilon^2 \left(C_0 v_2 + \frac{1}{2} T_0[v_1, v_1, \cdot] \right) + O(\varepsilon^3), \tag{142}$$

$$\frac{\partial L_2}{\partial v}(v) = B_0 + \varepsilon E_0 v_1 + O(\varepsilon^2), \qquad \frac{\partial L_3}{\partial v}(v) = D_0 + O(\varepsilon). \tag{143}$$

Using $M v_0 = p$, the momentum equation becomes

$$\begin{aligned} 0 = {} & \varepsilon (M v_1 + A_0) + \varepsilon^2 (M v_2 + B_0 + C_0 v_1) \\ & + \varepsilon^3 (M v_3 + D_0 + E_0 v_1 + C_0 v_2 + \tfrac{1}{2} T_0[v_1, v_1, \cdot]) + O(\varepsilon^4). \end{aligned} \tag{144}$$

So the coefficients are fixed order by order:

$$M v_1 + A_0 = 0, \tag{145}$$

$$M v_2 + B_0 + C_0 v_1 = 0, \tag{146}$$

$$M v_3 + D_0 + E_0 v_1 + C_0 v_2 + \frac{1}{2} T_0[v_1, v_1, \cdot] = 0. \tag{147}$$

Equations (145)–(147) fix the inversion coefficients order by order. Therefore

$$\boxed{\begin{aligned} v_1 &= -M^{-1}A_0, \\ v_2 &= M^{-1}C_0M^{-1}A_0 - M^{-1}B_0, \\ v_3 &= -M^{-1}D_0 + M^{-1}E_0M^{-1}A_0 + M^{-1}C_0M^{-1}B_0 \\ &\quad - M^{-1}C_0M^{-1}C_0M^{-1}A_0 - \frac{1}{2}M^{-1}T_0[M^{-1}A_0, M^{-1}A_0, \cdot]. \end{aligned}} \tag{148}$$

Equation (148) is the exact perturbative inversion needed through quartic order.

D.3 Exact quartic Hamiltonian identity

Now substitute (141) and (148) into the Legendre definition

$$H(p) = p \cdot v - L(v). \tag{149}$$

The Hamiltonian expansion takes the form

$$H = H_0 + \varepsilon H_1 + \varepsilon^2 H_2 + \varepsilon^3 H_3 + \varepsilon^4 H_4 + O(\varepsilon^5), \tag{150}$$

with zeroth-order piece

$$H_0 = \frac{1}{2}p^T M^{-1} p. \tag{151}$$

Carrying the quartic expansion through exactly gives

$$H_1 = -L_1(v_0), \tag{152}$$

$$H_2 = -L_2(v_0) + \frac{1}{2}A_0^T M^{-1} A_0, \tag{153}$$

$$H_3 = -L_3(v_0) + A_0^T M^{-1} B_0 - \frac{1}{2}A_0^T M^{-1} C_0 M^{-1} A_0, \tag{154}$$

and the exact 4PN compiler identity

$$\boxed{\begin{aligned} H_4 = &-L_4(v_0) + A_0^T M^{-1} D_0 + \frac{1}{2}B_0^T M^{-1} B_0 - B_0^T M^{-1} C_0 M^{-1} A_0 \\ &- \frac{1}{2}A_0^T M^{-1} E_0 M^{-1} A_0 + \frac{1}{2}A_0^T M^{-1} C_0 M^{-1} C_0 M^{-1} A_0 + \frac{1}{6}T_0[M^{-1}A_0, M^{-1}A_0, M^{-1}A_0]. \end{aligned}} \tag{155}$$

Equation (155) is the exact quartic analogue of the quadratic and cubic perturbative compiler identities used at lower order. It is the formula actually used to carry the ordinary 4PN data into the Hamiltonian chart and back again.

It is also useful to record the reduced COM specialization. In the normalized COM variables used later, the Newtonian quadratic block is the identity, so $M = I$ and therefore

$$v_0 = p, \qquad \begin{aligned} H_4 = &-L_4(p) + A_0 \cdot D_0 + \frac{1}{2}B_0 \cdot B_0 - B_0^T C_0 A_0 \\ &- \frac{1}{2}A_0^T E_0 A_0 + \frac{1}{2}A_0^T C_0 C_0 A_0 + \frac{1}{6}T_0[A_0, A_0, A_0]. \end{aligned} \tag{156}$$

This is the compact form checked by the local referee scripts before the later slot-by-slot COM compiler and fixed-chart local target comparison are built.

D.4 Practical meaning for the 4PN lift

The practical meaning of (155) is exactly the one that makes the local 4PN chain manageable. Once the lower-order 1PN/2PN/3PN ledger is frozen, the feedback tensors A_0, B_0, C_0, D_0, E_0, and T_0 are already known. So the genuinely new 4PN Hamiltonian content enters only through the explicit $-L_4(v_0)$ term, while the remaining quartic feedback terms are completely fixed by carried lower-order data.

That observation is what turns the local comparable-mass 4PN problem into a finite algebraic residual rather than a blind full derivation campaign. It is also why the compiler has to be frozen before the local fixed-chart target is translated: without (155), the ordinary-chart and Hamiltonian-chart comparisons are not rigidly related. With it, the Hamiltonian-first local lift becomes a theorem-level statement rather than an ansatz.

Finally, the same compiler applied to the exact one-body Schwarzschild Lagrangian gate reproduces the strict one-body 4PN Hamiltonian seed used in the main local theorem chain. So the one-body Hamiltonian gate is not an independent imported object. It is a direct consequence of the exact Schwarzschild Lagrangian gate together with the quartic compiler frozen here.

E Local generic-frame scaffold and COM projection

Section 5 used the raw local scaffold only through its headline counts. The purpose of the present appendix is to freeze that scaffold itself, together with the exact center-of-mass (COM) projection map that first reveals its structure. At this stage no fixed-chart target has yet been imported and no Hamiltonian-first quotient has yet been taken. The question is therefore purely kinematic: once the strict one-body content has been absorbed into the natural self/static seed (7), how large is the remaining exchange-symmetric generic-frame local interaction space, and what does COM projection determine about it exactly?

The answer is already very sharp. The raw local comparable-mass scaffold is blockwise COM-complete from the start. There is no early span shortage. What COM projection leaves behind is not a missing image direction but a very large COM-blind null family. This is exactly why the later local theorem chain must be completed by importing the fixed-chart Hamiltonian target and quotienting the null sector rather than by enlarging the raw scaffold itself.

E.1 Exchange-symmetric local interaction scaffold

Use the standard generic-frame invariants

$$a = v_A^2, \qquad b = v_B^2, \qquad c = \mathbf{v}_A \cdot \mathbf{v}_B, \qquad d = \mathbf{v}_A \cdot \mathbf{n}, \qquad e = \mathbf{v}_B \cdot \mathbf{n}, \tag{157}$$

with mass variables

$$p = m_A, \qquad q = m_B. \tag{158}$$

The raw local comparable-mass scaffold is defined relative to the natural self/static seed by

$$L_{4,\mathrm{loc}}^{\mathrm{raw}} = L_{4,\mathrm{seed}} + L_{4,\mathrm{res}}^{\mathrm{raw}}, \tag{159}$$

where

$$\begin{aligned} L_{4,\mathrm{res}}^{\mathrm{raw}} = {} & \frac{G m_A m_B}{r} \sum_{i=1}^{52} \alpha_i Q_i + \frac{G^2 m_A m_B}{r^2} \sum_{j=1}^{46} \beta_j T_j + \frac{G^3 m_A m_B}{r^3} \sum_{k=1}^{29} \gamma_k S_k \\ & + \frac{G^4 m_A m_B}{r^4} \sum_{\ell=1}^{10} \delta_\ell U_\ell + \frac{G^5 m_A m_B}{r^5} \sum_{m=1}^{2} \epsilon_m W_m. \end{aligned} \tag{160}$$

The basis families are generated algorithmically as exchange-symmetric monomials in (157)–(158) with blockwise velocity degrees

$$\deg_v(Q_i, T_j, S_k, U_\ell, W_m) = (8, 6, 4, 2, 0) \tag{161}$$

and additional mass degrees

$$\deg_m(Q_i, T_j, S_k, U_\ell, W_m) = (0, 1, 2, 3, 4). \tag{162}$$

Throughout this appendix, "raw" means that no contact/gauge quotient has yet been imposed.

The strict one-body gate is kept clean by requiring each residual basis element to vanish on the test-mass branch in which body A becomes the test particle and body B the fixed source. In the invariant variables this means

$$b = 0, \qquad c = 0, \qquad e = 0, \qquad p = 0, \qquad q = 1. \tag{163}$$

So every monomial retained in (160) vanishes on the strict one-body branch, and the one-body content stays entirely inside $L_{4,\text{seed}}$.

The exact block dimensions obtained by this construction are

$$\dim \mathcal{Q} = 52, \qquad \dim \mathcal{T} = 46, \qquad \dim \mathcal{S} = 29, \qquad \dim \mathcal{U} = 10, \qquad \dim \mathcal{W} = 2, \tag{164}$$

so the total raw local interaction scaffold contains

$$52 + 46 + 29 + 10 + 2 = 139 \tag{165}$$

exchange-symmetric directions before any fixed-chart conditions are imposed.

E.2 Exact COM reduction map and interaction-slot extraction

To read off the COM image, write the standard reduced COM variables as

$$\mathbf{v}_A = X_B \mathbf{v}, \qquad \mathbf{v}_B = -X_A \mathbf{v}, \qquad X_A + X_B = 1, \qquad \nu = X_A X_B, \tag{166}$$

with the usual mass-fraction parameterization

$$X_A = \frac{1+\Delta}{2}, \qquad X_B = \frac{1-\Delta}{2}, \qquad \Delta^2 = 1 - 4\nu. \tag{167}$$

Therefore the generic-frame invariants reduce to

$$a = X_B^2 v^2, \qquad b = X_A^2 v^2, \qquad c = -X_A X_B v^2 = -\nu v^2, \tag{168}$$

$$d = X_B \dot{r}, \qquad e = -X_A \dot{r}. \tag{169}$$

Because the raw basis is exchange-symmetric, the COM image is obtained by first substituting (168)–(169), then symmetrizing under $\Delta \mapsto -\Delta$, and finally eliminating all remaining even powers of Δ using (167). Every basis element then becomes a polynomial in ν multiplying one of the standard reduced COM monomials in v^2 and $\dot{r}^2$.

The induced COM interaction slots are exactly the fifteen slots already used in Section 5.1. Define the blockwise coefficient extraction maps by

$$\Pi_Q(F) \equiv \left([v^8]F_{\text{COM}}, [v^6\dot{r}^2]F_{\text{COM}}, [v^4\dot{r}^4]F_{\text{COM}}, [v^2\dot{r}^6]F_{\text{COM}}, [\dot{r}^8]F_{\text{COM}}\right)^T, \tag{170}$$

$$\Pi_T(F) \equiv \left([v^6]F_{\text{COM}}, [v^4\dot{r}^2]F_{\text{COM}}, [v^2\dot{r}^4]F_{\text{COM}}, [\dot{r}^6]F_{\text{COM}}\right)^T, \tag{171}$$

$$\Pi_S(F) \equiv \left([v^4]F_{\text{COM}}, [v^2\dot{r}^2]F_{\text{COM}}, [\dot{r}^4]F_{\text{COM}}\right)^T, \tag{172}$$

$$\Pi_U(F) \equiv \left([v^2]F_{\text{COM}}, [\dot{r}^2]F_{\text{COM}}\right)^T, \qquad \Pi_W(F) \equiv \left([1]F_{\text{COM}}\right). \tag{173}$$

So, in a chosen raw basis ordering, the exact COM projection matrices are

$$M_Q \in \mathrm{Mat}_{5\times 52}(\mathbb{Q}[\nu]), \qquad M_T \in \mathrm{Mat}_{4\times 46}(\mathbb{Q}[\nu]), \qquad M_S \in \mathrm{Mat}_{3\times 29}(\mathbb{Q}[\nu]), \tag{174}$$

$$M_U \in \mathrm{Mat}_{2\times 10}(\mathbb{Q}[\nu]), \qquad M_W \in \mathrm{Mat}_{1\times 2}(\mathbb{Q}[\nu]). \tag{175}$$

The audit establishes the exact blockwise ranks

$$\operatorname{rank} M_Q = 5, \qquad \operatorname{rank} M_T = 4, \qquad \operatorname{rank} M_S = 3, \qquad \operatorname{rank} M_U = 2, \qquad \operatorname{rank} M_W = 1, \tag{176}$$

with nullities

$$\begin{aligned} \dim\ker M_Q = 47, \qquad \dim\ker M_T = 42, \qquad \dim\ker M_S = 26,\\ \dim\ker M_U = 8, \qquad \dim\ker M_W = 1. \end{aligned} \tag{177}$$

Equivalently, the exact COM interaction-slot structure is

$$\begin{aligned} \#^{\mathrm{int}}_{\mathrm{COM}} = (5,4,3,2,1),\\ 5+4+3+2+1 = 15, \end{aligned} \tag{178}$$

so the total COM interaction rank is

$$5+4+3+2+1 = 15, \tag{179}$$

and the total COM-blind nullity is

$$47+42+26+8+1 = 124. \tag{180}$$

The interaction-slot count in (178) is the interaction analogue of the full 21-slot local 4PN reduced ordinary basis, whose remaining six slots are pure kinetic.

So the first exact conclusion is already visible before any target import:

$$\boxed{\text{the raw local generic-frame scaffold is blockwise COM-complete.}} \tag{181}$$

At this stage the only large ambiguity is the 124-dimensional COM-blind null family.

E.3 A minimal pivot-spanning subset

A convenient blockwise pivot-spanning subset is the following. For the G/r block one may take

$$\begin{aligned} \mathcal{P}_Q = \{a^2b^2, \quad a^2bd^2 + ab^2e^2, \quad a^2d^2e^2 + b^2d^2e^2,\\ ad^2e^4 + bd^4e^2, \quad d^4e^4\}. \end{aligned} \tag{182}$$

Their COM images are

$$\begin{aligned} \Pi_Q(\mathcal{P}_Q) = \{\nu^4v^8, \quad (\nu^2-4\nu^3+2\nu^4)v^6\dot r^2, \quad (\nu^2-4\nu^3+2\nu^4)v^4\dot r^4,\\ 2\nu^4v^2\dot r^6, \quad \nu^4\dot r^8\}, \end{aligned} \tag{183}$$

which already span the full five-dimensional COM image of the G/r block.

For the G^2/r^2 block one may take

$$\mathcal{P}_T = \{a^2bp + ab^2q, \quad a^2d^2p + b^2e^2q, \quad ad^2e^2p + bd^2e^2q, \quad d^3e^3p + d^3e^3q\}, \tag{184}$$

with COM images

$$\Pi_T(\mathcal{P}_T) = \{\nu^3v^6, \quad (\nu-5\nu^2+5\nu^3)v^4\dot r^2, \quad \nu^3v^2\dot r^4, \quad -\nu^3\dot r^6\}. \tag{185}$$

For the G^3/r^3 block one may take

$$\mathcal{P}_S = \{a^2p^2 + b^2q^2, \qquad ad^2p^2 + be^2q^2, \qquad d^2e^2p^2 + d^2e^2q^2\}, \tag{186}$$

with COM images

$$\Pi_S(\mathcal{P}_S) = \{(\nu^2 - 2\nu^3)v^4, \qquad (\nu^2 - 2\nu^3)v^2\dot{r}^2, \qquad (\nu^2 - 2\nu^3)\dot{r}^4\}. \tag{187}$$

For the G^4/r^4 block one may take

$$\mathcal{P}_U = \{ap^2q + bpq^2, \qquad d^2p^2q + e^2pq^2\}, \tag{188}$$

with COM images

$$\Pi_U(\mathcal{P}_U) = \{\nu^2v^2, \qquad \nu^2\dot{r}^2\}. \tag{189}$$

Finally, for the static G^5/r^5 block one may take

$$\mathcal{P}_W = \{p^2q^2\}, \qquad \Pi_W(\mathcal{P}_W) = \{\nu^2\}. \tag{190}$$

These pivot families are not unique, but they are especially convenient because they already display the full five-block COM image with the smallest possible number of representatives:

$$5 + 4 + 3 + 2 + 1 = 15. \tag{191}$$

E.4 Structural consequence

The structural interpretation is immediate. Already before any fixed-chart local target is imported, the raw constant-coefficient exchange-symmetric local 4PN scaffold is large enough to span the full COM interaction image block by block. So the first local 4PN bottleneck is not undercompleteness of the raw generic-frame basis. It is the opposite: a very large COM-blind null family. That is exactly why the next mathematically honest step is the one taken in Section 5.2: import the fixed local Hamiltonian chart, read off the true blockwise degree ceilings, and quotient the 124-dimensional COM-null sector by those fixed-chart conditions rather than by COM data alone.

The local scaffold audit therefore establishes the first local 4PN theorem front end:

$$\boxed{\text{there is no early kinematic shortage at local 4PN.}} \tag{192}$$

Everything that remains in the later local theorem chain is a chart, quotient, and seed-alignment problem, not a failure of the raw exchange-symmetric scaffold.

F Imported fixed local 4PN target and the Hamiltonian-first translation lesson

Section 5.2 used only the structural conclusions of the fixed-chart local target import. The purpose of the present appendix is to freeze that import itself, together with the exact quartic translation lesson that determines the correct order of the local generic-frame lift. The result is conceptually simple but technically decisive. The local instantaneous target is most naturally imported in the reduced center-of-mass (COM) *Hamiltonian* chart, where its blockwise mass-ratio degree ceilings match the constant- coefficient generic-frame scaffold exactly. After quartic translation to the ordinary chart, by contrast, the middle and upper local blocks acquire extra ν^4 tails that are not present in the Hamiltonian target and are not reachable from the raw constant-coefficient ordinary scaffold. This is the precise reason the local 4PN lift has to be done Hamiltonian-first.

F.1 Imported fixed local Hamiltonian target

The first point to freeze is the target logic itself. At conservative 4PN, the exact source split is

$$L_{\rm 4PN}^{\rm cons} = L_{\rm 4PN}^{\rm local} + L_{\rm 4PN}^{\rm tail}. \tag{193}$$

So the present appendix concerns only the *local instantaneous* target. Concretely, the reference local Hamiltonian is the fixed-chart ADM local 4PN Hamiltonian, imported consistently with the already-frozen local/nonlocal split. The harmonic/Fokker and EFT constructions play only the role of cross-check families; they are not used to redefine the target.

In reduced COM variables, the imported local Hamiltonian organizes into one pure free-kinetic slot and fifteen interaction slots,

$$1 \;\oplus\; 5 \;\oplus\; 4 \;\oplus\; 3 \;\oplus\; 2 \;\oplus\; 1, \tag{194}$$

corresponding respectively to the free block and the interaction blocks G/r, G^2/r^2, G^3/r^3, G^4/r^4, and G^5/r^5. This matches exactly the slot structure already seen in the raw local scaffold appendix.

The strict one-body check is equally exact. Applying the quartic Legendre compiler of Appendix D.3 to the frozen one-body Schwarzschild Lagrangian gives the reduced one-body 4PN Hamiltonian gate

$$\frac{7}{256}p^{10}, \qquad \frac{45}{128}\frac{p^8}{r}, \qquad \frac{13}{8}\frac{p^6}{r^2}, \qquad \frac{105}{32}\frac{p^4}{r^3}, \qquad \frac{105}{32}\frac{p^2}{r^4}, \qquad -\frac{1}{16}\frac{1}{r^5}. \tag{195}$$

The imported fixed-chart local Hamiltonian matches (195) exactly in the strict test-mass limit $\nu \to 0$.

The first structural theorem gate answered by the import is the blockwise ν-degree ceiling of the local Hamiltonian target. The outcome is

$$\deg_\nu^{\max}(\text{free}, G/r, G^2/r^2, G^3/r^3, G^4/r^4, G^5/r^5) = (4, 4, 3, 3, 2, 2). \tag{196}$$

In particular,

$$\deg_\nu^{\max}(G^4/r^4) = 2, \qquad \deg_\nu^{\max}(G^5/r^5) = 2, \tag{197}$$

so the imported fixed-chart local target does *not* force any ν^3 or ν^4 tails in the upper local interaction blocks.

This already removes one possible obstruction. At the level of the imported local Hamiltonian target, the upper interaction blocks are no more singular than the raw constant-coefficient scaffold had suggested.

F.2 Exact quartic COM translation and the ordinary local target

To make the chart lesson exact rather than heuristic, one translates the imported local Hamiltonian target through the full reduced COM quartic compiler. Use the full twenty-one-slot reduced COM ordinary basis,

$$6 \;\oplus\; 5 \;\oplus\; 4 \;\oplus\; 3 \;\oplus\; 2 \;\oplus\; 1, \tag{198}$$

corresponding to the free block and the five local interaction blocks. If the ordinary 4PN coefficients are denoted by $L_1, \ldots, L_{21}$, then the quartic reduced COM compiler is diagonal slot by slot:

$$h_i = F_i(\nu) - L_i, \qquad \frac{\partial h_i}{\partial L_j} = -\delta_{ij}. \tag{199}$$

So the local COM Hamiltonian-to-ordinary translation introduces no cross-slot mixing. The chart issue is not slot mixing; it is the new mass-ratio degree content created inside individual slots.

The translated exact ordinary local target is as follows.

Free block.

$$
\begin{aligned}
L_1^{\text{target}} &= \frac{7+71\nu-1135\nu^2+4082\nu^3-4497\nu^4}{256}, \\
L_2^{\text{target}} &= L_3^{\text{target}} = L_4^{\text{target}} = L_5^{\text{target}} = L_6^{\text{target}} = 0.
\end{aligned} \tag{200}
$$

So the free block remains a one-slot block after quartic translation.

G/r **block.**

$$L_7^{\text{target}} = \frac{150+512\nu-7292\nu^2+9285\nu^3+10070\nu^4}{256}, \tag{201}$$

$$L_8^{\text{target}} = \frac{\nu(2414\nu^3-1723\nu^2+346\nu-16)}{64}, \tag{202}$$

$$L_9^{\text{target}} = \frac{3\nu^2(270\nu^2-191\nu+30)}{128}, \tag{203}$$

$$L_{10}^{\text{target}} = -\frac{5\nu^3(34\nu-13)}{64}, \tag{204}$$

$$L_{11}^{\text{target}} = \frac{35\nu^3(2\nu-1)}{256}. \tag{205}$$

G^2/r^2 **block.**

$$L_{12}^{\text{target}} = -\frac{7504\nu^4+22415\nu^3+3297\nu^2+340\nu-944}{256}, \tag{206}$$

$$L_{13}^{\text{target}} = -\frac{\nu(17648\nu^3+17503\nu^2-12188\nu+1872)}{256}, \tag{207}$$

$$L_{14}^{\text{target}} = -\frac{\nu(21888\nu^3-5105\nu^2+6915\nu-2916)}{768}, \tag{208}$$

$$L_{15}^{\text{target}} = \frac{\nu(5760\nu^3-2673\nu^2+11510\nu-2952)}{1280}. \tag{209}$$

G^3/r^3 **block.**

$$
\begin{aligned}
L_{16}^{\text{target}} = \frac{1}{3686400}\Big(&30412800\nu^4+128822400\nu^3+(258968192-3987675\pi^2)\nu^2 \\
&-(76341312+1294650\pi^2)\nu+23385600\Big),
\end{aligned} \tag{210}
$$

$$L_{17}^{\text{target}} = \frac{\nu}{153600}\Big(6144000\nu^3+15650400\nu^2+(5897600+281025\pi^2)\nu-7914912+166050\pi^2\Big), \tag{211}$$

$$L_{18}^{\text{target}} = \frac{\nu}{245760}\Big(5836800\nu^3+4899200\nu^2+(579825\pi^2-13348224)\nu-11250\pi^2+3289536\Big). \tag{212}$$

G^4/r^4 **block.**

$$
\begin{aligned}
L_{19}^{\text{target}} = -\frac{1}{1228800}\Big(&614400\nu^4+1382400\nu^3+(40000704-3916025\pi^2)\nu^2 \\
&+(22640704-3160350\pi^2)\nu-1190400\Big),
\end{aligned} \tag{213}
$$

$$
\begin{aligned}
L_{20}^{\text{target}} = -\frac{\nu}{3686400}\Big(&27648000\nu^3+89856000\nu^2+(21708480+7681425\pi^2)\nu \\
&-5340450\pi^2+114494656\Big).
\end{aligned} \tag{214}
$$

G^5/r^5 block.

$$L_{21}^{\rm target} = -\frac{1}{230400}\Big((555225\pi^2 - 6430720)\nu^2 + (1403325\pi^2 - 16243104)\nu - 14400\Big). \tag{215}$$

The translated ordinary target still satisfies the strict one-body gate exactly:

$$\begin{aligned} &L_1 \to \frac{7}{256}, && L_7 \to \frac{75}{128}, && L_{12} \to \frac{59}{16}, \\ &L_{16} \to \frac{203}{32}, && L_{19} \to \frac{31}{32}, && L_{21} \to \frac{1}{16}, \qquad (\nu \to 0), \end{aligned} \tag{216}$$

with every subleading slot vanishing in that same limit. So the translation preserves the strict one-body content exactly. The problem it introduces is not a one-body mismatch but new comparable-mass degree content in the ordinary interaction blocks.

F.3 The Hamiltonian-first lesson

The comparison that matters is now exact. From the raw local generic-frame scaffold one already knows that the constant-coefficient interaction basis has blockwise COM degree ceilings

$$\deg_\nu^{\max}(G/r, G^2/r^2, G^3/r^3, G^4/r^4, G^5/r^5) = (4,3,3,2,2). \tag{217}$$

The imported local Hamiltonian target has exactly the same interaction ceilings, so in the Hamiltonian chart the target and scaffold are degree-matched block by block.

The translated ordinary target, however, has

$$\deg_\nu^{\max}(G/r, G^2/r^2, G^3/r^3, G^4/r^4, G^5/r^5) = (4,4,4,4,2). \tag{218}$$

So the quartic compiler itself creates extra ν^4 tails in the ordinary chart precisely in the middle and upper local interaction blocks,

$$G^2/r^2, \qquad G^3/r^3, \qquad G^4/r^4, \tag{219}$$

while leaving the top static G^5/r^5 block at the same ceiling as before.

This gives the exact chart lesson:

$$\boxed{\text{The fixed local generic-frame lift at 4PN must be performed in the Hamiltonian chart first.}} \tag{220}$$

The reason is not stylistic. It is structural. In the Hamiltonian chart, the imported local target matches the constant-coefficient generic-frame scaffold blockwise and can therefore be lifted directly before any further translation. In the ordinary chart, the translated target contains new ν^4 tails that are absent from the fixed Hamiltonian target and are not reachable from the raw constant-coefficient ordinary scaffold.

So the correct local-first program is

$$\boxed{\begin{array}{c}\text{fixed local Hamiltonian target} \;\longrightarrow\; \text{generic-frame Hamiltonian lift} \\ \longrightarrow \text{ordinary-chart translation only afterward}\end{array}} \tag{221}$$

The tail bridge remains separate throughout. Nothing in the Hamiltonian-first lesson relaxes the local/nonlocal split, and nothing here changes the fact that the hereditary coefficient must be handled independently through the quadrupole bridge of Section 6.

At the same time, the lesson should not be overstated. It does *not* yet settle the full local ordinary generic-frame representative. It settles the order in which the local problem must be attacked. The actual Hamiltonian-chart generic-frame residual, its canonical slice, the ordinary residual sign-flip translation, and the aligned-seed correction are subsequent steps. What this appendix establishes is the chart logic that makes all of those later steps both necessary and mathematically clean.

G Hamiltonian-chart generic-frame lift

Section 5.3 used only the headline conclusion of this stage: once the fixed local target is imported in the Hamiltonian chart, the generic-frame local 4PN comparable-mass lift has no remaining span obstruction. The purpose of the present appendix is to freeze that result in full. Relative to the raw scaffold of Appendix E and the imported target of Appendix F, the exact question is the following: after subtracting the natural one-body/self-static seed, does the constant-coefficient exchange-symmetric generic-frame scaffold span the full fixed-chart local center-of-mass (COM) Hamiltonian residual, or does a further local obstruction survive?

The answer is completely sharp. In the Hamiltonian chart, the answer is yes. The free slot is absorbed exactly by the natural Hamiltonian seed, the interaction residual is polynomially reachable block by block, and the only remaining ambiguity is a 92-direction COM-blind null family. In particular, there is no Hamiltonian-chart analogue of the ordinary-chart ν^4-tail obstruction. The only nontrivial task left after the present appendix is to choose a canonical quotient representative inside the null family; that is the role of Appendix H.

G.1 Natural Hamiltonian seed and exact free-slot absorption

The Hamiltonian-chart lift begins from the exact one-body 4PN Hamiltonian gate already frozen in Appendix F. In reduced COM variables, write

$$X_A = \frac{1+\Delta}{2}, \qquad X_B = \frac{1-\Delta}{2}, \qquad X_A + X_B = 1, \qquad X_A X_B = \nu. \tag{222}$$

Then the natural reduced COM Hamiltonian self/static seed is

$$K^{\text{seed}} = \frac{7}{256}(X_A^9 + X_B^9), \tag{223}$$

$$Q_1^{\text{seed}} = \frac{45}{128}(X_A^8 + X_B^8), \qquad T_1^{\text{seed}} = \frac{13}{8}(X_A^7 + X_B^7), \qquad S_1^{\text{seed}} = \frac{105}{32}(X_A^6 + X_B^6), \tag{224}$$

$$U_1^{\text{seed}} = \frac{105}{32}(X_A^5 + X_B^5), \qquad W_1^{\text{seed}} = -\frac{1}{16}(X_A^4 + X_B^4), \tag{225}$$

with all subleading Hamiltonian slots initially set to zero.

This seed has one immediate exact consequence: it already absorbs the entire free 4PN COM Hamiltonian slot. Writing the fixed local COM Hamiltonian target as

$$H_{4,\text{loc}}^{\text{target,COM}} = K^{\text{target}} + H_{4,\text{loc}}^{\text{int,target,COM}}, \tag{226}$$

the free residual vanishes identically,

$$\Delta K \equiv K^{\text{target}} - K^{\text{seed}} = 0. \tag{227}$$

So the Hamiltonian-chart generic-frame lift problem is purely an interaction problem. No extra Hamiltonian free-slot repair survives beyond the natural one-body seed.

G.2 Exact polynomial coefficient-space formulation

The generic-frame Hamiltonian scaffold uses the same formal block families as the raw local scaffold,

$$\frac{Gpq}{r}\,\mathcal{Q}, \qquad \frac{G^2pq}{r^2}\,\mathcal{T}, \qquad \frac{G^3pq}{r^3}\,\mathcal{S}, \qquad \frac{G^4}{r^4}\,\mathcal{U}, \qquad \frac{G^5}{r^5}\,\mathcal{W}, \tag{228}$$

with the same raw block dimensions

$$\dim\mathcal{Q} = 52, \qquad \dim\mathcal{T} = 46, \qquad \dim\mathcal{S} = 29, \qquad \dim\mathcal{U} = 10, \qquad \dim\mathcal{W} = 2. \tag{229}$$

The only change is interpretive: in the present appendix these are read as Hamiltonian-side monomials in the Newtonian-order momentum invariants rather than as ordinary-chart velocity monomials.

Because the imported local Hamiltonian target has degree ceilings

$$\deg_\nu^{\max}(G/r,\, G^2/r^2,\, G^3/r^3,\, G^4/r^4,\, G^5/r^5) = (4,3,3,2,2), \tag{230}$$

exactly as recorded in Appendix F, the strict lift question is stronger than slot counting alone. One must ask whether the raw Hamiltonian scaffold spans the full *polynomial coefficient space* of the imported COM target. The blockwise coefficient-space dimensions are therefore

$$\begin{aligned}
\frac{G}{r}&: \quad 5 \text{ slots} \times (\nu,\nu^2,\nu^3,\nu^4) && \Rightarrow 20 \text{ coefficients},\\
\frac{G^2}{r^2}&: \quad 4 \text{ slots} \times (\nu,\nu^2,\nu^3) && \Rightarrow 12,\\
\frac{G^3}{r^3}&: \quad 3 \text{ slots} \times (\nu,\nu^2,\nu^3) && \Rightarrow 9,\\
\frac{G^4}{r^4}&: \quad 2 \text{ slots} \times (\nu,\nu^2) && \Rightarrow 4,\\
\frac{G^5}{r^5}&: \quad 1 \text{ slot} \times (\nu,\nu^2) && \Rightarrow 2.
\end{aligned} \tag{231}$$

Hence the full interaction coefficient space has total dimension

$$20 + 12 + 9 + 4 + 2 = 47. \tag{232}$$

This formulation is the correct exact lift test. The issue is not whether the scaffold spans fifteen reduced interaction slots after COM projection. The issue is whether it spans all forty-seven independent mass-ratio coefficients of the fixed local COM Hamiltonian residual.

G.3 Blockwise exact rank test and the Hamiltonian lift theorem

With the coefficient-space formulation fixed, the blockwise Hamiltonian-chart rank test is exact:

$$\begin{aligned}
\frac{G}{r}&: \quad 52 \to 20, \quad \text{nullity } 32,\\
\frac{G^2}{r^2}&: \quad 46 \to 12, \quad \text{nullity } 34,\\
\frac{G^3}{r^3}&: \quad 29 \to 9, \quad \text{nullity } 20,\\
\frac{G^4}{r^4}&: \quad 10 \to 4, \quad \text{nullity } 6,\\
\frac{G^5}{r^5}&: \quad 2 \to 2, \quad \text{nullity } 0.
\end{aligned} \tag{233}$$

So the interaction coefficient-space rank is maximal in every block, and the combined totals are

$$\text{rank} = 47, \qquad \text{nullity} = 32 + 34 + 20 + 6 + 0 = 92. \tag{234}$$

The local Hamiltonian lift theorem is then immediate.

Theorem 2 (Hamiltonian-chart generic-frame lift theorem)**.** *Let the fixed-chart local COM Hamiltonian target be imported with the natural Hamiltonian seed* (223)–(225) *and let the* *interaction residual be defined by subtracting that seed. Then the constant-coefficient exchange-symmetric generic-frame Hamiltonian scaffold spans the full local comparable-mass interaction residual. Equivalently, there exists at least one generic-frame Hamiltonian representative*

$$\Delta H^{\mathrm{gen}}_{4,\mathrm{loc}} = \frac{Gpq}{r} Q_{\mathrm{gen}} + \frac{G^2pq}{r^2} T_{\mathrm{gen}} + \frac{G^3pq}{r^3} S_{\mathrm{gen}} + \frac{G^4}{r^4} U_{\mathrm{gen}} + \frac{G^5}{r^5} W_{\mathrm{gen}} \tag{235}$$

such that

$$\Pi_{\mathrm{COM}}\left[\Delta H^{\mathrm{gen}}_{4,\mathrm{loc}}\right] = \Delta H^{\mathrm{target,COM}}_{4,\mathrm{loc}} \tag{236}$$

slot by slot and coefficient by coefficient. No further Hamiltonian seed refinement is required before the generic-frame comparable-mass lift is performed.

Proof. Equation (227) removes the free slot, so only the interaction residual remains. The interaction coefficient space has dimension 47 by (232). The exact blockwise rank test (233) shows that the generic-frame Hamiltonian scaffold attains the full coefficient-space rank in every block. Hence the image of the scaffold coincides with the entire imported local COM Hamiltonian residual, which proves existence of at least one solution of (236). Because maximal rank is already achieved with the natural Hamiltonian seed fixed, no additional Hamiltonian-side seed dressing can be necessary at the level of comparable-mass existence. The only remaining freedom is the 92-direction kernel recorded in (234). □

Two consequences are worth stating explicitly. First, the previous ordinary-chart obstruction is not a fundamental local span failure. It is a chart artifact created by quartic translation. Second, the Hamiltonian-chart lift theorem is stronger than the earlier COM slot-counting statement: the lift closes not merely at the level of fifteen reduced interaction slots, but at the level of all forty-seven independent mass-ratio coefficients carried by those slots.

G.4 Why span is solved but quotienting is not

The present appendix closes the Hamiltonian-chart existence question and nothing more. The rank theorem shows that the generic-frame scaffold is large enough, but it also shows that the lift is highly nonunique. The exact residual family contains a 92-direction COM-blind null sector, so after (236) is solved one still has to choose a distinguished representative. That is not a span problem; it is a quotient problem.

This is precisely why the next appendix is logically separate. The task of Appendix H is to identify the COM-null ideals, show that the null family is purely algebraic rather than physical, and select one exact sparse canonical quotient representative. The present appendix does not need that construction. Its theorem is already complete once the rank test is finished.

Finally, the chart lesson can now be stated in fully exact form. The generic- frame local lift has to be done in the Hamiltonian chart because only there do the imported degree ceilings match the constant-coefficient scaffold exactly. The translated ordinary target carries extra ν^4 tails in the middle and upper blocks, whereas the Hamiltonian target does not. So the correct local- first route is

$$\boxed{\begin{array}{c}\text{fixed local Hamiltonian target} \longrightarrow \text{generic-frame Hamiltonian lift} \\ \text{canonical quotient slice} \longrightarrow \text{ordinary translation only afterward}\end{array}} \tag{237}$$

That is the precise carry-forward conclusion of the present appendix.

H Canonical comparable-mass local residual slice

Appendix G established the exact Hamiltonian-chart lift theorem: after subtracting the natural one-body/self-static seed, the full local 4PN comparable-mass Hamiltonian residual lies in the image of the constant-coefficient exchange-symmetric generic-frame scaffold, block by block. What remained open was no longer span, but quotient choice. The COM projection forgets a large family of generic-frame directions, so one must decide how to choose a clean representative inside that COM-blind null family.

The present appendix freezes that choice. It identifies the relevant COM-null ideals, records the exact 92-direction ambiguity, and then fixes one canonical representative by the simplest exact quotient rule available: in the Gauss–Jordan lift of each block, set every null coordinate equal to zero. This produces a sparse generic-frame Hamiltonian representative of the full local comparable-mass residual. The representative is exact, reduces to the imported fixed-chart COM target in every slot, and is the canonical starting point for the subsequent ordinary-chart translation.

H.1 COM-null ideals and the 92-direction ambiguity

We use the same formal generic-frame invariant symbols a, b, c, d, e, p, q as in Appendix G. The COM identities are

$$\mathcal{C}_1 = pa + qc, \qquad \mathcal{C}_2 = pc + qb, \qquad \mathcal{C}_3 = pd + qe, \tag{238}$$

$$\mathcal{C}_4 = ab - c^2, \qquad \mathcal{C}_5 = ae - cd, \qquad \mathcal{C}_6 = bd - ce. \tag{239}$$

These are the same algebraic COM-blind generators already familiar from the 3PN analysis. At 4PN, however, the null family is substantially larger.

From Appendix G, the blockwise Hamiltonian-chart nullities are

$$Q : 32, \qquad T : 34, \qquad S : 20, \qquad U : 6, \qquad W : 0, \tag{240}$$

so the total COM-blind ambiguity is

$$32 + 34 + 20 + 6 + 0 = 92. \tag{241}$$

The exact ideal-membership statement is sharper than the raw nullity count:

$$\mathcal{I}_{\mathrm{COM}}^{\mathrm{full}} = \langle \mathcal{C}_1, \mathcal{C}_2, \mathcal{C}_3, \mathcal{C}_4, \mathcal{C}_5, \mathcal{C}_6 \rangle, \qquad \mathcal{I}_{\mathrm{COM}}^{\mathrm{lin}} = \langle \mathcal{C}_1, \mathcal{C}_2, \mathcal{C}_3 \rangle. \tag{242}$$

Theorem 3 (Exact COM-null ideal theorem at local 4PN)**.** *For the Hamiltonian-chart local 4PN generic-frame scaffold, every Q-block null vector lies in the full COM ideal $\mathcal{I}_{\mathrm{COM}}^{\mathrm{full}}$, while every T-, S-, and U-block null vector already lies in the linear-momentum ideal $\mathcal{I}_{\mathrm{COM}}^{\mathrm{lin}}$. The W-block has zero nullity and is therefore fixed completely once the COM target is fixed.*

This theorem is the precise algebraic meaning of the statement that the Hamiltonian lift ambiguity is *purely* COM-blind. No further span defect or hidden upper-block freedom survives. The only remaining task is to pick one representative in the quotient by these ideals.

H.2 Exact Gauss–Jordan quotient slice

Let $B \in \{Q, T, S, U, W\}$ denote one interaction block. After choosing a raw generic-frame scaffold basis for that block, the exact coefficient-space lift is an affine linear system

$$M_B x_B = y_B, \tag{243}$$

where M_B is the blockwise COM image matrix and y_B is the imported COM Hamiltonian residual written in coefficient space. Because the ranks are maximal in every block, the solution set has the exact form

$$x_B = x_B^{\mathrm{part}} + \sum_{\alpha=1}^{n_B} \lambda_{B,\alpha}\, n_{B,\alpha}, \tag{244}$$

where x_B^{part} is one particular solution, $\{n_{B,\alpha}\}$ is a basis of the null space, and

$$n_Q = 32, \qquad n_T = 34, \qquad n_S = 20, \qquad n_U = 6, \qquad n_W = 0. \tag{245}$$

The canonical quotient slice used throughout the local 4PN chain is the simplest exact choice:

$$\lambda_{B,\alpha} = 0 \qquad \text{for every block } B \text{ and every null coordinate } \alpha. \tag{246}$$

Equivalently, one takes the exact Gauss–Jordan lift and discards every purely COM-blind null direction. This is not claimed as the only possible quotient choice; it is claimed to be the cleanest canonical one available once the Hamiltonian chart and the fixed COM target are frozen.

A useful structural consequence is sparsity. The resulting canonical representative uses only

$$Q:11, \qquad T:12, \qquad S:9, \qquad U:4, \qquad W:2 \tag{247}$$

nonzero scaffold directions. So the quotient choice removes the entire 92-direction COM-blind family while preserving one exact sparse representative of the same COM target.

H.3 Explicit canonical representative

Write the local comparable-mass Hamiltonian residual as

$$\Delta H^{\text{can}}_{4,\text{loc}} = \frac{Gpq}{r} Q_{\text{can}} + \frac{G^2pq}{r^2} T_{\text{can}} + \frac{G^3pq}{r^3} S_{\text{can}} + \frac{G^4}{r^4} U_{\text{can}} + \frac{G^5}{r^5} W_{\text{can}}. \tag{248}$$

Then one exact canonical quotient-slice representative is the following.

G/r block.

$$\begin{aligned} Q_{\text{can}} = &-\frac{11}{64}a^2b^2 + \frac{5}{256}(a^2bc + ab^2c) - \frac{3}{32}(a^2bd^2 + ab^2e^2) + \frac{1}{64}(a^2bde + ab^2de) \\ &- \frac{27}{64}(a^2c^2 + b^2c^2) - \frac{9}{64}(a^2d^2e^2 + b^2d^2e^2) + \frac{3}{128}(a^2de^3 + b^2d^3e) + \frac{3}{64}(a^2e^4 + b^2d^4) \\ &- \frac{5}{32}(ad^2e^4 + bd^4e^2) + \frac{5}{64}(ad^3e^3 + bd^3e^3) - \frac{35}{256}(d^5e^3 + d^3e^5). \end{aligned} \tag{249}$$

G^2/r^2 block.

$$\begin{aligned} T_{\text{can}} = &-\frac{87}{128}(a^2bp + ab^2q) - \frac{2227}{256}(a^2bq + ab^2p) + \frac{63}{64}(a^2cq + b^2cp) \\ &+ \frac{49}{16}(a^2d^2p + b^2e^2q) - \frac{435}{64}(a^2dep + b^2deq) + \frac{2435}{256}(a^2e^2p + b^2d^2q) \\ &- \frac{183}{16}(ad^2e^2p + bd^2e^2q) - \frac{8305}{768}(ad^2e^2q + bd^2e^2p) + \frac{889}{192}(ad^3eq + bde^3p) \\ &- \frac{5343}{1280}(d^3e^3p + d^3e^3q) + \frac{325}{128}(d^4e^2q + d^2e^4p) - \frac{369}{160}(d^5eq + de^5p). \end{aligned} \tag{250}$$

G^3/r^3 **block.**

$$
\begin{aligned}
S_{\text{can}} = {} & \left(-\frac{3307423}{28800} + \frac{40483\pi^2}{16384}\right)(a^2p^2 + b^2q^2) + \left(-\frac{211189}{19200} + \frac{2749\pi^2}{8192}\right)(a^2pq + b^2pq) \\
& + \left(-\frac{184897}{900} + \frac{34985\pi^2}{8192}\right) abpq + \left(\frac{139241}{1200} - \frac{12507\pi^2}{2048}\right)(ad^2p^2 + be^2q^2) \\
& + \left(\frac{63347}{1600} - \frac{1059\pi^2}{1024}\right)(ad^2pq + be^2pq) + \left(-\frac{716971}{9600} + \frac{10389\pi^2}{2048}\right)(adep^2 + bdeq^2) \\
& + \left(-\frac{16727}{384} - \frac{35655\pi^2}{16384}\right)(d^2e^2p^2 + d^2e^2q^2) + \left(-\frac{51193}{960} - \frac{36405\pi^2}{8192}\right) d^2e^2pq \\
& + \left(\frac{23533}{1280} - \frac{375\pi^2}{8192}\right)(d^3eq^2 + de^3p^2).
\end{aligned} \tag{251}
$$

G^4/r^4 **block.**

$$
\begin{aligned}
U_{\text{can}} = {} & \left(\frac{930047}{9600} - \frac{551243\pi^2}{49152}\right)(ap^2q + bpq^2) + \left(\frac{500761}{19200} - \frac{21837\pi^2}{8192}\right)(apq^2 + bp^2q) \\
& + \left(\frac{274387}{1600} - \frac{62047\pi^2}{49152}\right)(d^2p^2q + e^2pq^2) + \left(\frac{3401779}{57600} - \frac{28691\pi^2}{24576}\right)(d^2pq^2 + e^2p^2q).
\end{aligned} \tag{252}
$$

G^5/r^5 **block.**

$$
W_{\text{can}} = \left(-\frac{169799}{2400} + \frac{6237\pi^2}{1024}\right)(p^3q + pq^3) + \left(-\frac{609427}{3600} + \frac{44825\pi^2}{3072}\right) p^2q^2. \tag{253}
$$

These formulas are best read as a theorem witness. They are not a heuristic guess and they are not an approximate sparse fit. They are the exact output of the canonical quotient rule (246) applied to the full Hamiltonian-chart lift family.

H.4 Exact verification and status boundary

The decisive verification is immediate: after COM reduction, the representative (248) reproduces the imported fixed-chart local COM Hamiltonian residual in every block and every slot:

$$
\Pi_{\text{COM}}\left[\Delta H^{\text{can}}_{4,\text{loc}}\right] = \Delta H^{\text{target,COM}}_{4,\text{loc}}. \tag{254}
$$

More explicitly, all five Q-slots, all four T-slots, all three S-slots, both U-slots, and the single W-slot match exactly.

Theorem 4 (Canonical comparable-mass local residual slice)**.** *Within the Hamiltonian chart, the local 4PN comparable-mass residual admits an exact sparse generic-frame representative obtained by setting all COM-null coordinates to zero in the blockwise Gauss–Jordan lift. The resulting representative is given by* (248)–(253)*, contains only the nonzero scaffold counts* (247)*, and reduces exactly to the imported fixed-chart local COM Hamiltonian residual.*

What this appendix settles is therefore precise. The Hamiltonian-chart local existence problem is no longer open; it has been quotiented, canonically sliced, and frozen in explicit form. What it does *not* settle is the stronger question of fixed-chart uniqueness against the true full generic-frame ADM Hamiltonian target. For the purposes of the local-first program, however, that stronger issue is no longer the next bottleneck. The next clean move is the ordinary-chart translation of the canonical representative, with the hereditary tail bridge kept separate.

I Ordinary translation and aligned-seed correction

Appendices F, G, and H already settled the local 4PN problem in the chart in which it is naturally posed. The fixed instantaneous target was imported in the reduced COM Hamiltonian chart, the constant-coefficient exchange-symmetric generic-frame scaffold was shown to span its full comparable-mass interaction residual, and one sparse canonical Hamiltonian representative was selected by quotienting the 92-direction COM-blind null family. What remained open after those steps was not the Hamiltonian-side existence problem, but the exact relation between that canonical Hamiltonian representative and the ordinary local chart.

This appendix freezes that relation. There are two logically separate claims. First, once the lower-order ledger is held fixed and the ordinary 4PN seed is chosen in the *Hamiltonian-aligned* convention, the quartic comparable-mass residual translates by an exact sign flip,

$$\boxed{\Delta L^{\rm can}_{4,\rm loc} = -\Delta H^{\rm can}_{4,\rm loc}.} \tag{255}$$

Second, the earlier ordinary-chart obstruction is therefore not a residual-lift obstruction at all. It is the separate chart problem carried by the *seed-alignment correction*

$$\delta L_{4,\rm seed} = L^{(\rm aligned)}_{4,\rm seed} - L^{(\rm natural)}_{4,\rm seed}. \tag{256}$$

In particular, the true local ordinary obstruction is seed-sector in origin and remains completely separate from the hereditary/tail bridge.

I.1 Exact quartic residual compiler theorem

The quartic perturbative Legendre transform of Appendix D.3 has the schematic structure

$$H_4 = F[L_1, L_2, L_3](v_0) - L_4(v_0), \qquad v_0 = M^{-1}p, \tag{257}$$

where the feedback functional F depends only on the already frozen lower-order ordinary data. Therefore if the ordinary 4PN block is split as

$$L_4 = L_4^{\rm seed} + \Delta L_4, \tag{258}$$

and the seed is chosen to be the one obtained by translating the fixed local Hamiltonian seed through the exact quartic compiler, then the residual is the only genuinely new quartic object and it obeys an exact sign flip.

Theorem 5 (Quartic ordinary residual sign-flip theorem)**.** *Let the local 4PN ordinary chart be defined relative to the Hamiltonian-aligned seed. Then the exact comparable-mass local residual satisfies*

$$\boxed{\Delta H_4 = -\Delta L_4.} \tag{259}$$

In particular, for the canonical sparse representative selected in Appendix H,

$$\boxed{\Delta L^{\rm can}_{4,\rm loc} = -\Delta H^{\rm can}_{4,\rm loc}.} \tag{260}$$

Proof. Equation (257) is affine in the new quartic ordinary block L_4 and all lower-order feedback is frozen. If the quartic block is decomposed as in (258), then the seed is chosen precisely so that its quartic counterimage reproduces the fixed local Hamiltonian self/static package. Subtracting the seeded relation from the full quartic relation leaves only the residual piece, and the dependence on ΔL_4 enters linearly with coefficient -1. Hence (259) follows exactly, and (260) is its canonical local specialization. □

The practical consequence is immediate. The canonical sparse Hamiltonian representative

$$\Delta H^{\mathrm{can}}_{4,\mathrm{loc}} = \frac{Gpq}{r}Q_{\mathrm{can}} + \frac{G^2pq}{r^2}T_{\mathrm{can}} + \frac{G^3pq}{r^3}S_{\mathrm{can}} + \frac{G^4}{r^4}U_{\mathrm{can}} + \frac{G^5}{r^5}W_{\mathrm{can}} \tag{261}$$

constructed in Appendix H already determines a canonical sparse ordinary representative,

$$\Delta L^{\mathrm{can}}_{4,\mathrm{loc}} = -\frac{Gpq}{r}Q_{\mathrm{can}} - \frac{G^2pq}{r^2}T_{\mathrm{can}} - \frac{G^3pq}{r^3}S_{\mathrm{can}} - \frac{G^4}{r^4}U_{\mathrm{can}} - \frac{G^5}{r^5}W_{\mathrm{can}}. \tag{262}$$

So the comparable-mass local residual itself does not generate any new ordinary uncertainty once the aligned seed convention has been fixed.

I.2 Exact COM verification of the translated residual

The translated canonical ordinary residual must be compared with the exact fixed local COM ordinary target reconstructed in Appendix F. The essential point is that the comparison is not made relative to the earlier natural one-body ordinary seed, but relative to the *ordinary seed aligned with the chosen Hamiltonian self/static seed.* With that aligned seed, the COM comparison is exact block by block:

$$\frac{G}{r}:\ 5 \text{ slots match}, \qquad \frac{G^2}{r^2}:\ 4 \text{ slots match}, \qquad \frac{G^3}{r^3}:\ 3 \text{ slots match}, \tag{263}$$

$$\frac{G^4}{r^4}:\ 2 \text{ slots match}, \qquad \frac{G^5}{r^5}:\ 1 \text{ slot matches}. \tag{264}$$

So the comparable-mass residual survives quartic translation intact. The exact ordinary residual is already COM-correct before any further generic-frame rescue is attempted.

It is useful to record the degree ceilings of the translated residual in the same block language. Writing the five local interaction blocks in the order Q, T, S, U, W, the ordinary residual has ceilings

$$Q : (4,4,4,4,3), \qquad T : (3,3,3,3), \qquad S : (3,3,3), \qquad U : (2,2), \qquad W : (2). \tag{265}$$

Equivalently, the quartic ordinary translation introduces no new comparable-mass polynomial degree beyond what the Hamiltonian-chart lift already spans. This is another way of saying that the residual itself is not the source of the earlier ordinary-chart confusion.

I.3 The exact seed decomposition of the local ordinary target

Once the residual is isolated, the exact local ordinary target decomposes as

$$\boxed{L^{\mathrm{loc}}_{4,\mathrm{target}} = L^{\mathrm{natural}}_{4,\mathrm{seed}} + \delta L_{4,\mathrm{seed}} + \Delta L^{\mathrm{can}}_{4,\mathrm{loc}},} \tag{266}$$

with

$$\delta L_{4,\mathrm{seed}} \equiv L^{(\mathrm{aligned})}_{4,\mathrm{seed}} - L^{(\mathrm{natural})}_{4,\mathrm{seed}}. \tag{267}$$

Here $L^{(\mathrm{natural})}_{4,\mathrm{seed}}$ means the direct symmetric promotion of the exact one-body ordinary gate, while $L^{(\mathrm{aligned})}_{4,\mathrm{seed}}$ means the ordinary self/static seed selected by Hamiltonian-first quartic translation of the fixed local seed. Equation (266) is the clean post-translation statement of the local ordinary problem.

The free block is already fixed exactly by the aligned seed. In the twenty-one slot reduced COM ordinary basis of Appendix F, one has

$$L^{(\mathrm{aligned})}_1 = L^{\mathrm{target}}_1, \qquad L^{(\mathrm{aligned})}_2 = L^{(\mathrm{aligned})}_3 = L^{(\mathrm{aligned})}_4 = L^{(\mathrm{aligned})}_5 = L^{(\mathrm{aligned})}_6 = 0. \tag{268}$$

So the free local ordinary sector does not need a separate generic-frame repair. The only local ordinary mismatch not already absorbed by the translated residual is the seed-alignment correction itself.

At COM level, the *interaction* part of $\delta L_{4,\mathrm{seed}}$ is nonzero in the four lower interaction blocks and vanishes in the top static block:

$$Q_\delta \neq 0, \qquad T_\delta \neq 0, \qquad S_\delta \neq 0, \qquad U_\delta \neq 0, \qquad W_\delta = 0. \tag{269}$$

The pure-kinetic correction is separate and is not part of the local interaction lift. For the interaction part, the decisive blockwise mass-ratio ceilings are

$$\deg_\nu^{\max}(Q_\delta, T_\delta, S_\delta, U_\delta, W_\delta) = (4, 4, 4, 4, 0). \tag{270}$$

This statement should be read together with the fixed-target import of Appendix F. The point is not that the G/r block reaches degree 4—that was already allowed. The point is that the aligned-seed correction newly injects *ordinary-chart* ν^4 content into the middle and upper local blocks

$$\frac{G^2}{r^2}, \qquad \frac{G^3}{r^3}, \qquad \frac{G^4}{r^4}, \tag{271}$$

which is precisely the pattern that looked like an ordinary-chart obstruction in the earlier direct translation audit.

Corollary 6 (The ordinary obstruction is seed-sector only)**.** *The quartic ordinary-chart obstruction that appears after direct translation of the local target is not a failure of the comparable-mass generic-frame lift. It is exactly the seed-alignment correction $\delta L_{4,\mathrm{seed}}$ of* (267)*. Equivalently, after the Hamiltonian-chart comparable-mass residual has been translated by the sign-flip theorem, the only remaining local ordinary chart problem is the generic-frame realization of the aligned ordinary seed.*

Proof. The translated residual already matches the exact fixed-chart ordinary target in all interaction blocks when measured relative to the aligned seed, by (263)–(264). The aligned seed already fixes the entire free block by (268). Therefore the exact difference between the natural-seed decomposition and the aligned-seed decomposition is carried solely by $\delta L_{4,\mathrm{seed}}$. Its support and degree pattern are given by (269) and (270), so there is no second local ordinary obstruction beyond that seed-sector object. ☐

I.4 What this leaves for the next appendix

The local-first 4PN program is therefore split sharply into three parts:

$$\boxed{\text{(A) Hamiltonian-derived generic-frame residual: solved,}} \tag{272}$$

$$\boxed{\text{(B) ordinary seed-alignment correction: the remaining local chart problem,}} \tag{273}$$

$$\boxed{\text{(C) tail / hereditary quadrupole bridge: still completely separate.}} \tag{274}$$

Part (A) is finished by the Hamiltonian-chart lift and the exact sign-flip translation theorem. Part (C) belongs to the hereditary bridge and is untouched by the present appendix. The only local chart work that remains is therefore Part (B): realizing $\delta L_{4,\mathrm{seed}}$ as an exact generic-frame object.

That is the content of the next local appendix. Once the aligned-seed correction is lifted in generic-frame form and added back to the natural seed, the already translated canonical residual can be inserted without further local ambiguity. The ordinary-chart story then closes, and the remaining 4PN conditionality is pushed entirely onto the separate passive/outgoing quadrupole normalization problem.

J Structured generic-frame lift of the aligned-seed correction

Appendix I already isolated the exact local ordinary-chart problem. Once the lower-order ledger is frozen and one works relative to the Hamiltonian-aligned seed, the canonical comparable-mass local residual translates by exact sign flip and is therefore no longer the source of the ordinary-chart obstruction. What remains is the separate *seed-alignment correction*

$$\delta L_{4,\mathrm{seed}} = L_{4,\mathrm{seed}}^{(\mathrm{aligned})} - L_{4,\mathrm{seed}}^{(\mathrm{natural})}. \tag{275}$$

The purpose of the present appendix is to freeze the exact generic-frame lift of that correction. The conclusion is completely sharp: the earlier ordinary-chart obstruction is real, but it is narrowly seed-sector in origin, and it is removed exactly by a small structured enlargement of the seed spaces. Once that lift is performed, the natural generic-frame local seed, the lifted aligned-seed correction, and the already solved canonical ordinary residual combine to give one exact generic-frame ordinary candidate for the entire local instantaneous 4PN sector.

J.1 The actual local ordinary problem

The local interaction part of (275) lives in the same block language used throughout the local 4PN chain:

$$\delta L_{4,\mathrm{seed}}^{\mathrm{loc}} = \frac{G}{r}Q_\delta + \frac{G^2}{r^2}T_\delta + \frac{G^3}{r^3}S_\delta + \frac{G^4}{r^4}U_\delta + \frac{G^5}{r^5}W_\delta. \tag{276}$$

At COM level this correction is nonzero in the lower four interaction blocks and vanishes in the top static block:

$$Q_\delta \neq 0, \qquad T_\delta \neq 0, \qquad S_\delta \neq 0, \qquad U_\delta \neq 0, \qquad W_\delta = 0. \tag{277}$$

Its decisive mass-ratio degree pattern is equally simple. The induced COM interaction correction has degree 4 in ν in each of the nonvanishing blocks,

$$Q_\delta : \deg_\nu = 4, \qquad T_\delta : \deg_\nu = 4, \qquad S_\delta : \deg_\nu = 4, \qquad U_\delta : \deg_\nu = 4, \qquad W_\delta = 0, \tag{278}$$

while the pure-kinetic COM correction remains separate and plays no role in the local interaction lift.

This is the precise sense in which the remaining local ordinary issue was never another comparable-mass residual-lift problem. It was the additional ordinary chart correction induced by aligning the local seed with the Hamiltonian-first translation.

J.2 Why the old ordinary scaffold failed

The ordinary comparable-mass interaction scaffold inherited from the earlier local span audit consists of the constant-coefficient exchange-symmetric block families

$$\mathcal{Q}, \qquad \mathcal{T}, \qquad \mathcal{S}, \qquad \mathcal{U}, \qquad \mathcal{W}, \tag{279}$$

with mass degrees $0, 1, 2, 3, 4$ respectively in the invariant variables (a, b, c, d, e, p, q). That scaffold is sufficient for the Hamiltonian-chart comparable-mass lift and, after Appendix I, also for the translated canonical ordinary residual. It does *not*, however, lift the seed-alignment correction.

The exact blockwise verdict is

$$Q_\delta \in \mathcal{Q}, \qquad W_\delta = 0, \tag{280}$$

but

$$T_\delta \notin \mathrm{Im}(\mathcal{T}), \qquad S_\delta \notin \mathrm{Im}(\mathcal{S}), \qquad U_\delta \notin \mathrm{Im}(\mathcal{U}). \tag{281}$$

So the earlier ordinary-chart obstruction was genuine, but it was already much narrower than a new local existence problem: it sat entirely in the seed sector.

J.3 Exact structured seed-sector lift theorem

The exact cure is to enlarge only the seed spaces, and only in a very specific way.

Theorem 7 (Structured generic-frame lift theorem for the aligned-seed correction)**.** *Let $\delta L^{\mathrm{loc}}_{4,\mathrm{seed}}$ be the local ordinary seed-alignment correction defined by (275). Then the correction is exactly generic-frame liftable in the structured seed spaces*

$$Q_\delta \in \mathcal{Q}, \qquad T_\delta \in \mathcal{T} \oplus (pq)\,\mathcal{T}, \qquad S_\delta \in \mathcal{S} \oplus (pq)\,\mathcal{S}, \tag{282}$$

$$U_\delta \in \mathcal{U} \oplus (p^2q^2)\,\mathcal{U}, \qquad W_\delta = 0. \tag{283}$$

Moreover, the enlarged blockwise coefficient-space ranks are full:

$$Q : \mathrm{rank} = 20/20, \qquad T : \mathrm{rank} = 16/16, \qquad S : \mathrm{rank} = 12/12, \qquad U : \mathrm{rank} = 8/8. \tag{284}$$

Proof. The starting point is the failure statement (281). The ordinary seed-alignment correction has the same exchange symmetry and strict one-body separation properties as the earlier local scaffold, so the correct repair must preserve the block families rather than replacing them by unrelated new monomials. The exact audit shows that adjoining one pq-dressed copy of the old $\mathcal{T}$ family and one pq-dressed copy of the old $\mathcal{S}$ family, together with one (p^2q^2)-dressed copy of the old $\mathcal{U}$ family, is already sufficient to make the image ranks maximal. No enlargement is required in $\mathcal{Q}$, and the top block vanishes identically. Hence the correction is exactly liftable in the structured seed spaces listed in (282)– (283), with the full-rank statement of (284) following blockwise. □

The conceptual consequence is the important one. The seed-alignment correction is not an inexplicable extra obstruction. It is itself an exact generic-frame object, but only after the seed sector is dressed in the structured way above. This is why the ordinary-chart obstruction is sharp, narrow, and local rather than diffuse.

J.4 Explicit canonical representative of the aligned-seed correction

Using the exact Gauss–Jordan lift with all null coordinates set to zero yields one sparse canonical representative of the aligned-seed correction. In the structured spaces of Theorem 7, the nonzero-direction counts are

$$Q_\delta : 13, \qquad T_\delta : 14, \qquad S_\delta : 12, \qquad U_\delta : 8, \qquad W_\delta : 0. \tag{285}$$

One exact canonical representative is the following.

G/r block.

$$\begin{aligned} Q_\delta = -\frac{1}{32}\Big(&214a^3c - 698a^2b^2 - 476a^2bc - 122a^2bd^2 - 290a^2bde - 181a^2be^2 + 16a^2c^2 - 8a^2cd^2 \\ &- 18a^2d^2e^2 - 54a^2de^3 - 27a^2e^4 \\ &- 476ab^2c - 181ab^2d^2 - 290ab^2de - 122ab^2e^2 + 30ad^3e^3 + 15ad^2e^4 + 214b^3c \\ &+ 16b^2c^2 - 8b^2ce^2 - 27b^2d^4 - 54b^2d^3e - 18b^2d^2e^2 \\ &+ 15bd^4e^2 + 30bd^3e^3\Big). \end{aligned} \tag{286}$$

G^2/r^2 **block.**

$$T_\delta = -\frac{1}{96}\Big(2814a^2bp^2q + 1065a^2bp - 4725a^2bq + 2256a^2cq - 3495a^2d^2p^2q + 408a^2d^2p \\ + 8031a^2dep^2q - 1782a^2dep \\ + 2814ab^2pq^2 - 4725ab^2p + 1065ab^2q - 80ad^3eq + 2736ad^2e^2p^2q \\ - 592ad^2e^2p + 80ad^2e^2q + 2256b^2cp \\ + 8031b^2depq^2 - 1782b^2deq - 3495b^2e^2pq^2 + 408b^2e^2q \\ + 2736bd^2e^2pq^2 + 80bd^2e^2p - 592bd^2e^2q - 80bde^3p \\ + 432d^3e^3p^2q + 432d^3e^3pq^2 \\ + 576d^3e^3p + 576d^3e^3q\Big). \tag{287}$$

G^3/r^3 **block.**

$$S_\delta = \frac{1}{192}\Big(3672a^2p^3q + 4464a^2p^2q^2 \\ - (10660 + 3\pi^2)a^2p^2 + (1220 - 3\pi^2)a^2pq - 11304ad^2p^3q - 7464ad^2p^2q^2 \\ - (2460 - 9\pi^2)ad^2p^2 + (9\pi^2 - 2292)ad^2pq + 4464b^2p^2q^2 + 3672b^2pq^3 \\ + (1220 - 3\pi^2)b^2pq - (10660 + 3\pi^2)b^2q^2 \\ - 7464be^2p^2q^2 - 11304be^2pq^3 + (9\pi^2 - 2292)be^2pq \\ - (2460 - 9\pi^2)be^2q^2 \\ + 960d^3eq^2 \\ - 11848d^2e^2p^3q - 19136d^2e^2p^2q^2 \\ - 8512d^2e^2p^2 - 11848d^2e^2pq^3 \\ - 8512d^2e^2q^2 + 960de^3p^2\Big). \tag{288}$$

G^4/r^4 **block.**

$$U_\delta = -\frac{pq}{96}\Big(372ap^3q^2 + 108ap^2q^3 + (30\pi^2 - 1799)ap + (9\pi^2 - 1200)aq + 108bp^3q^2 + 372bp^2q^3 \\ + (9\pi^2 - 1200)bp + (30\pi^2 - 1799)bq \\ + 7740d^2p^3q^2 + 2340d^2p^2q^3 - (6953 + 96\pi^2)d^2p - (2688 + 27\pi^2)d^2q \\ + 2340e^2p^3q^2 + 7740e^2p^2q^3 - (2688 + 27\pi^2)e^2p - (6953 + 96\pi^2)e^2q\Big). \tag{289}$$

G^5/r^5 **block.**

$$W_\delta = 0. \tag{290}$$

These formulas are not meant to be unique. They are one exact canonical choice inside the enlarged seed spaces. What matters for the theorem chain is that a representative exists and that its COM reduction reproduces the exact aligned seed correction slot by slot.

J.5 Natural seed, aligned seed, and one full local generic-frame candidate

The natural generic-frame local ordinary seed is the direct symmetric promotion of the exact one-body 4PN gate:

$$Q_{\text{nat}} = \frac{75}{128}(a^4 + b^4), \qquad T_{\text{nat}} = \frac{59}{16}(qa^3 + pb^3), \tag{291}$$

$$S_{\mathrm{nat}} = \frac{203}{32}(q^2a^2 + p^2b^2), \qquad U_{\mathrm{nat}} = \frac{31}{32}(q^3a + p^3b), \qquad W_{\mathrm{nat}} = \frac{1}{16}(q^4 + p^4). \tag{292}$$

Thus the aligned local seed is simply

$$L^{\mathrm{aligned,loc}}_{4,\mathrm{seed}} = L^{\mathrm{natural,loc}}_{4,\mathrm{seed}} + \delta L^{\mathrm{loc}}_{4,\mathrm{seed}}, \tag{293}$$

with $\delta L^{\mathrm{loc}}_{4,\mathrm{seed}}$ built from (276) and (286)– (290).

Combining that aligned seed with the already solved canonical ordinary residual of Appendix I yields one exact local ordinary generic-frame candidate,

$$\boxed{L^{(\mathrm{gen})}_{4,\mathrm{loc}} = L^{(\mathrm{natural,gen})}_{4,\mathrm{seed}} + \delta L^{(\mathrm{gen})}_{4,\mathrm{seed}} + \Delta L^{(\mathrm{can,gen})}_{4,\mathrm{loc}}.} \tag{294}$$

Its COM reduction reproduces the full fixed-chart local ordinary 4PN target block by block. Therefore the local-first 4PN program no longer has a local ordinary existence bottleneck.

More precisely, the logical chain is now complete:

1. the Hamiltonian-chart comparable-mass residual lift was solved in Appendices G and H;
2. the translated canonical ordinary residual was fixed exactly in Appendix I;
3. the remaining seed-alignment correction is solved by the structured seed-sector lift of the present appendix; and
4. the sum (294) reproduces the full fixed-chart local ordinary target.

So the only genuinely open local question left is no longer existence, but the sharper uniqueness/interpretation question for the aligned-seed sector itself.

K Exact local referee reconstruction

The previous appendices already froze all of the ingredients needed for the local 4PN theorem chain, but they did so in the logically correct order rather than as one end-to-end replay. Appendix F imported the fixed-chart local ADM Hamiltonian target and showed that the local problem must be handled *Hamiltonian-first*. Appendix G then solved the generic-frame Hamiltonian comparable-mass lift, and Appendix H froze one exact canonical representative of that Hamiltonian residual. Appendix I showed that the translated canonical ordinary residual obeys the exact quartic sign-flip rule, while Appendix J proved that the remaining ordinary-chart seed-alignment correction is itself exactly liftable in a minimal structured enlargement of the seed sector.

The purpose of the present appendix is to assemble those already-frozen pieces into one exact referee reconstruction of the *entire local instantaneous* ordinary 4PN sector. The conclusion is sharper than a consistency check. There is now one explicit generic-frame ordinary representative whose center-of-mass reduction reproduces the full fixed-chart local ordinary target slot by slot. So the local 4PN program is no longer blocked by local existence. What remains open after the present appendix is the separate aligned-seed uniqueness / interpretation question and the tail / hereditary bridge.

K.1 Exact input data reused by the referee reconstruction

The referee reconstruction uses four exact inputs and nothing else.

First, let

$$L^{\mathrm{ord,target}}_{4,\mathrm{loc}} \tag{295}$$

be the fixed-chart local ordinary 4PN target obtained from the imported local ADM Hamiltonian target by the exact quartic Hamiltonian-to-ordinary compiler of Appendix D. By construction, this is the object whose local COM slots must be reproduced.

Second, write the natural and Hamiltonian-aligned local ordinary seeds as

$$L_{4,\mathrm{seed}}^{(\mathrm{natural})}, \qquad L_{4,\mathrm{seed}}^{(\mathrm{aligned})}, \tag{296}$$

and define their difference by

$$\delta L_{4,\mathrm{seed}} = L_{4,\mathrm{seed}}^{(\mathrm{aligned})} - L_{4,\mathrm{seed}}^{(\mathrm{natural})}. \tag{297}$$

The aligned seed already fixes the entire free block of the translated local ordinary target: the unique nonzero free slot matches exactly, while the remaining free slots vanish identically. So the free local sector does not require any additional generic-frame repair.

Third, let

$$\Delta H_{4,\mathrm{loc}}^{\mathrm{can}} \tag{298}$$

be the canonical generic-frame Hamiltonian comparable-mass residual frozen in Appendix H. Appendix I then gives the exact translated ordinary residual

$$\Delta L_{4,\mathrm{loc}}^{\mathrm{can}} = -\Delta H_{4,\mathrm{loc}}^{\mathrm{can}}. \tag{299}$$

So the true comparable-mass local residual is already settled in the ordinary chart: there is no remaining comparable-mass lift obstruction after the Hamiltonian-first step.

Fourth, Appendix J proved that the local seed-alignment correction (297) is exactly liftable in the structured seed spaces

$$Q, \qquad T \oplus (pq)T, \qquad S \oplus (pq)S, \qquad U \oplus (p^2q^2)U, \qquad W = 0. \tag{300}$$

Write one such exact generic-frame lift as

$$\delta L_{4,\mathrm{seed}}^{(\mathrm{gen})}. \tag{301}$$

This is the only place where the old ordinary-chart obstruction ever lived.

K.2 The referee reconstruction formula

With the exact pieces above in hand, define the generic-frame local ordinary candidate by

$$L_{4,\mathrm{loc}}^{(\mathrm{gen})} = L_{4,\mathrm{seed}}^{(\mathrm{natural,gen})} + \delta L_{4,\mathrm{seed}}^{(\mathrm{gen})} + \Delta L_{4,\mathrm{loc}}^{(\mathrm{can,gen})}. \tag{302}$$

This is the exact local referee reconstruction formula.

Its interpretation is transparent.

1. The natural generic-frame seed carries the exact one-body / self / static local 4PN content in the ordinary chart.
2. The lifted seed-alignment correction repairs the fact that the ordinary target must be aligned with the Hamiltonian-first translation rather than with the raw natural seed.
3. The translated canonical residual inserts the already-solved comparable-mass local content.

Because the three pieces are exact and because the quartic compiler is linear in the 4PN slot once the lower-order ledger is frozen, the only remaining question is direct slot-by-slot verification after COM reduction.

K.3 Exact local referee reconstruction theorem

The local ordinary target has a six-slot free block together with fifteen interaction slots distributed by block as

$$Q:5, \qquad T:4, \qquad S:3, \qquad U:2, \qquad W:1. \tag{303}$$

The referee reconstruction theorem is therefore most cleanly stated directly in that slot language.

Theorem 8 (Exact local referee reconstruction). *Let $L_{4,\mathrm{loc}}^{\mathrm{ord,target}}$ be the fixed-chart local ordinary 4PN target defined by (295), and let $L_{4,\mathrm{loc}}^{(\mathrm{gen})}$ be the generic-frame candidate assembled in (302). Then the center-of-mass reduction of $L_{4,\mathrm{loc}}^{(\mathrm{gen})}$ reproduces the full local ordinary target exactly:*

$$(L_{4,\mathrm{loc}}^{(\mathrm{gen})})_{\mathrm{COM}} = L_{4,\mathrm{loc}}^{\mathrm{ord,target}}. \tag{304}$$

Equivalently,

$$\Delta Q = 0, \qquad \Delta T = 0, \qquad \Delta S = 0, \qquad \Delta U = 0, \qquad \Delta W = 0, \tag{305}$$

where each difference denotes the induced COM blockwise slot mismatch between (302) and the fixed-chart local ordinary target. In particular, the entire local instantaneous ordinary 4PN sector admits an exact generic-frame representative inside the declared hierarchy.

Proof. The free block is already fixed by the aligned seed, so there is nothing further to solve there. For the interaction blocks, split the target as in Appendix I into natural seed, seed-alignment correction, and canonical comparable-mass residual. The natural seed is carried unchanged by $L_{4,\mathrm{seed}}^{(\mathrm{natural,gen})}$. The seed-alignment correction is carried exactly by $\delta L_{4,\mathrm{seed}}^{(\mathrm{gen})}$ because of the structured-lift theorem of Appendix J. The remaining comparable-mass local residual is carried exactly by $\Delta L_{4,\mathrm{loc}}^{(\mathrm{can,gen})}$ because the Hamiltonian-chart canonical representative exists by Appendix H and its ordinary translation is fixed by the exact sign-flip rule (299). By linearity of the COM slot map, the sum of the three reconstructed pieces therefore reproduces the COM image of the full target in each block. This gives (304) and hence the blockwise vanishing statement (305). □

The theorem is stronger than a mere existence statement in coefficient space. The exact referee audit checks the blockwise COM image directly. The local ordinary 4PN target is reproduced in all five interaction blocks with the exact slot counts of (303); no unresolved local coefficient remains after the assembly.

K.4 Blockwise replay of the exact referee checks

The end-to-end audit may be read as the following sequence of exact replay checks.

Free block. The aligned ordinary seed reproduces the entire free local block: the unique nonzero free coefficient matches the translated target exactly, and the remaining five free slots are identically zero. So the free ordinary local sector is already closed before any interaction lift is attempted.

Q **block.** The natural seed supplies the exact one-body G/r ordinary content, the structured seed lift contributes the required aligned-seed correction already inside Q, and the translated canonical residual supplies the remaining comparable-mass G/r content. The referee replay verifies that all five COM slots of the reconstructed Q block match the fixed-chart target exactly.

T **block.** The natural seed supplies the one-body G^2/r^2 content. The seed-alignment correction is not contained in the old constant-coefficient T family, but it is exactly liftable in the enlarged space $T \oplus (pq)T$. After inserting that lifted correction and the translated canonical residual, all four COM T slots match exactly.

S **block.** The same pattern repeats in the G^3/r^3 sector. The natural seed carries the exact one-body block, the aligned-seed correction is lifted in $S \oplus (pq)S$, and the translated canonical residual completes the comparable-mass part. The resulting S block matches the three target COM slots exactly.

U **block.** At G^4/r^4 the seed-alignment correction is lifted in the minimal enlarged space $U \oplus (p^2q^2)U$. Once that correction and the translated canonical residual are added to the natural seed, both COM U slots match the fixed-chart target exactly.

W **block.** The aligned-seed correction vanishes identically in the top static block, so no additional lift is required there. The natural static seed together with the translated canonical residual already reproduces the unique W-slot exactly.

So the referee reconstruction is genuinely end to end: every local interaction block is checked directly after COM reduction, and every block passes.

K.5 Canonical sparsity of the lifted representative

The end-to-end theorem above is an existence statement for the full local ordinary sector, but the referee archive also records one concrete sparse realization of the lifted seed-alignment correction. In the structured spaces (300), the exact Gauss–Jordan slice with all null coordinates set to zero yields the nonzero-direction counts

$$Q : 13, \qquad T : 14, \qquad S : 12, \qquad U : 8, \qquad W : 0, \tag{306}$$

inside the basis sizes

$$Q : 52, \qquad T : 92, \qquad S : 58, \qquad U : 20, \qquad W : 0. \tag{307}$$

These numbers do not by themselves prove uniqueness of the aligned-seed sector, but they do show that the local theorem is not merely formal. One explicit, sparse, exact representative exists and has been checked directly against the fixed-chart target.

K.6 What this appendix settles and what it leaves open

The result of the present appendix should be read carefully.

It *does* settle the local 4PN existence problem in the ordinary chart. The exact imported local target is reconstructed from the Hamiltonian-first compiler chain; the generic-frame comparable-mass local residual is already solved in the Hamiltonian chart and translates back by exact sign flip; the remaining aligned-seed correction is exactly liftable in the minimal structured seed spaces; and the combined generic-frame candidate reproduces the full local ordinary target after COM reduction.

It *does not* settle two sharper questions. First, it does not yet provide a final uniqueness / interpretation theorem for the aligned-seed sector beyond the exact sparse slice used by the referee archive. Second, it does not touch the separate hereditary bridge. That bridge remains isolated by construction and is controlled by the same passive / outgoing STF quadrupole normalization that already survived the 2.5PN analysis.

Accordingly, the correct post-referee status of the local chain is the compact statement

$$\boxed{L^{\mathrm{ord,target}}_{4,\mathrm{loc}} \text{ is already reconstructed exactly in the generic frame.}} \tag{308}$$

From this point onward, the mathematically clean open problem is no longer local existence. It is the separate tail / normalization interface.

L Hereditary/tail bridge and coefficient relation

This appendix records the detailed coefficient arithmetic behind the hereditary 4PN bridge. The local instantaneous 4PN sector is already fixed by the local referee chain, so the only remaining nonlocal issue is the coefficient of the time-symmetric tail functional. The point of the appendix is threefold. First, we freeze the standard GR hereditary structure and its local logarithmic shadow. Second, we show that the hereditary coefficient is tied exactly to the same canonically normalized STF quadrupole coefficient that already controls our carried 2.5PN Burke–Thorne channel. Third, we isolate the only remaining model-side ambiguity as a single scalar transport factor rather than a new tensor channel, a new generic-frame existence problem, or a new independent normalization datum.

L.1 GR reference structure

The conservative 4PN dynamics in GR splits into a local instantaneous sector plus a time-symmetric hereditary sector,

$$L^{\mathrm{cons}}_{4\mathrm{PN}} = L^{\mathrm{local}}_{4\mathrm{PN}} + L^{\mathrm{tail}}_{4\mathrm{PN}}. \tag{309}$$

On the action side the hereditary term may be written as

$$S_{\mathrm{tail}} = \frac{G^2 M}{5c^8}\,\mathrm{Pf}_{2s/c} \iint \frac{dt\,dt'}{|t-t'|} I^{(3)}_{ij}(t) I^{(3)}_{ij}(t'), \tag{310}$$

while the corresponding time-symmetric nonlocal Hamiltonian term is

$$H_{\mathrm{tail}}(t) = -\frac{G^2 M}{5c^8} I^{(3)}_{ij}(t)\,\mathrm{Pf}_{2s/c} \int_{-\infty}^{+\infty} \frac{dv}{|v|} I^{(3)}_{ij}(t+v). \tag{311}$$

So the universal GR hereditary coefficient is

$$C^{\mathrm{GR}}_{\mathrm{tail}} = \frac{G^2 M}{5c^8}. \tag{312}$$

At the level relevant for the local logarithmic shadow, the same hereditary sector contributes the local functional

$$F_{\mathrm{tail}}(t) = \frac{2}{5}\frac{G^2 M}{c^8}\left(I^{(3)}_{ij}(t)\right)^2. \tag{313}$$

This is already enough to see that the hereditary side is structurally much smaller than the local lift: it is one fixed STF-quadrupole functional with one scalar coefficient.

L.2 Universal Newtonian quadrupole shadow

Because the hereditary coefficient first enters at 4PN, its local logarithmic shadow is fixed by the Newtonian order-reduced STF quadrupole,

$$I_{ij} = \mu\,\mathrm{STF}(x_i x_j), \qquad M \equiv m_A + m_B, \qquad \mu \equiv \frac{m_A m_B}{M}, \qquad \nu \equiv \frac{\mu}{M}. \tag{314}$$

The exact third derivative is

$$I_{ij}^{(3)} = -\frac{2GM\mu}{r^3}\left(4x_{\langle i}v_{j\rangle} - 3\frac{\dot r}{r}x_{\langle i}x_{j\rangle}\right), \tag{315}$$

so that

$$(I_{ij}^{(3)})^2 = \frac{8}{3}\frac{G^2M^2\mu^2}{r^4}\left(12v^2 - 11\dot r^2\right). \tag{316}$$

Substituting (316) into (313) gives the exact local logarithmic tail shadow

$$\boxed{\frac{F_{\rm tail}}{\mu} = \frac{16}{15}\,\nu\,\frac{U^4}{c^8}\left(12v^2 - 11\dot r^2\right), \qquad U \equiv \frac{GM}{r}.} \tag{317}$$

Equation (317) shows that the hereditary shadow lives only in the degree-2 local G^4/r^4 block. That is the precise reason the tail problem is much smaller than the local generic-frame lift problem.

L.3 Exact bridge to the carried 2.5PN quadrupole coefficient

The 2.5PN program already isolated the correct canonical language for the surviving radiative branch. Let

$$\gamma_{\rm quad}^{\rm eff} = \mathcal{N}_Q\,\frac{a_{\rm th}^5}{27c_s^5} = \overline{\Gamma}_5 = 9\frac{\overline{K}_2^{5/2}}{\overline{K}_0^{3/2}} \tag{318}$$

be the canonically normalized STF quadrupole coefficient on that branch. In GR the corresponding Burke–Thorne coefficient is

$$\gamma_{\rm GR} = \frac{2G}{5c^5}. \tag{319}$$

The key arithmetic observation is that the GR hereditary coefficient is related exactly to the Burke–Thorne coefficient by

$$\boxed{C_{\rm tail}^{\rm GR} = \frac{GM}{2c^3}\,\gamma_{\rm GR}.} \tag{320}$$

Indeed,

$$\frac{GM}{2c^3}\,\frac{2G}{5c^5} = \frac{G^2M}{5c^8} = C_{\rm tail}^{\rm GR}.$$

So the conservative hereditary coefficient is not merely analogous to the carried 2.5PN odd quadrupole coefficient. It is fixed by the same canonical quadrupole coefficient through an exact bridge.

This upgrades immediately to the carried toy-model branch. The unique compatible conservative hereditary coefficient is

$$\boxed{C_{\rm tail} = \frac{GM}{2c^3}\,\gamma_{\rm quad}^{\rm eff}.} \tag{321}$$

Equation (321) is the central coefficient relation of the appendix.

L.4 Toy-model branch forms and the scalar transport factor

It is useful to isolate explicitly the only model-side ambiguity that can still survive at this stage. Let

$$\gamma_{\rm toy} \equiv \hat m_0^2\Gamma_5 \tag{322}$$

be the canonically normalized STF quadrupole coefficient inherited from the 2.5PN branch analysis. Then one may parameterize the reduced hereditary transport by a single scalar factor $\Theta_{\rm tail}$ via

$$\boxed{\alpha_{\rm tail}^{\rm toy} = \Theta_{\rm tail}\frac{GM}{2c_s^3}\gamma_{\rm toy}.} \tag{323}$$

On the canonical branch with

$$c_s = c, \qquad \gamma_{\rm toy} = \gamma^{\rm eff}_{\rm quad}, \tag{324}$$

Eq. (323) reduces to (321) when

$$\Theta_{\rm tail} = 1. \tag{325}$$

So the only possible remaining mismatch on the hereditary side is a single scalar monopole-scattering transport coefficient. No new tensor channel is needed, and no new generic-frame existence solve is needed.

Using (318), the same bridge may be written in the compact toy-model form

$$C^{\rm toy}_{\rm tail} = \frac{GM}{2c^3}\,\gamma^{\rm eff}_{\rm quad} = \frac{GM}{2c^3}\,\overline{\Gamma}_5 = \frac{9GM}{2c^3}\,\frac{\overline{K}_2^{5/2}}{\overline{K}_0^{3/2}}. \tag{326}$$

Thus the entire hereditary 4PN bridge is controlled by the same normalized quadrupole branch data as the 2.5PN odd channel.

L.5 Why no new independent 4PN normalization datum opens

The appendix may now be summarized as the following coefficient theorem inside the declared hierarchy.

Coefficient theorem. If the canonically normalized orbital/worldtube STF quadrupole branch satisfies the passive/outgoing low-frequency relations already isolated by the 2.5PN audit, then

1. the unique compatible conservative hereditary coefficient is (321),
2. the hereditary 4PN sector introduces no new independent quadrupole normalization datum, and
3. full GR matching of the hereditary coefficient is equivalent to the already-known 2.5PN normalization target

$$\gamma^{\rm eff}_{\rm quad} = \frac{2G}{5c^5}. \tag{327}$$

The moving-throat normalization language used later in the paper rewrites the same condition as

$$\mathcal{N}_Q^{\rm target} = \frac{54Gc_s^5}{5a_{\rm th}^5 c^5}, \tag{328}$$

and, on the source-map notation of the reduced PDE program, equivalently as

$$\hat{m}_0^2 P_0 = \frac{54Gc_s^5}{5a_{\rm th}^5 c^5}. \tag{329}$$

Substituting (327) into (321) gives

$$C_{\rm tail} = \frac{GM}{2c^3}\,\frac{2G}{5c^5} = \frac{G^2M}{5c^8} = C^{\rm GR}_{\rm tail}. \tag{330}$$

So once the already-isolated passive/outgoing STF quadrupole normalization is fixed on the natural moving-throat branch, the 4PN hereditary coefficient closes automatically.

What remains open is therefore maximally narrow. The present appendix does not claim a first-principles moving-throat PDE derivation of the passive/outgoing quadrupole normalization, nor the final insertion of the normalized tail module into a single end-to-end referee/master script. But it does show that the full remaining 4PN hereditary gap is exactly the same passive/outgoing quadrupole normalization theorem already inherited from the 2.5PN program.

M Verification artifact ledger and reference run summaries

This appendix records the supplementary referee archive for the present 4PN paper. Its role is narrower than the analytic derivation in the main text. The archive is not presented as a black-box generator of the theorem. Instead, it serves as an executable and symbolic backstop for the exact one-body gate, the quartic compiler, the local Hamiltonian-first lift, the aligned-seed repair, the local referee reconstruction, and the hereditary bridge formulas used in the paper. The archive therefore verifies the paper *within the declared closure hierarchy*; it does not replace any analytic step, and it does not by itself upgrade the present work into an assumption-free theorem of a fully solved moving-throat bulk PDE.

The verification logic at 4PN is naturally split into three layers. The first layer is the inherited lower-order archive carried forward from the parent 4D, 1PN, 2PN, 2.5PN, and 3PN packages. The second layer is the dedicated dual-CAS 4PN archive, whose job is to check the new theorem chain specific to the present paper: the strict one-body gate, the quartic ordinary / Hamiltonian compiler, the local scaffold and fixed-chart import, the Hamiltonian-first generic-frame lift, the aligned-seed correction, the local referee reconstruction, and the hereditary bridge. The third layer is the conditional end-to-end replay, which ties the already-frozen 2.5PN quadrupole narrowing to the exact local 4PN closure and the hereditary bridge, and thereby verifies that the only remaining 4PN interface is the same passive/outgoing STF quadrupole normalization already isolated on the 2.5PN side.

M.1 Reading rule for the archive

The archive should be read with the same claim-status firewall used throughout the paper. Exact symbolic statements are checked as exact identities. Reduced and chart-level statements are checked inside the same Hamiltonian-first and worldtube / small-body conventions declared in the main text. Closure-level statements are checked only inside that same hierarchy. So the archive is not being asked to prove a different theorem from the one stated in the paper. It is being asked to verify that the paper's own exact, reduced, and closure-dependent claims close where they are claimed to close.

M.2 Inherited reference masters

The 4PN archive is not standalone in the historical sense. It sits on top of four already-frozen referee layers.

1. **Parent 4D and full conservative 1PN archive.** This is the inherited lower-order freeze that carries the exact projection backbone, Newtonian mass-dressing coefficient $\kappa_\rho = 1$, optical matching $n = 5$, the added-inertia ledger $\kappa_{\rm add} = 1/2$, the adiabatic $\kappa_{\rm PV} = 3/2$ closure, the parity-even wake coefficients, and the exact conservative EIH equality.

2. **Full conservative 2PN archive.** This is the inherited referee archive that freezes the one-body denominator closure, the self / static seed, the comparable-mass residual solve, and the exact equality to the standard generic-frame ADM Hamiltonian within the declared hierarchy.

3. 2.5**PN master audit.** This is the inherited audit that narrowed the universal dissipative bridge to the canonically normalized orbital / worldtube STF quadrupole branch and reduced the remaining theorem gap to the passive/outgoing quadrupole normalization.

4. **Full conservative 3PN referee master.** This is the carried archive that closes the grouped real P_2 middle block, the geometry-side static completion, and the pure-kinetic collapse beneath the same 2.5PN outgoing-normalization bridge.

These inherited archives matter because the present 4PN package is not trying to reopen lower-order viability conditions. It is trying to prove that no new independent normalization datum opens at 4PN beyond the already-known 2.5PN quadrupole gate.

M.3 Dedicated 4PN stage ledger

The dedicated 4PN publication bundle is organized around the following named stage audits. For each artifact we record both its theorem role and the short reference summary against which reruns should be checked.

1. **`4pn_onebody_audit.py`.** *Role:* exact Schwarzschild one-body gate through order c^{-8}. *Reference summary:* the strict one-body target forces the minimal repair ledger

$$\mu_{\rho,4} = \frac{1}{8}, \qquad d_4 = \frac{205}{16}, \qquad s_{34} = -\frac{15}{32}, \qquad s_{26} = -\frac{1}{16}.$$

2. **`4pn_quartic_legendre_audit.py`.** *Role:* exact quartic perturbative Legendre-transform compiler. *Reference summary:* the audit verifies the exact coefficients H_1 through H_4 and freezes the quartic ordinary / Hamiltonian map used everywhere later in the local chain.

3. **`4pn_local_scaffold_audit.py`.** *Role:* raw local generic-frame interaction scaffold and COM completeness test. *Reference summary:* the raw exchange-symmetric block sizes are

$$52, \quad 46, \quad 29, \quad 10, \quad 2,$$

for the G/r, G^2/r^2, G^3/r^3, G^4/r^4, and G^5/r^5 blocks, respectively, while COM image ranks are already maximal,

$$5, \quad 4, \quad 3, \quad 2, \quad 1,$$

with total COM-nullity 124. So there is no early local span shortage.

4. **`4pn_local_target_import_audit.py`.** *Role:* fixed-chart local ADM target import. *Reference summary:* the strict one-body Hamiltonian gate matches the imported local target exactly, the local COM Hamiltonian has one free slot plus interaction slot structure $5+4+3+2+1$, and the upper local Hamiltonian blocks stop at ν^2 in the G^4/r^4 and G^5/r^5 sectors.

5. **`4pn_local_hamiltonian_to_ordinary_audit.py`.**
Role: exact reduced COM Hamiltonian-to-ordinary translation. *Reference summary:* the quartic compiler translates the fixed local Hamiltonian target into the exact local ordinary target, but the ordinary chart develops extra ν^4 tails in the G^2/r^2, G^3/r^3, and G^4/r^4 blocks. This freezes the Hamiltonian-first lesson.

6. **`4pn_hamiltonian_chart_generic_frame_lift_audit.py`.**
Role: Hamiltonian-chart generic-frame lift of the local comparable-mass residual. *Reference summary:* the exact blockwise coefficient-space ranks are maximal,

$$20, \quad 12, \quad 9, \quad 4, \quad 2,$$

so there is no Hamiltonian-side span obstruction.

7. **`4pn_hamiltonian_chart_canonical_slice_audit.py`.**
Role: canonical comparable-mass local residual slice. *Reference summary:* the audit quotients the 92-direction COM-blind null family and records one sparse canonical representative with blockwise nonzero counts

$$Q:11, \qquad T:12, \qquad S:9, \qquad U:4, \qquad W:2,$$

together with exact COM verification.

8. **4pn_generic_frame_ordinary_translation_audit.py.** *Role:* translation of the canonical Hamiltonian residual back to the ordinary chart. *Reference summary:* the canonical comparable-mass local residual satisfies the exact sign-flip theorem

$$\Delta H_4 = -\Delta L_4,$$

and the translated target is measured relative to the Hamiltonian-aligned ordinary seed.

9. **4pn_generic_frame_aligned_seed_lift_audit.py.**
Role: generic-frame lift of the aligned-seed ordinary correction. *Reference summary:* the old ordinary obstruction is shown to be a seed-sector problem only, and the aligned-seed correction lifts exactly in the structured spaces

$$Q, \qquad T \oplus (pq)T, \qquad S \oplus (pq)S, \qquad U \oplus (p^2q^2)U, \qquad W = 0.$$

10. **4pn_local_referee_master_sympy_audit.py.** *Role:* end-to-end referee reconstruction of the full local ordinary target. *Reference summary:* the assembled generic-frame local ordinary candidate reproduces the fixed-chart local ordinary target slot by slot in the Q, T, S, U, and W blocks, so the entire local instantaneous 4PN sector is closed exactly inside the declared hierarchy.

11. **4pn_tail_bridge_audit.py.** *Role:* universal Newtonian quadrupole shadow of the conservative GR tail. *Reference summary:* the order-reduced STF quadrupole shadow is confined to the degree-2 G^4/r^4 block, so the hereditary problem is structurally much smaller than the local generic-frame lift.

12. **4pn_tail_hereditary_bridge_audit.py.** *Role:* exact coefficient bridge from the normalized STF quadrupole to the hereditary 4PN coefficient. *Reference summary:* the audit proves

$$C_{\text{tail}} = \frac{GM}{2c^3}\,\gamma_{\text{quad}}^{\text{eff}},$$

so the hereditary coefficient introduces no new independent normalization datum.

13. **4pn_conditional_referee_master_audit.py.** *Role:* conditional full conservative 4PN replay. *Reference summary:* the final replay ties the inherited 2.5PN quadrupole narrowing to the exact local 4PN closure and the hereditary bridge, and reduces the full remaining theorem gate to

$$\gamma_{\text{quad}}^{\text{eff}} \overset{?}{=} \frac{2G}{5c^5}, \qquad \hat{m}_0^2 P_0 \overset{?}{=} \frac{54Gc_s^5}{5a_{\text{th}}^5 c^5}.$$

M.4 Reference master reruns

For a referee-facing rerun, the shortest load-bearing replay is the following.

1. Run the carried `2_5pn_master_session_sympy_audit.py` archive and verify that the universal higher-order bridge is still narrowed to the canonically normalized orbital / worldtube STF quadrupole branch, with one passive/outgoing normalization gap left open.

2. Run `4pn_local_referee_master_sympy_audit.py` and verify that the full fixed-chart local ordinary target is reproduced exactly by one generic-frame local ordinary representative assembled from the natural seed, the aligned-seed lift, and the canonical comparable-mass residual.

3. Run `4pn_tail_hereditary_bridge_audit.py` and verify the bridge relation

$$C_{\text{tail}} = \frac{GM}{2c^3}\,\gamma_{\text{quad}}^{\text{eff}}.$$

4. Run `4pn_conditional_referee_master_audit.py` and verify that the only remaining global gate is the same passive/outgoing quadrupole normalization already inherited from the 2.5PN package.

These are the publication reference runs because they reproduce the paper's sharpest theorem-status statement in the smallest number of reruns. More local reruns are still useful when one wants to inspect a specific stage in detail, but the four-step replay above is the minimal end-to-end referee path.

For a second-engine replay, run the seven Mathematica scripts listed in `mathematica/wl_notes.txt` sequentially with `math-script`. Those grouped audits mirror the same theorem chain at coarser granularity and carry embedded `Output:` blocks so that the publication archive contains their live reference transcripts as well as the stagewise SymPy records.

M.5 What the archive verifies

The archive verifies the load-bearing symbolic statements of the paper sector by sector.

First, the front-end audits verify the exact one-body and compiler backbone. They check the strict Schwarzschild 4PN target, confirm that the carried 3PN self-sector packaging is insufficient by itself, solve the minimal repair ledger, and freeze the exact quartic perturbative Legendre-transform identity.

Second, the local middle audits verify the entire Hamiltonian-first comparable- mass closure chain. They check the raw local scaffold and COM completeness, the fixed local ADM target import, the chart-level degree-ceiling lesson, the Hamiltonian-chart generic-frame lift, the canonical residual slice, the exact ordinary sign flip, and the structured liftability of the aligned-seed correction. In the language of the main text, this is the stage chain that justifies the statement that the whole local instantaneous 4PN sector has an exact generic-frame ordinary representative inside the declared hierarchy.

Third, the local referee master verifies the end-to-end local assembly itself. It rebuilds the reduced COM local ordinary target from the imported local ADM Hamiltonian target, verifies the one-body gate and natural seed, replays the canonical residual translation and aligned-seed correction, and confirms that the assembled generic-frame candidate reduces back to the full fixed-chart local ordinary target slot by slot.

Fourth, the tail-side audits verify the narrow hereditary bridge. They confirm that the GR tail contributes a universal Newtonian quadrupole shadow in the local logarithmic block and that, on the canonically normalized STF quadrupole branch, the hereditary coefficient is exactly

$$C_{\text{tail}} = \frac{GM}{2c^3}\,\gamma_{\text{quad}}^{\text{eff}}.$$

So the hereditary side is not an independent 4PN coefficient hunt. It is an interface theorem tying the conservative hereditary coefficient to the same quadrupole normalization that already controls the 2.5PN odd sector.

Finally, the conditional master replay verifies the paper's sharpest global status statement: once the inherited 2.5PN narrowing, the exact local 4PN closure, and the hereditary bridge are replayed together, the only remaining full-4PN gate is the already-known passive/outgoing quadrupole normalization. In other words, the executable archive verifies the same conclusion as the analytic paper: no new independent normalization datum opens at 4PN.

M.6 What the archive does not verify

The archive does *not* replace the analytic argument in the main text, and it does not solve the full moving-throat PDE. In particular, it does not by itself derive the passive/outgoing

STF quadrupole normalization on the actual moving-throat branch; it does not prove from first principles that

$$\gamma_{\rm quad}^{\rm eff} = \frac{2G}{5c^5} \qquad \text{or} \qquad \hat{m}_0^{\,2} P_0 = \frac{54 G c_s^5}{5 a_{\rm th}^5 c^5}$$

holds; and it does not upgrade the present work into an assumption-free theorem of the full bulk dynamics.

It also does not attempt to settle every deeper microscopic interpretation issue that remains outside the present 4PN scope. The archive is not a proof of the full moving-throat origin of every grouped conservative datum, it is not a final uniqueness theorem for every aligned-seed local representative, and it is not a general radiative or higher-PN completion package. Those questions remain outside the stated theorem target of the paper.

So the correct reading rule is simple. The referee archive verifies that the paper's exact local theorem chain and hereditary bridge close where claimed, and that the whole conservative 4PN problem has been reduced to one already-known quadrupole-normalization gate. What it does not verify is the final PDE-side realization of that gate. That distinction is exactly the one on which the conditional status of the present 4PN theorem rests.